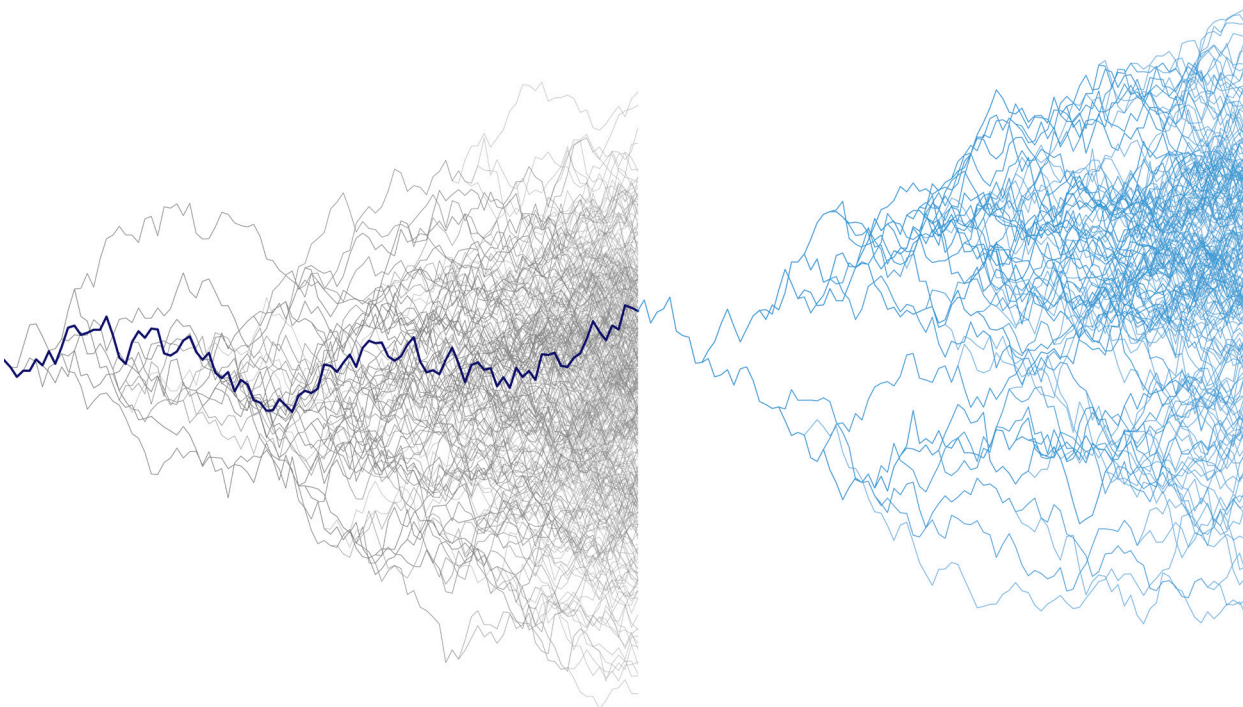
Doctoral Thesis in Machine Design

Automated driving in road freight transport

On system-level impacts, policy implications and the role of uncertainty

ALBIN ENGHOLM

KTH ROYAL INSTITUTE OF TECHNOLOGY



Automated driving in road freight transport

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ALBIN ENGHOLM

Academic Dissertation which, with due permission of the KTH Royal Institute of Technology, is submitted for public defence for the Degree of Doctor of Philosophy on Wednesday the 11th of December 2024, at 1:15 p.m. in Kollegiesalen, Brinellvägen 6, Stockholm.

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Dedicated to Rakel and Tage

I. Abstract

The freight transport system is expected to face significant changes driven by emerging technologies, increasing transport demand, and the need for rapid decarbonization. Automated driving systems and their application to road freight in the form of driverless trucks is one such technology that may influence the development. Driverless trucks could potentially enable cost-efficient, safe, and flexible transport solutions, provided that technical, regulatory, and operational challenges are overcome. However, there is significant uncertainty regarding their development trajectory, future use, and system-level impacts such as changes in transport costs, freight patterns, and mode shifts, as well as their implications for sustainable freight transport.

This thesis explores potential long-term system-level impacts of driverless trucks and implications for planning, policy, and sustainability, with a focus on the Swedish freight transport system. Four objectives are addressed. First, future scenarios for the freight transport system are developed, and an analysis of the Swedish innovation system for driverless trucks is performed. The results suggest that plausible initial deployments of driverless trucks are within confined areas, short-distance repetitive flows, and for highway driving between logistics facilities. The innovation process of driverless trucks is characterized by cooperation among a broad set of actors, and it is possible that driverless trucks will disrupt the value chain of road freight transport.

Second, the potential impacts on road transport costs are modeled, showing that driverless trucks could reduce costs by 20% or more, largely determined by the extent to which total labor costs can be reduced.

Third, system-level impacts are analyzed for a large set of introduction scenarios, using national freight transport modeling. The change in cost structure could lead to increased demand for road transport and shifts from rail and sea to road, which may have implications for infrastructure planning, policymaking, and environmental sustainability. Furthermore, driverless

trucks capable of operating on highways and strategically chosen access roads can address a substantial amount of freight demand and generate significant system impacts.

Finally, this thesis explores how model-based analysis of driverless trucks' cost performance, system-level impacts, and climate policy implications under uncertainty can be enhanced using exploratory modeling and methods for decision-making under deep uncertainty. The application of such methods demonstrates that they can contribute to a broader understanding of potential impacts and policy robustness. Several challenges for their introduction in national transport planning are identified, including the need for more flexible and faster models, and managing fundamental differences in approach compared to the current prediction-based planning paradigm.

This thesis contributes with research on the potential system-level impacts of driverless trucks, which may be of relevance for the freight transport industry as well as planners and policymakers at the national level. The research offers an initial, broad examination of a topic for which literature is scarce. Several areas for future research are identified, including the relationship between driverless trucks, electrification, and freight decarbonization; improving modeling of costs and operations of driverless trucks at the vehicle and fleet levels; as well as developing tools to support exploratory modeling and planning to handle uncertainty about the future.

Keywords

Driverless trucks, Freight modeling, System-level impacts, Exploratory modeling, Electrification, Deep uncertainty

II. Sammanfattning

Godstransportsystemet förväntas stå inför betydande förändringar, drivna av ny teknik, ökande transportefterfrågan och behovet av en snabb omställning till fossilfria transporter. Automatiserade körsystem, och deras tillämpning i form av förarlösa lastbilar är en sådan teknologi som kan påverka utvecklingen under de kommande årtiondena. Förarlösa lastbilar kan ha potential att möjliggöra kostnadseffektiva, säkra och flexibla transportlösningar, förutsatt att tekniska, regulatoriska och operativa utmaningar hanteras. Det finns dock betydande osäkerhet kring teknikens framtida utvecklingsriktning och användning, samt kring systemeffekter såsom förändrade transportkostnader, transportmönster och val av transportslag, och dess påverkan på omställningen till ett hållbart transportsystem.

Denna avhandling utforskar potentiella, långsiktiga systemeffekter av förarlösa lastbilar och implikationer för planering, policy och hållbarhet med fokus på det svenska godstransportsystemet. Avhandlingen fokuserar på fyra områden. För det första utvecklas framtidsscenarier för godstransportsystemet och en analys av innovationssystemet för förarlösa lastbilar genomförs. Resultaten tyder på att förarlösa lastbilar initialt kan komma att implementeras inom avgränsade områden, för korta repetitiva flöden och för motorvägskörning mellan logistikfaciliteter. Innovationsprocessen för förarlösa lastbilar kännetecknas av samarbete mellan en bred uppsättning aktörer och det är möjligt att förarlösa lastbilar kommer att förändra värdekedjan för vägtransporter.

För det andra modelleras potentiella effekter av förarlösa lastbilar på vägtransportkostnader. Resultaten visar att förarlösa lastbilar kan minska kostnaderna med 20 % eller mer, där magnituden till stor del beror på i vilken utsträckning de totala arbetskostnaderna kan reduceras.

För det tredje analyseras systemeffekter i ett stort antal introduktionsscenarier, med hjälp av nationell godstransportmodellering.

Förändringen i kostnadsstrukturen kan leda till ökad efterfrågan på vägtransporter och överflyttning från järnväg och sjöfart till väg, vilket kan ha konsekvenser för infrastrukturplanering, klimatstyrmedel och miljömässig hållbarhet. Vidare kan förarlösa lastbilar som kan köra på motorvägar samt strategiskt valda tillfartsvägar tillgodose en stor andel av transportefterfrågan.

Slutligen utforskar avhandlingen hur modellbaserad analys av förarlösa lastbilars kostnadsprestanda, systemeffekter och klimatpolitiska implikationer under osäkerhet kan förbättras genom användning av explorativ modellering och metoder för beslutsfattande under djup osäkerhet. Tillämpandet av sådana metoder visar att de kan bidra till en bredare förståelse av möjliga effekter och robustheten för olika styrmedel. Identifierade utmaningar för att införa sådana metoder i nationell transportplanering inkluderar behovet av mer flexibla och snabbare modeller samt att hantera grundläggande skillnader i angreppssätt jämfört med det nuvarande prognosbaserade planeringsparadigmet.

Avhandlingen bidrar med forskning om potentiella systemeffekter av förarlösa lastbilar, som kan vara relevant för transportindustrin samt för planering och policy på nationell nivå. Forskningen utgör en tidig, bred analys av ett område där litteraturen är begränsad. Flera områden för framtida forskning identifieras, vilka inkluderar relationen mellan förarlösa lastbilar, elektrifiering och fossilfria godstransporter; förbättrad modellering av kostnader, planering och användning av förarlösa lastbilar på fordons- och flottnivå; samt utveckling av verktyg som stödjer explorativ modellering och planering för att hantera osäkerhet om framtiden.

Nyckelord
Förarlösa lastbilar, Godsmodellering, Systemeffekter, Explorativ modellering, Elektrifiering

III. Acknowledgments

This is the last page I'll write as a student. I can’t think of a better way to use it than to express my gratitude to the people who have contributed to this final page being part of a PhD thesis. It wouldn’t have happened without you.

I have had three fantastic supervisors. Thank you, Anna Pernestål, for endless encouragement, inspiration, opportunities and for helping me focus on the end goal. Ida Kristoffersson, your guidance and patience has been invaluable; I am grateful for your support. Sofia Ritzén, thank you for thoughtful advice and perspective on the research. I also want to thank the advance reviewer, Zhenliang Ma, who provided valuable comments to improve the kappa.

I was lucky to spend the last three years at Einride among some of the smartest and most ambitious people I've ever met. Mehdi Akbarian, thank you for believing in me, guiding me, and challenging me intellectually. You are a role model of authenticity, friendliness, and curiosity. Thank you, Andreas Allström, for all your support and for bringing the best out of every situation; you are a rock. To everyone else at Einride, including Benjamin Kickhöfer, Anton Zackrisson, and many more - thanks for letting me learn from you; you are amazing.

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Most of my research has been funded by Trafikverket, and I am grateful for that support. I want to thank Peter Smeds for the friendly, patient, and open-minded collaboration, which I have enjoyed. I also want to express my gratitude to everyone else at Trafikverket who kindly contributed with data, knowledge, and support throughout the years.

Now I am at the final part of the final page, which I want to dedicate to my family. To my parents Lotta and Tomas for the foundation of love and trust upon which everything else rests. Dad, you have always been my biggest supporter, which has made me dare to take on challenging things. Mom, I believe you gave me the grit to push through these challenges, including this PhD. To my siblings Oskar, Hugo, Johanna, and Simon. I am a very proud brother, and I look up to each one of you. Thank you, Linnéa, for being patient with me. I love you and I am so fortunate to get to spend my life with you. Till Rakel och Tage: nu är boken klar, tänk att vi klarade det här tillsammans. Ingenting kan stoppa oss!

Albin Engholm
Stockholm
November 2024

IV. List of publications

The thesis includes seven academic papers, denoted Paper A to Paper G, hereafter referred to as the *thesis papers*. The papers are listed below along with a description of my author contribution using the Contributor Roles Taxonomy (CRediT)[1].

A. Pernestål, A., Engholm, A., Bemler, M., & Gidofalvi, G. (2020).
How will digitalization change road freight transport? Scenarios tested in Sweden.
Sustainability, 13(1), 304. *https://doi.org/10.3390/su13010304*
Albin Engholm: Conceptualization, formal analysis, investigation, writing - original draft.

B. Engholm, A., Björkman, A., Joelsson, Y., Kristoffersson, I., & Pernestål, A. (2020).
The emerging technological innovation system of driverless trucks.
Transportation Research Procedia, 49, 145–159.
https://doi.org/10.1016/j.trpro.2020.09.013
Albin Engholm: Conceptualization, methodology, investigation, writing - original draft.

C. Engholm, A., Pernestål, A., & Kristoffersson, I. (2020).
Cost analysis of driverless truck operations.
Transportation Research Record: Journal of the Transportation Research Board, 2674(9), 511–524. https://doi.org/10.1177/0361198120930228

[1]More information available here: https://www.elsevier.com/researcher/author/policies-and-guidelines/credit-author-statement

Albin Engholm: Conceptualization, methodology, formal analysis, investigation, data curation, writing - original draft, visualization.

D. Engholm, A., Kristoffersson, I., & Pernestål, A. (2021).
Impacts of large-scale driverless truck adoption on the freight transport system.
Transportation Research Part A: Policy and Practice, 154, 227–254.
https://doi.org/10.1016/j.tra.2021.10.014
Albin Engholm: Conceptualization, methodology, formal analysis, investigation, data curation, writing - original draft, visualization.

E. Engholm, A., Allström, A., & Akbarian, M. (2024).
Exploring cost performance tradeoffs and uncertainties for electric and autonomous electric trucks using computational experiments.
European Transport Research Review, 16(1), 41.
https://doi.org/10.1186/s12544-024-00662-0
Albin Engholm: Conceptualization, methodology, software, formal analysis, investigation, data curation, writing - original draft, visualization.

F. *Engholm, A., Frölander, S., Johansson, M., Kristofersson, F., & Kristoffersson, I. (Submitted).*
Impacts of electric and driverless heavy-duty trucks on the future decarbonized freight transport system: analyzing techno-economic uncertainty using exploratory modeling and analysis.
Submitted to Transportation Research Part A: Policy and Practice
Albin Engholm: Conceptualization, methodology, software, formal analysis, investigation, data curation, writing - original draft, visualization, project administration, funding acquisition.

G. Engholm, A., & Kristoffersson, I. (Under review).
Exploring many-objective robust decision making for managing uncertainty in climate policy analysis for the transport sector.
Under review at Transportation Research Interdisciplinary Perspectives.
Albin Engholm: Conceptualization, methodology, formal analysis, investigation, data curation, writing - original draft, visualization, project administration, funding acquisition.

V. Additional publications

In addition to the thesis papers, there are several papers and publications that I have contributed to during my PhD project relevant for the scope of the thesis.

- Engholm, A., Pernestål, A., & Kristoffersson, I. (2018). System-level impacts of self-driving vehicles: Terminology, impact frameworks and existing literature syntheses. Research report, KTH Royal Institute of Technology. https://www.diva-portal.org/smash/get/diva2:1268871/FULLTEXT02.pdf
- Engholm, A., Kristoffersson, I., & Pernestål Brenden, A. (2019). Is the driverless future sustainable? Strategic uncertainties and system impacts. Conference paper presented at the World Conference on Transport Research (WCTR) 2019, Mumbai, 26–31 May 2019. https://www.wctrs-society.com/wp-content/uploads/abstracts/mumbai/WCTR2019_Mumbai_240.pdf
- Pernestål, A., Engholm, A., Kristoffersson, I., & Jussila Hammes, J. (2020). The impacts of automated vehicles on the transport system and how to create policies that target sustainable development goals. In A. Paulsson & C. H. Sørensen (Eds.), Shaping smart mobility futures: Governance and policy instruments in times of sustainability transitions (pp. 37–53). Emerald Publishing Limited. https://doi.org/10.1108/978-1-83982-650-420201003
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- Stenemo, E., Raoofi, Z., Engholm, A., & Pernestål, A. (2021). Prestudy on system-level impacts of automation, electrification and digitalization for long-term transport analysis and planning. Research report, KTH Royal Institute of Technology. https://urn.kb.se/resolve?urn=urn:nbn:se:kth:diva-339573
- Raoofi, Z., Stenemo, E., Engholm, A., & Pernestål, A. (2023). How can we structure the future development of automation, electrification, and

digitalization in the transportation sector by using morphological analysis. Transportation Research Procedia, 72, 1808–1815. Conference paper presented at Transportation Research Arena 2022. https://doi.org/10.1016/j.trpro.2023.11.657

- Engholm, A., & Kristoffersson, I. (2024). MUST—Managing deep uncertainty in planning for sustainable transport. Project report: Phase 1. Research report, KTH Royal Institute of Technology. https://urn.kb.se/resolve?urn=urn:nbn:se:kth:diva-343507
- Zackrisson, A., Engholm, A., & Tang, O. (Manuscript). Data-driven analysis of strategic-operational interfaces in freight electrification under deep uncertainty. Under review at Transportation Research Part D: Transport and Environment.

VI. List of abbreviations

ADA	Aggregate-disaggregate-aggregate
CST	Charging strategy type
D-ET	Driverless electric truck
D-ICET	Driverless internal combustion engine truck
DT	Driverless truck
DMDU	Decision making under deep uncertainty
EMA	Exploratory modeling and analysis
ET	Electric truck
EU	European Union
GHG	Greenhouse gas emissions
ICET	Internal combustion engine truck
IL	Intuitive logics
LHS	Latin hypercube sampling
M-ET	Manual electric truck
M-ICET	Manual internal combustion engine truck
MOEA	Multi-objective evolutionary algorithm
MORDM	Many objective robust decision making

MT	Manually driven truck
ODD	Operational design domain
PRIM	Patient rule induction method
RDM	Robust decision making
SAV	Shared autonomous vehicles
STA	Swedish Transport Administration
TCO	Total cost of ownership
TIS	Technological innovation system
tkm	Tonne-kilometer
VKT	Vehicle kilometers travelled
ZET	Zero-emission truck

VII. Preface

This PhD thesis is a compilation thesis. This means that the thesis is composed of a set of research papers and this document, the thesis *kappa*. The research papers are academic articles that have either been published, or are in the process of being published, in scientific journals or have been presented at scientific conferences. The kappa is a different type of document and serves multiple purposes: to introduce the overarching aim and research problems addressed in the thesis, to discuss theoretical and methodological considerations, to outline how the research papers complement each other in addressing the research objectives, to summarize the research papers, and to discuss the results and implications of the conducted research.

I want to stress that the kappa complements the thesis papers and that both are integral parts of the thesis. Thus, reading only the kappa will not provide a complete and detailed understanding of the thesis and its findings. While the kappa is designed to stand alone by providing a comprehensive overview of the research and its implications, it is most effective when read in conjunction with the detailed analyses presented in the thesis papers.

I have learned many skills during my PhD. However, upon completing the kappa, it is evident that I can improve on writing concisely. Regardless of whether you read the whole thesis or just parts of it, I am glad to have you as a reader and I hope you will find the thesis interesting.

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1 Introduction

This thesis contributes to the development of knowledge and analysis methods to address uncertainty about the future impacts of automated driving technology and its application to road freight transport in the form of driverless trucks. The remainder of this chapter motivates the need for this research, the aim and objectives of the thesis, and describes the overall research design and delimitations.

1.1 The need for research on driverless truck impacts

Road freight transport is a large industry constituting about 3% of Swedish GDP, but operates at low profit margins, with driver costs typically accounting for approximately 40% of total costs (Rementum, 2024). Basic economic rationales thus suggest that trucks that do not require an on-board driver to operate would be an attractive technology. This is arguably one, although far from the only, motivation behind the ongoing efforts to develop and deploy driverless trucks. Driverless trucks could potentially enable cost-effective, safe, and flexible transport solutions, provided that technical, regulatory, and operational challenges are overcome.

However, there is significant uncertainty regarding driverless trucks' development trajectory, future use, and impacts on the transport system such, as changes in transport costs, freight patterns, and mode shifts, as well as their implications for the transition to a sustainable transport system. Despite the significant industry interest and an array of potential impacts of policy relevance, research on impacts of driverless trucks on the transport system is scarce. Given that this is a technology still in development, to be deployed sometime in the future in a system that is expected to undergo significant change, any such research is tainted with uncertainty.

1.1.1 Change and uncertainty in future freight transport

The freight transportation system is under pressure to change. This pressure comes from multiple sources, including technological developments within digitalization and automation (Dong et al., 2021; Stenemo et al., 2021), increasing global freight demand (International Transport Forum, 2023), and shifting trade patterns and consumer preferences (L. A. Tavasszy, 2020). At the same time, the urgent need to decarbonize the freight transport sector to limit global warming to levels stipulated by the Paris Agreement requires drastic changes in freight transport (European Scientific Advisory Board on Climate Change, 2023). This change includes a rapid large-scale rollout of battery electric trucks (ETs) and other zero-emission truck (ZET) technologies (Börjesson & Eliasson, 2024; Rogstadius et al., 2024).

Freight transport can be understood as a complex sociotechnical system shaped by decisions of multiple types of actors operating with different objectives and on varying timescales (Auvinen & Tuominen, 2014; L. Tavasszy & de Jong, 2014). Understanding change in sociotechnical systems comes with inherent uncertainties (Holtz et al., 2015; Köhler et al., 2019), and with uncertainty comes the need for appropriate analysis methods and decision support to assist stakeholders and policy makers to govern the system toward a sustainable pathway (Holtz et al., 2015; Lempert et al., 2003).

While it is plausible, or even likely, that the freight transport sector will face substantial change during the coming decades, it is uncertain how these changes will unfold and what their effects will be. The many uncertain trends, their interaction with unpredictable responses to climate and sustainability policies, and the structural uncertainties associated with technological change and transitions in sociotechnical systems all contribute to uncertainty about the future of freight transport (Holtz et al., 2015; Köhler et al., 2019; N. Rosenberg, 1998; Trosvik et al., 2023).

1.1.2 Analyzing future freight transport under uncertainty

When the future is highly uncertain, the relevance for planning approaches relying on predicting the most likely development becomes questionable. When several distinct development trajectories are plausible, it may not be possible for analysts and experts to agree on a probabilistic assessment of which trajectories are more or less likely to occur – this is a state referred to as *deep uncertainty* (Kwakkel et al., 2010; Walker et al., 2003; Walker, Haasnoot, et al., 2013). Facing deep uncertainty requires a different conceptual understanding of uncertainty and the use of alternative approaches for decision making compared to situations with well-characterized shallow uncertainty (Dittrich et al., 2016; Lempert et al., 2003; Maier et al., 2016; Marchau et al., 2019b). During the last decades, there has been a growing research interest in analysis methods for supporting planning and policy analysis under deep uncertainty within the area of decision making under deep

uncertainty (DMDU) (Marchau et al., 2019b). In general, approaches for dealing with deep uncertainty account for multiple plausible futures, focus on achieving robust policies, and support adaptive and flexible policy approaches in contrast to conventional prediction-based or probabilistic approaches (Lyons & Davidson, 2016; Maier et al., 2016; Malekpour et al., 2016). While DMDU approaches have gained popularity for other infrastructure systems such as water management, their use in transportation has been relatively limited so far (Stanton & Roelich, 2021).

Transport models have long been important tools for informing planning and policy by forecasting future demand and providing impact assessments for infrastructure investments or policy interventions (Curtis et al., 2021; Jonsson et al., 2011). However, there is a growing recognition in the transport literature that conventional model-based transport planning practices have limited capabilities for handling deep uncertainty in foresight, planning and policy analysis, as they typically rely on predictive approaches that formalize assumptions to produce a single or a limited set of predictions about the future (Lyons & Davidson, 2016; Marsden & McDonald, 2019; Witzell et al., 2022). Emerging transport trends and technologies not only introduce uncertainty about what forecast assumptions to use, but also uncertainty about the predictive ability of models, as these technologies might substantially affect cost structures, transport patterns, and behaviors in ways that do not align with current model structures or parameter estimations (Köhler et al., 2018; Marsden & McDonald, 2019; Muratori et al., 2020; L. A. Tavasszy, 2020).

Transport scholars have identified numerous research needs and proposed changes to tools and practices to deal with such uncertainties. A general topic is the need to put greater emphasis on long-term system-level foresight (Auvinen & Tuominen, 2014), and to increase the use of alternative planning approaches, such as scenario planning, to enable exploration of multiple futures and to test the robustness of plans and policies against these scenarios (Lyons et al., 2021; Witzell et al., 2022). Scholars have also stressed the need for an increased focus on system-level perspectives in freight modeling, including linkages and feedback between system layers within the transport system, couplings to other systems, and the dynamics of system change (Browne et al., 2023; Ghisolfi et al., 2022; Muratori et al., 2020; Raoofi et al., 2024; L. A. Tavasszy, 2020). There is also a need to adapt and improve transport models to better represent the characteristics of emerging technologies to enable out-of-sample extrapolation by incorporating new structural elements and functional properties of the technologies (Muratori et al., 2020; L. A. Tavasszy, 2020). Furthermore, it has been proposed that modeling efforts focused on emerging technologies should prioritize conceptualizing the future and how different assumptions affect system outcomes rather than seeking to predict specific impacts (Curtis et al., 2021). A related area is the proposed use of exploratory modeling as a complement to

conventional predictive modeling to support decision making under deep uncertainty (Engelberg, 2024; Lempert et al., 2022; Milkovits et al., 2019).

1.1.3 Driverless trucks: a high-impact, high-uncertainty technology

Driverless trucks could offer various benefits to the transport industry by mitigating driver shortages (ITF, 2017), reducing operating costs (Bray & Cebon, 2022b; Ghandriz et al., 2020; C. Lee et al., 2023; Wadud, 2017) and allowing for increased truck utilization and more efficient operational strategies (Bray & Cebon, 2022a, 2022b; Hong et al., 2023; Sindi & Woodman, 2021). Studies have shown that many logistics and transport professionals regard driverless trucks as a feasible and potentially attractive technology with the potential to significantly impact logistics, business models, and transport operations (Fritschy & Spinler, 2019; Melander et al., 2019).

Trials with driverless trucks operating without on-board drivers have been conducted for a few years, primarily in the US and parts of Asia and Europe, and initial commercial operations are planned to commence within the coming years for specific market segments and transport routes (Bishop Consulting, 2024). Studies and industry roadmaps have suggested that driverless trucks might see widespread deployment post-2030, with less demanding use cases, such as operations in confined areas, and hub-to-hub operations relying on automated driving on highways, being feasible earlier (ERTRAC, 2019; Melander et al., 2019; Paddeu et al., 2019; Slowik & Sharpe, 2018). However, such timelines are volatile, and it is uncertain if, when, and for what use cases large-scale deployments of driverless trucks will take place. A nexus of technical challenges, regulatory issues, and the search for viable, scalable business models remains unsolved (Engström et al., 2019; Le & Thiel, 2023; Sindi & Woodman, 2021). Nonetheless, significant investments have been made in the driverless truck industry, and progress have arguably been made towards commercialization (Jungblut et al., 2024; Le & Thiel, 2023).

Despite the considerable industry interest and the potentially significant impacts on logistics and freight transport, driverless trucks have so far received limited attention in the literature. A recent literature review identified 125 publications on driverless or autonomous trucks, of which fewer than ten were published until 2017 (Hasiri & Kermanshah, 2024), the year in which the work on this thesis commenced. These numbers can be contrasted with the broader research area of automated or self-driving vehicles, primarily focusing on passenger transport, which has grown substantially over the last decade (Gandia et al., 2019). It is estimated that the current annual number of publications in this area is around 5000, although a majority of these are within various engineering disciplines related to automated driving technology rather than within transportation research (Almlöf, 2024).

From a transport planning and policy perspective, the lack of research on driverless trucks and the limited knowledge about their potential impacts on

the transport system are problematic, as they may affect several important planning and policy characteristics of the freight transport system. The potential reductions in road freight operating costs may increase road freight demand through mode shifts and changes in trade patterns (Bao & Mundy, 2018; Huang et al., 2024; Huang & Kockelman, 2020). Moreover, some types of driverless truck use cases might bring structural changes to how road freight is spatially organized, with driverless trucks operating certain parts of the road network with manually driven trucks as a complement for other areas (Monios & Bergqvist, 2019). Furthermore, driverless trucks may result in new business models focusing on value creation through efficient operations and orchestration of transport networks using driverless and electric trucks rather than through the truck technology itself (Fritschy & Spinler, 2019; Monios & Bergqvist, 2020).

Moreover, driverless trucks may be of importance for the decarbonization transition, and the policies needed to achieve it. For instance, a decrease in costs of road transport relative to other modes could limit the feasibility of achieving policies aimed at modal shifts from road to sea and rail. This may be problematic since policy makers often consider modal shifts as an important instrument for reducing greenhouse gas (GHG) emissions and other external costs of road freight (Johansson et al., 2024).

The uncertain trajectory and timeline for the development and large-scale deployment of driverless trucks, in combination with limited research on their potential impacts on logistics and the transport system, result in a challenging position for planners and policymakers. The prospects of driverless trucks constitute an additional source of uncertainty in an already complex planning environment. Conventional predictive modeling and planning approaches have limited capacity for handling such uncertainties. Therefore, there is a need for more knowledge about the potential impacts of driverless trucks, and for exploring new approaches and tractable analysis methods for managing deep uncertainty. Addressing these needs is important for better understanding the potential impacts of driverless trucks and for guiding the freight transport system toward sustainable pathways. These needs constitute the main motivation for this thesis.

1.2 Aim and objectives

This thesis addresses two broad research needs for improving the understanding of how driverless trucks may impact the freight transport system in the future. The first need is for more knowledge about plausible developments of driverless trucks and their relation to other technological and societal trends affecting freight transport, and how driverless trucks may affect freight transport system outcomes relevant to planning, policy and sustainability. The second need is for robust planning approaches and

tractable analysis methods to manage deep uncertainty associated with driverless trucks in model-based analysis.

Based on these two research needs, the thesis aim is formulated as follows:

The aim of this thesis is to explore long-term system-level impacts of driverless trucks and implications for planning, policy, and sustainability.

The concept of *system-level impacts* refers to changes in system outcomes resulting from mechanisms that form a causal link from the introduction and use of driverless trucks to changes in aggregate system outcomes. The notion of *long-term* system-level impacts signifies that the focus is on a time-horizon during which driverless trucks might be widely deployed and relevant system adaptations have occurred. It is uncertain when, or if ever, such a state will be reached, which is reflected in that the temporal focus applied in the thesis papers varies, see Section 1.4.4. The *implications for planning and policy* are focused on national-level long-term planning and policy making for the Swedish freight transport system, and *sustainability* is primarily considered from a climate- and economic perspective.

The thesis aim is operationalized into four research objectives.

Objective 1a: Develop and analyze scenarios for the future freight transport system in Sweden.

Objective 1b Analyze the Swedish innovation system of driverless trucks and identify key uncertainties for its development.

The first objective is to understand the ongoing development of driverless trucks and their context in terms of other emerging technologies and trends in the freight transport system. This objective provides a basis for the thesis by identifying relevant factors and dynamics that may shape the development of driverless trucks going forward. The objective is split into two parts, 1a and 1b. The first part (1a) is to explore the plausible future development of the Swedish freight transport system from a broader perspective than solely the prospects of driverless trucks. The intention is that future scenarios for the freight transport system can guide model-based analysis of driverless trucks and help place results from such analyses in a broader context. The intention of Objective 1b is to provide an understanding of the ongoing development of driverless trucks and the prospects for a future introduction in the Swedish freight transport system by analyzing the innovation system of driverless trucks.

Objective 2: Model and analyze impacts of driverless trucks on the production cost of road freight transport.

The second objective focuses on how driverless trucks may affect the production costs of road freight transport services. System-level impacts, such

as changes in mode split, transport demand, and system costs, are ultimately the aggregate outcomes of decisions by a large number of actors on the freight transport market (Ben-Akiva & de Jong, 2013). Transport cost is a key determinant for freight transport and logistics decisions, in particular for choice of transport service and transport chain (Flodén et al., 2017; Holguín-Veras et al., 2021; Lundberg, 2006). Given that driverless trucks might substantially affect the operating costs of road freight transport (Ghandriz et al., 2020; Wadud, 2017), analyzing how the cost performance of these technologies might develop under various scenarios and applications is important for supporting assessments of system-level impacts.

Objective 3a: Develop appropriate model representations of driverless trucks for use in national freight transport modeling.

Objective 3b: Provide a plausible outcome space of system-level impacts of driverless trucks using national freight transport modeling.

The adoption and use of emerging road freight technologies such as driverless trucks imply a change in the supply of transport services, which then affect the logistics and transport decisions of actors in the transport market (L. Tavasszy & de Jong, 2014). The third objective covers the modeling and analysis of changes to these decisions in different scenarios for the deployment and use of driverless trucks. The focus is on analyzing how these changes affect various system outcomes on a national level for Sweden. The objective is split into two parts, 3a and 3b, where the former focuses on enabling such modeling by developing model-representations of driverless trucks and their characteristics for use in national freight transport models. Objective 3b focuses on performing model-based analysis of system-level impacts using insights, cost estimates, and model representations from Objectives 1, 2, and 3a.

Objective 4: Assess and demonstrate methods for managing deep uncertainty in model-based foresight and policy analysis for driverless trucks.

The future development and system-level impacts of driverless trucks are associated with deep uncertainty. The fourth objective is to explore methods for handling deep uncertainty to improve foresight and policy analysis. The focus is on exploring how such approaches can be applied to existing modeling tools and on assessing benefits and challenges of such approaches from a national planning and policy analysis perspective.

The primary focus of the thesis is on driverless trucks. However, attention is also given to battery electric trucks and, to a lesser extent, broader developments within digitalization affecting freight transport. The scope is the Swedish freight transport system, which is further elaborated in Section 1.4.3.

1.3 Research design and contributions by the thesis papers

Each of the thesis objectives is addressed by one or multiple thesis papers. An overview of the thesis papers' contribution to the objectives and their relationships are summarized in Figure 1.

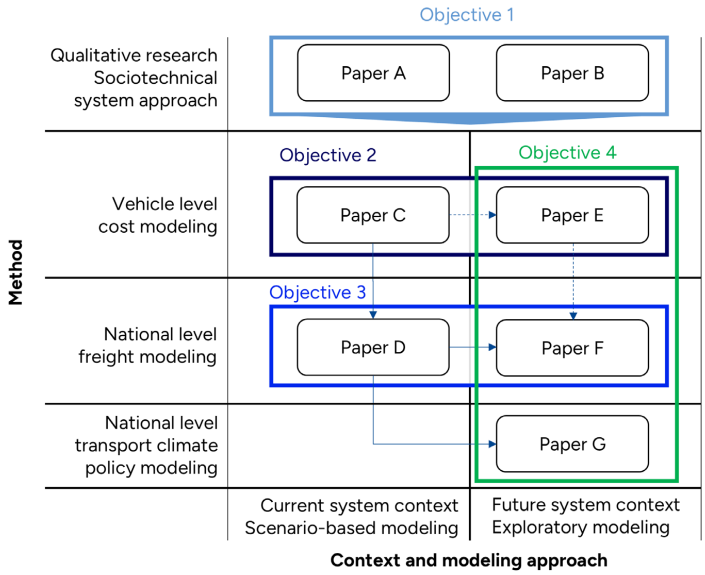


Figure 1 Overview of relationship between papers and how they map to the thesis objectives. For the model-based papers (C-G) it is also indicated whether the modeling is done for the current system context and with scenario-based modeling, or whether a future system context and exploratory modeling is used.

Objective 1 is addressed by Paper A and Paper B, which are based on qualitative data gathered from workshops and interviews with experts and stakeholders related to the Swedish freight transport system. Objective 1a is primarily addressed by Paper A, which analyses how digitalization, including automation, digitized information flows, data sharing and artificial intelligence, may affect the Swedish freight transport system during the coming two decades. Objective 1b is addressed by Paper B, in which the technological niche of driverless trucks is studied using the Technological Innovation Systems (TIS) framework (Bergek et al., 2008). The paper provides a mapping of the structures and processes of the driverless truck innovation system, highlights critical uncertainties for the future development, and identifies forces influencing the direction of development for driverless trucks.

Objective 2 is addressed by Paper C and Paper E, which both analyze driverless trucks in terms of how they may affect total cost of ownership and thus the production costs of road freight transport. However, these papers differ in

scope and approaches. In Paper C, a conventional total cost of ownership (TCO) model for heavy-duty trucks is used to analyze the impacts of automated driving technology by comparing the TCO of driverless trucks with corresponding manually driven trucks. The analysis considers driverless internal combustion engine trucks (ICETs) and manually driven ICETs. Paper E focuses on electrification and how the relationship between the characteristics of the transport application, techno-economic assumptions, and charging strategy affects the cost performance of driverless ETs compared to manually driven ETs and ICETs. Computational experiments are used to identify and analyze conditions under which manually driven ETs and driverless ETs are cost competitive compared to manually driven ICETs. In contrast to Paper C, the cost analysis does not focus on changes from a fixed manually driven ICET baseline but rather on relative cost performance under various scenario assumptions.

Objective 3 is addressed by Paper D and Paper F, which both analyze system-level impacts of driverless trucks based on the national Swedish freight transport model Samgods. Both papers build on the results and insights from Objective 2 for the model representations of driverless trucks. However, the approach, scope and methodology differ substantially between paper D and paper F, and while both papers contribute both to Objective 3a, and Objective 3b, they do so in different ways.

In paper D, the focus is on analyzing how a large-scale introduction of driverless trucks would affect system outcomes given the conditions of the current freight transport system[2]. Eight driverless truck scenarios combining three levels of driverless truck cost assumptions derived in Paper C, and variants of two driverless truck operating models are modeled. Paper D can be regarded as an extension of Paper C in that both papers analyze the impacts driverless trucks may have on freight transport given current conditions with Paper D analyzing how the cost impacts results in Paper C may propagate as system-level impacts in terms of modal split, total system costs and the use of different types of transport chains.

[2] A calibrated baseline scenario for 2017 is used as a proxy for “current conditions”.

Paper F also concerns national-level freight transport modeling, uses the same underlying freight transport model, and focuses on assessing a similar set of system outcomes. However, there are central differences in scope and approach compared to Paper D.

- Driverless trucks are analyzed in a future system context, considering decarbonization and the modeling of battery electric trucks.
- The model is extended from one to three groups of trucks: manually driven ICETs, manually driven ETs, and driverless ETs. This enables analysis of how driverless trucks may complement and compete with other truck types.
- A larger variety of driverless truck operating models are modeled: five instead of two.
- Exploratory Modeling and Analysis (EMA) is used to analyze over 300 scenarios with varying assumptions about the development of electrification and automated driving technology.

Thus, Paper D and Paper F complement each other. Paper D provides a foundation for how driverless trucks can be modeled and provides an initial estimate of system impacts through an approach in which the effects of automated driving are isolated. Paper F builds on Paper D and makes a more comprehensive analysis covering a broader range of scenarios and uncertainties and performs the analysis in a future system setting.

Objective 4 is addressed by Paper E, Paper F, and Paper G, which explore how various approaches for EMA and DMDU can be applied to various model-based analyses of driverless trucks and their impacts. In Paper E, computational experiments are used to analyze the conditions under which driverless ETs and manually driven ETs might be cost competitive compared to manually driven ICETs. It thus explores how EMA can be applied to vehicle-level cost modeling to deal with uncertainty and the challenging multidimensional aspects of assessing different types of transportation tasks. In Paper F, computational experiments are used to analyze system-level impacts of driverless trucks, considering uncertainty related to automated driving technology and electrification. It thus explores how EMA might be a complement to conventional prediction-based foresight and decision support for national-level transport planning. Paper G focuses on handling uncertainty in transport climate policy analysis and explores the use of Many Objective Robust Decision Making (MORDM). Thus, Objective 4 is addressed by testing various approaches for handling deep uncertainty for different types of applications: vehicle level cost performance modeling, national level freight modeling, and transport climate policy analysis.

1.4 Key definitions and delimitations

These following sections present important delimitations and definitions of key terms used for the thesis.

1.4.1 Vehicle technologies

The term *driverless* does not have a generally accepted definition and has been used to describe a wide range of automated driving-related concepts (Shladover, 2018). Therefore, an elaboration on its definition, related terms, and a discussion of its implications for the thesis scope is provided.

A driverless truck is defined in this thesis as:

> *A truck equipped with an automated driving system with a restricted or unrestricted operational design domain operated without an onboard driver.*

Driverless trucks are equipped with an Automated Driving System (ADS) corresponding to SAE level 4 or level 5 (SAE International, 2018). For this thesis, a modified version of the ADS definition provided by the Knowledge Base on Connected and Automated Driving is used[3] (Connected Automated Driving Europe, 2024)

> *Automated driving system refers to hardware and software that are collectively capable of performing the entire dynamic driving task on a sustained basis in a specific operational design domain.*

Operational design domain (ODD) refers to the specific conditions under which an ADS is designed to operate safely considering factors such as geographical area, road type, weather and traffic conditions.

In addition, the term *automated driving technology* is used as a general term for the technologies utilized for automated driving systems. The term *automated truck* denotes a truck with any level of automated driving capability and is generally used in this thesis to contrast with *driverless truck* to denote trucks equipped with an ADS but not being driverless.

The definition of driverless trucks does not distinguish between those with a highly limited ODD only used for a specific transport application (SAE level 4, e.g. for a specific route or traffic environment) and those with an unrestricted ODD that can be used a broad range of transport applications comparable to a manually driven truck of an equivalent configuration (SAE level 5). Moreover, the definition allows driverless trucks to use remote support to enhance

[3] The original definition by The Knowledge Base on Connected and Automated Driving contains an addendum stating that SAE level 3,4 and 5 systems are considered as an ADS. However, for the purpose of this thesis, this part of the definition is excluded since this thesis focuses on SAE level 4 and 5.

transport operations or extend the operational domain (Ma et al., 2024; Zhao et al., 2020).

In this thesis, the term zero-emission trucks (ZETs) denotes heavy-duty trucks without an internal combustion engine and that do not produce tailpipe GHG emissions[4]. ZETs thus include battery electric trucks (ETs), fuel-cell trucks, and some other hydrogen-powered trucks. Trucks with internal combustion engines are denoted as ICETs.

The terminology used for driverless trucks and propulsion technologies differs between the thesis papers. To simplify the kappa, the terminology is harmonized using the terms and abbreviations in Table 1. Note that terms may appear in full or abbreviated form.

Table 1 Truck technology terminology used in the kappa. Note that this terminology may differ from the terminology used in the thesis papers.

Truck	***Driverless***	***Propulsion technology***
Driverless truck (DT)	Yes	Unspecified
Manually driven truck (MT)	No	Unspecified
Internal combustion engine truck (ICET)	Unspecified	Internal combustion engine
Zero emission truck (ZET)	Unspecified	Zero emission technology
Electric truck (ET)	Unspecified	Battery electric
Driverless electric truck (D-ET)	Yes	Battery electric
Manual electric truck (M-ET)	No	Battery electric
Driverless internal combustion engine truck (D-ICET)	Yes	Internal combustion engine
Manual internal combustion engine truck (D-ICET)	No	Internal combustion engine

The thesis papers differ in terms of the propulsion technology considered. In Paper A and Paper B, there is no focus on any specific propulsion technology. In paper C and paper D, the analysis is made for driverless ICETs. This choice was made to "isolate" impacts from automated driving and to simplify

[4] This definition differs slightly from EU's current definition of zero-emission heavy-duty vehicles which also include trucks with internal combustion engines with emissions lower than 3g CO_2 per tkm (EU Regulation 2019/1242 (Amended 2024), 2019).

comparing TCO (Paper C) and system outcomes (Paper D) by using the current freight transport system based on ICETs as a baseline. Moreover, when these studies were conceptualized (2018-2019), the prospects of ZETs in general, and ETs in particular, were unclear.

During the second part of the thesis, covering Paper E, Paper F and Paper G, driverless ETs have been the focus. The key reason is that it has become clear that by the timeframe for which a large-scale introduction of driverless trucks may take place (possibly the 2030s-2040s), EU regulations will stipulate that the vast majority of new trucks will be ZETs (Börjesson & Eliasson, 2024). Hence it is of importance to understand driverless trucks from a ZET perspective. The choice to focus on ETs and not on other ZET technologies is primarily due to that several trends and projections suggest that ETs will be a main, if not the dominating, ZET technology, due to a combination of techno-economic advantages and feasibility for scaling-up production volumes (Basma & Rodríguez, 2023; Link et al., 2024; Parviziomran & Bergqvist, 2023; Plötz, 2022; Rogstadius et al., 2024). The focus is on ETs using static charging solutions while dynamic charging, such as electric road systems, is not considered.

1.4.2 Transport applications and freight transport activities

The thesis focuses on heavy trucks, i.e., trucks and truck combinations of more than 3.5 tonnes of maximum permissible gross combination weight (GCW). Heavy trucks constitute a heterogeneous set of vehicle types (ACEA, 2016) and the focus of this thesis is on road freight transport involving the transportation of goods on the public road network. Thus, only certain types of vehicle configurations are relevant. However, it is recognized and discussed in Paper B, that applications in closed-off areas may be important steppingstones towards driverless truck applications on public roads.

While road freight transport is the focus of the analysis, interactions with other transport modes are considered and modeled in Paper D and Paper F, and to some extent in Paper G. This is an important aspect of analyzing impacts on the freight transport system, since the various transport modes have both competing and complementary relationships (Reis & Macário, 2019). In Paper D and Paper F, the modeling covers road, rail, sea, and air traffic and includes unimodal, multimodal, and intermodal transport chains. In Paper G, only road transport is modeled, but changes in demand for road transport are modeled using own-price elasticities derived from multimodal transport system models. In Paper G, both freight and passenger transport are considered, which differs from the other papers, as these only cover freight transport.

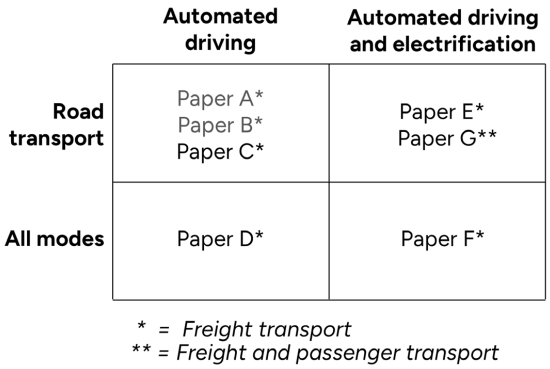


Figure 2 Overview of the scope of the thesis papers in terms of technology focus and whether only road transport or all modes are modeled.

The focus of this thesis is on the automated driving aspect of driverless trucks, not on the automation of non-driving tasks or similar operational activities such as loading, securing and unloading of cargo, charging, document and customs handling. However, it is acknowledged that the introduction of driverless trucks will result in changes and adaptations needed to handle such tasks (Flämig, 2016).

1.4.3 Geographical scope

This thesis focuses on Sweden and the Swedish freight transport system. This general focus corresponds to different specific delimitations and focus areas for the thesis papers depending on their scope and approach.

In Paper A, the Swedish freight transport sector is considered, which includes tangible assets of the freight transport system, such as infrastructure, vehicles, and terminals, but also political, technical, social, and economic factors. The international political, technological, and economical context affecting Swedish freight transport are considered in the scenario analysis.

In Paper B, the Swedish innovation system of driverless trucks is in focus, which is defined as:

> The sociotechnical innovation system focused on the generation, diffusion, and utilization of driverless trucks in Sweden *(Engholm, 2021)*

While actors operating in Sweden are the focus of the analysis, many of these actors are international companies, and the Swedish driverless truck innovation system is naturally affected by, and sometimes integrated within, international developments, institutions, and structures.

For Paper C and Paper E, the Swedish focus means that trucks purchased on the Swedish market by Swedish companies and operated mainly in Sweden under Swedish laws and regulations are considered in the modeling through the selection of parameter values and ranges.

In Paper D, Paper F and Paper G, the focus is primarily on freight transport activity performed at Swedish territory and the associated system outcomes. However, a large share of territorial freight transport is generated by import and export flows to and from Sweden, and thus, the resulting Swedish territorial flows are not just the outcome of the transport system within Sweden. In Paper D and Paper F, the "international part" of import and export flows is also modeled to account for the full freight flow, and not just the "Swedish part" of such flows.

1.4.4 Temporal scope

The general temporal focus of the thesis can be summarized as a period in the future when driverless trucks may be widely available and in use, except for Objective 1B and Paper B, which analyses the current state of the Swedish innovation system for driverless trucks. The thesis is based on a loose assumption that such an introduction may occur sometime during the 2030s or 2040s. The thesis papers differ in that some papers state a specific timeframe, while others do not.

In Paper A, the developed scenarios are set in 2040. In Paper C and Paper D, no specific year is stated for the analysis. Baseline data for the analysis in Paper C is based on truck operating costs for 2014 from ASEK[5] 6.1. In Paper D, the baseline scenario is based on the base year scenario for 2017 from the Swedish reference forecast (Trafikverket, 2020b). Cost and performance assumptions for driverless trucks are made for different technology performance levels, which may be interpreted to represent different future points in time.

In Paper E, an exploratory analysis approach is taken, where the intention is not to forecast timelines for the development of techno-economic factors, but instead to explore the conditions under which M-ETs and D-ETs would be cost competitive to M-ICETs. Thus, no specific timeframe is set for the analysis, but the parameter ranges generally account for plausible technological improvements up to approximately 2040. Paper F and Paper G model scenarios for specific years. In Paper F, 2045 is the scenario year, while 2040 is the scenario year for Paper G. In both papers, the analysis is based on the Swedish Transport Administration's (STA) reference forecasts for these years.

[5] ASEK is the STA's set of guidelines and standardized values for economic analysis in the Swedish transport sector.

1.4.5 Definition of deep uncertainty

Uncertainty can be understood as existing along a spectrum ranging from *the unachievable ideal of complete determinism* on one end to *total ignorance* on the other side (Marchau et al., 2019b). Deep uncertainty refers to situations where it is difficult for decision makers and stakeholders to assign probabilities to different events or predict outcomes due to complexity, lack of information, or inherent unpredictability of a system. It is thus a type of uncertainty that resides towards the *total ignorance* end of the uncertainty spectrum. Uncertainty towards the other end of the uncertainty spectrum can be referred to as *shallow uncertainty*.

For this thesis, a definition of deep uncertainty provided by Lempert et al. (2003) is used:

> *Deep uncertainty exists when analysts do not know, or the parties to a decision cannot agree on, (1) the appropriate models to describe the interactions among a system's variables, (2) the probability distributions to represent uncertainty about key variables and parameters in the models, and/or (3) how to value the desirability of outcomes. (Lempert et al., 2003, pp. 3–4)*

Further elaboration on (deep) uncertainty, including a framework of uncertainty in model-based analysis and an introduction to decision making under deep uncertainty, is provided in Section 3.2.

1.5 Research timeline and development of thesis

In hindsight, this PhD project can be separated in three phases, see Figure 3. The first year was spent searching for a clear research direction, with the hypothesis of combining scenario planning, Systems Thinking and System Dynamics for high-level impact analysis of driverless vehicles in passenger transport. During the initial literature review, I noticed the lack of research on freight transport applications, which was surprising given the growing industry interest. I decided to focus my research on the impacts of driverless trucks since I saw an opportunity to add marginal value. This decision marked the start of the second phase.

The focus during this phase was to use the Swedish national freight transport model Samgods to analyze system-level impacts of driverless trucks. This work resulted in two thesis papers: Paper C, covering the cost-inputs to the modeling, and Paper D, covering the model implementation and results. In parallel, I was involved in a separate project to develop scenarios for how digitalization, including automated driving, may affect the Swedish freight transport system. This work resulted in Paper A. During the same period, the work for Paper B was carried out. This paper is based on data from a small-set of expert interviews I performed as preparatory work for my modeling study, and a larger-set of interview data collected during a master thesis I supervised (Björkman & Joelsson, 2019). This phase culminated with my Licentiate thesis, defended in June 2021 (Engholm, 2021).

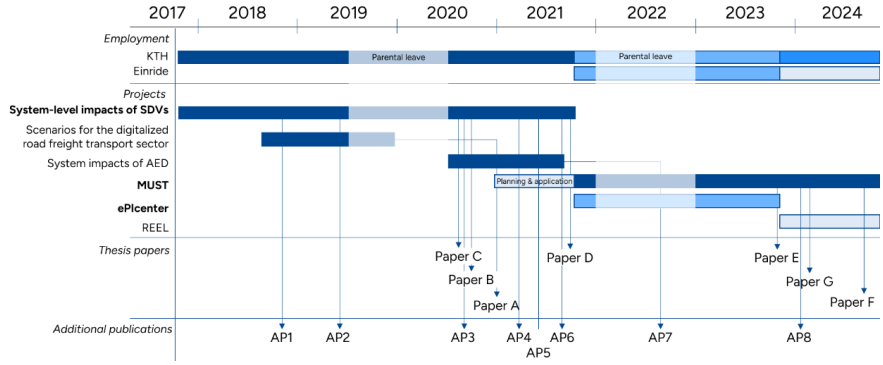


Figure 3 Research timeline with employment, research projects, thesis papers, and additional publications (AP).

A key insight during the second phase was that conventional scenario and modeling approaches relying on a small set of scenarios specified prior to the (model) analysis are poorly suited for analysis of uncertain technological development. During the development of Paper C and Paper D, I struggled with the *curse of dimensionality* (de Jong et al., 2002) when trying to create a small set of scenarios for driverless trucks to model by having to make

assumptions for many uncertain, largely independent scenario inputs. Similarly, the scenario development process for Paper A was challenging in its attempt to project the uncertainty into a few dimensions. I went into the literature for guidance, found Exploratory Modeling and Analysis (EMA) and Decision Making under Deep Uncertainty (DMDU). I decided to focus the remainder of my PhD research, phase three, on exploring these approaches for analyzing driverless truck impacts. To enable this, I initiated and developed an application for the project *Managing deep Uncertainty in planning for Sustainable Transport (MUST)*[6] which was granted funding from the STA. Both Paper F and Paper G were developed within the MUST project.

Since 2021 I have split my time and research between ITRL and the Swedish freight technology company Einride, which I joined to do research on driverless electric trucks within the EU project *ePIcenter*[7]. Paper E was developed within this project as an effort to apply EMA to enhance cost performance modeling of driverless electric trucks. My time at Einride has directed my focus towards electrification and provided me with valuable data and learnings on the logistics context of truck operations and a better understanding of real-world challenges related to driverless trucks and electrification.

[6] https://www.itrl.kth.se/research/ongoingprojects/must-1.1146492
[7] https://epicenterproject.eu/

2 Driverless trucks: an overview and literature

This chapter provides an overview of the current state, development, and trends for driverless trucks in Section 2.1. In Section 2.2, the literature on impacts of driverless trucks is surveyed. This chapter is partly based on Engholm (2021) but updated to include more recent developments and literature.

2.1 Driverless trucks: an overview

Various forms of driverless vehicles have been used for logistics applications in confined areas for several decades. Examples include automated guided vehicles within industrial and logistics facilities and port areas, and driverless haul trucks used in mines, quarries, and other raw material sites (Flämig, 2016; Parreira & Meech, 2011). There have been ideas about how automated driving technology could be applied to truck transport on public roads for a relatively long time, for instance as various forms of platooning applications (Shladover, 2010). During the last decade, the prospects for driverless truck applications on public roads have become more feasible due to technological advancements in automated driving technology (Daduna, 2020).

Automated driving, as it is typically applied to driverless trucks, can be described as four main tasks performed continuously by the ADS to perform the driving task (Jungblut et al., 2024):

- Sensing: is the gathering, pre-processing and fusion of raw data from the vehicle's surroundings using a variety of sensors.
- Perception: is the interpretation of raw sensor data to identify objects, track and predict their motion, and recognize static and dynamic elements of the traffic environment.

- Planning: involves the planning and decision making for the driving on longer-term (route choice) and shorter term (path planning).
- Control: is the control of the vehicle's steering, throttle, brakes, and other actuators to follow the planned path.

During the late 2010s and the first years of the 2020s, development efforts for driverless trucks accelerated, and several rapidly growing startup companies emerged. Substantial investments were made in the industry, with about 10 billion USD invested in driverless truck companies solely during 2020 and 2021 (Jungblut et al., 2024; Le & Thiel, 2023). During this period, there was a seemingly high pace of technological progress (Ackerman, 2021) and an increasing number of pilot projects were initiated or announced (Zarif et al., 2021).

In the last few years, the driverless truck industry has faced multiple challenges, accompanied by a general slowdown in the investment climate resulting in that several companies had to either scale-down or go out of business, with TuSimple and Embark as well-known examples (Le & Thiel, 2023). Despite these challenges, progress towards commercialization has been made, including substantial volumes of real-world testing. In particular, there has been progress for long-haul trucking in the US by companies such as Aurora and Kodiak, but also for short-distance facility-to-facility operations, for instance by the Swedish company Einride (Bishop Consulting, 2024). Several US driverless truck companies have claimed readiness to launch commercial operations without an in-vehicle driver during this or the coming years (Bishop Consulting, 2024). As of now, the industry is described as having matured and now being in a phase of ecosystem experimentation, facing strong pressures to find sustainable business models and successful commercialization strategies (Le & Thiel, 2023).

One overarching distinction between alternative deployment approaches for driverless trucks is provided by Chan (2017) who outlines two pathways towards achieving driverless vehicles with unrestricted ODD. One is denoted a "somethings everywhere" approach, which is an evolutionary pathway that involves a gradual development and deployment of automated driving technology. This starts with driver assistance to support and enhance the human driver who is still intended to be present in the vehicle to perform the driving task (Kirschbaum, 2015). This is followed by an introduction of ADS with an increasing level of automation, aiming to ultimately eliminate the need for a human driver at some point in the future when automated driving technology has matured significantly.

The other approach is described as an "everything somewhere" approach, where the focus is on trucks without the need for an onboard driver that, initially, are only capable of operating in a limited ODD. Over time, when automated driving technologies develop, the ODD can be expanded, and driverless trucks could be applied to more transport applications. Based on

Engström et al. (2019), the efforts to develop driverless trucks within the industry, and in particular by the established OEMs, can be understood as initially having been focused on pursuing the evolutionary approach, but that the focus has somewhat shifted towards the everything somewhere approach.

It is hard to forecast timelines for large-scale driverless truck deployment. It may take several decades before automated driving systems capable of safely allowing driverless trucks to operate in all environments at all times (i.e. SAE level 5) are developed, if they ever are (Shladover, 2018). Furthermore, the introduction of driverless trucks is not only a matter of achieving sufficient technological maturity, but also requires adaptations to legislation, such as aligning traffic law to allows for driverless truck operations and specifying testing procedures for vehicle certification (Litman, 2021), development of the business ecosystem (Le & Thiel, 2023) and adjustment of a wide range of logistics processes (Flämig, 2016). However, several trends suggest that it is plausible that driverless trucks at SAE level 4 may become commercially deployed for certain transport applications on public roads within the coming decade, given stated industry timelines (Jungblut et al., 2024).

ODDs for automated driving in industrial areas and on highways and similar environments are considered to be less demanding compared to driving in urban environments, which present more complex and unpredictable traffic conditions (Engström et al., 2019). This is a view expressed in a wide range of academic and grey literature (Gittleman & Monaco, 2020; Hu et al., 2020; Jungblut et al., 2024; Meldert & Boeck, 2016; Monios & Bergqvist, 2019; Müller & Voigtländer, 2019; Viscelli, 2018).

Two SAE level 4 applications following the “everything somewhere” approach are often discussed as potential initial use cases, as they are regarded as technically and operationally achievable, commercially attractive, and feasible from a legislative perspective in the early stages. The first application is to use driverless trucks for short-distance, repetitive transport flows between logistics facilities such as factories, warehouses, and harbors, where the route takes place in relatively simple traffic environments, for instance within industrial areas. A roadmap by ERTRAC (2019) expects such applications to be attainable around 2027. This use case typically involves flows between fixed locations where processes for loading, unloading, and handling of the goods can be standardized and adjusted to the fact that no driver is present to assist. A few pilot projects for this application have been performed in Sweden and the US, for instance by the Swedish company Einride (Bishop Consulting, 2024; Kristensson, 2019).

The other type of application is driverless trucks operating on highways or other major roads, making them feasible for long-haulage applications. ERTRAC (2019) expects such applications to be attainable around 2030. One example of this application could be line-haul transport between terminals in parcel and logistics networks with a hub-spoke character. Several solutions

have been proposed to handle flows with origins or destinations not located in direct connection to the highway network and may therefore not be reachable through automated driving. One approach is to use remote operations (Viscelli, 2018) while another option is for manually driven trucks to tow trailers to/from driverless truck transfer terminals located near the highway network, where a trailer swap is performed between the manually driven truck and a driverless truck, which then performs the long-haulage leg (Hu et al., 2020; C. Lee et al., 2023; Monios & Bergqvist, 2019; Viscelli, 2018; Zarif et al., 2021). Several driverless truck technology companies have stated that they primarily pursue highway operations (Ackerman, 2021) and this domain has been prioritized by the industry during the last year with substantial testing taking place in the US and China (Bishop Consulting, 2024). In Sweden, Scania has trialed such routes with a safety driver on board for a few years and announced in early 2024 that they are launching a commercial pilot program for "hub-to-hub transport" (Scania, 2024).

All in all, the development suggests that the pathway to a large-scale introduction of driverless trucks is highly uncertain, but may happen stepwise, with an initial focus on applications in relatively simple traffic environments and specific routes where a straightforward business case and operational setup can be achieved. Over time, more complex operations and business models may be deployed (Engström et al., 2019; ITF, 2017; Viscelli, 2018). While there are rather clearly stated short-term timelines for initial commercialization, these apply to specific use cases and markets (e.g. specific freight lanes in the US) and it is unclear to what extent this information can be used to infer realistic timelines for a more widespread large-scale roll out for other use cases and markets, such as in Sweden.

2.2 Literature on impacts of driverless trucks

The scientific literature studying driverless trucks from a transport system or societal perspective is relatively scarce. A scientometric analysis of the literature on driverless vehicles by Gandia et al. (2019) shows that the field has been dominated by technical subjects, with only a few percent of the publications addressing non-technical issues. Within research focused on impacts of automated driving, freight transport is not a common topic. Cavoli et al. (2017) performed a systematic literature review on the societal impacts of automated vehicles and found that less than 5% of approximately 400 identified publications covered freight transport aspects. This scarcity is confirmed by more recent research that identified 125 publications on driverless or autonomous trucks published until 2023, of which more than 70% were published in 2020 or later (Hasiri & Kermanshah, 2024).

Before reviewing the academic literature on driverless trucks, it is noted that driverless trucks have been the subject of many consulting and industry reports (Bishop Consulting, 2024; Chottani et al., 2018; DHL, 2014; Keese et

al., 2018; Manyika et al., 2013; Nowak et al., 2018; Roland Berger, 2016; Shanker et al., 2013; Zarif et al., 2021). These reports provide insights by speculating on potential developments, applications, impacts, and barriers for driverless trucks and reporting on the developments in the industry. However, there are reasons to take a critical position towards claims and views expressed in these publications as these actors have economic interests that may shape their viewpoints (Shladover, 2018).

Besides the consulting and industry reports, there is a body of non-peer reviewed grey literature comprising policy reports and technical reports that provide valuable overviews of opportunities and challenges associated with automated and driverless trucks (Bao & Mundy, 2018; Center for Global Policy Solutions, 2017; Després et al., 2018; Gittleman & Monaco, 2020; Groshen et al., 2018; ITF, 2017; Jaller et al., 2020; Kulmala et al., 2019; Le & Thiel, 2023; Litman, 2021; Mehta & Levy, 2020; Meldert & Boeck, 2016; Short & Murray, 2016; Slowik & Sharpe, 2018; Trafikanalys, 2019; Viscelli, 2018; Waschik et al., 2021). There are also a few publications summarizing discussions and learnings from conferences on automated driving that provide insights into recent developments within the industry (Engström et al., 2019; Gold et al., 2024; Poorsartep & Stephens, 2015; Shladover et al., 2024).

The remainder of this chapter summarizes relevant academic literature on driverless trucks, complemented by findings from a selection of grey literature.

2.2.1 Perceptions on driverless trucks and their impacts among freight transport actors

One topic in the literature is how various actors in logistics and freight transport perceive the prospects of driverless trucks and how they might affect their business.

Melander et al. (2019) used a Delphi approach to derive scenarios for the freight transport system in Sweden in 2050. Experts assessed the probability, impact, and desirability of sixteen projections for 2050. The projection that "all trucks are driverless" was given an average probability of 50%, and ranked second in terms of impact, and fifth in terms of desirability. Additional comments by the study participants included expectations of large cost savings, long-haulage as a likely use-case for driverless trucks, Sweden having a larger adoption compared to the global average, but also skepticism about that all transport will become automated.

Kristoffersson and Pernestål (2018) assessed perceived benefits, costs, and barriers for driverless trucks among freight and logistics experts in Sweden by developing societal scenarios for driverless vehicles. The findings suggest that experts foresee a potential for driverless trucks to reduce transport costs and increase vehicle utilization but raise concern over cyber-security issues and how to deal with cargo handling and delivery processes. Long-distance

highway transport between major logistics nodes is seen as a plausible initial application although issues with international cross-border transport were highlighted as a potential challenge.

Other studies conducted interview-based research with different stakeholders within freight and logistics. Müller and Voigtländer (2019) interviewed high-level managers at German logistics and freight companies in their role as potential future buyers and users of automated and driverless trucks. A key finding is that driverless trucks are regarded as significantly more attractive compared to automated trucks that would still have an onboard driver. This is primarily because driverless trucks can target important challenges for the industry by alleviating the driver shortage, reduce costs, and thus increase profit margins.

Sindi and Woodman (2021) studied the views of 76 freight transport professionals in the UK on driverless trucks through semi-structured interviews and scoring questionnaires. The findings largely align with previous literature by identifying these central topics: impacts on direct and indirect fleet costs, how to deal with operational aspects of not having an onboard driver, and impacts on safety and lead time. However, participants disagreed about whether driverless trucks with or without the capability to be operated by a human driver are the most attractive option. Those favoring the option without the capability for human drivers saw benefits in being able to automate the whole transport process and avoid costs associated with a driver's cabin while those favoring a truck with a driver's cabin saw benefits with having more operational flexibility. Furthermore, most participants expected their organizations to adopt driverless trucks if a credible business case in terms of total cost of ownership could be made, although many participants doubted that driver cost savings and increased fuel efficiency would be sufficient to compensate for additional technology costs.

Another research topic concerns potential impacts on business models for various actors in the freight transport and logistics industry and how the type and nature of collaborations may change. Fritschy and Spinler (2019) performed a Delphi study among German freight and automotive experts on how driverless trucks will impact business models in their industries. The results show that driverless trucks are expected to be associated with higher degrees of industry cooperation, both vertically between suppliers, OEMs, and logistics companies to develop driverless truck solutions, but also horizontally between logistics companies to increase resource utilization by sharing driverless truck capacity. Other possible developments include OEMs owning driverless truck fleets and offering transport capacity as a service, driverless trucks becoming a highly standardized product, and a possible consolidation of the logistics industry.

Impacts on business models and the structure of the freight transport sector are further discussed by Monios and Bergqvist (2020). They reason that

driverless trucks may lead to the disappearance of traditional business models for truck OEMs focused on selling trucks to road carriers, which could be replaced by a transport-as-a-service model. This development could require a new type of actor and business model focused on operating transport networks with driverless trucks and providing transport services to the transport buyer. This new network operator role could be taken by either existing truck OEMs, logistic service providers, or new entrants. Furthermore, the authors claim that this development will change the role of several key actors in the current road freight transport system. For instance, road carriers may become outcompeted in certain market segments by the network operator.

Implications of changing roles are further studied by Lind and Melander (2021) from the perspective of truck OEMs. They expect that driverless trucks and electrification will require OEMs to shift towards more networked business models, meaning that OEMs cannot independently govern all necessary resources, activities, and relationships required for developing, producing, and marketing their products or services. Instead, multiple firms and actors, such as IT firms, energy providers, and logistics companies need to collaborate within a network to sustain the business model.

2.2.2 Market adoption

Another area of research concerns the potential adoption patterns of driverless trucks. Forecasts of the market uptake of driverless trucks are rare, although some publications in the grey literature provide estimates for the "penetration rate" of driverless trucks over time, usually to enable analysis of potential impacts (ITF, 2017; Kulmala et al., 2019; Litman, 2021; Waschik et al., 2021). In the academic literature, limited attention has been given to modeling the market uptake of driverless trucks. Simpson et al. (2019) used the theory of diffusion of innovations to model the adoption of driverless trucks among freight transport companies using data for a county in the State of Tennessee. Model parameters were estimated based on adoption patterns of previous road freight innovations. Depending on the assumptions for how driverless truck technology improves over time, the public perception, and market factors, the estimated market penetration of driverless trucks 20 years after market introduction ranges from less than 20% to more than 95%. In a subsequent paper, Simpson and Mishra (2020) proposed an alternative methodology based on modeling peer effects, which allows for simulating uptake of driverless trucks resulting from competition and communication in networks of freight transport actors.

A related area of research concerns preferences for driverless trucks and the identification of determinants for future adoption among different types of freight transport companies. Anderhofstadt and Spinler (2020) conducted a choice-based conjoint analysis on preferences for driverless trucks and alternative fuel-powered trucks among freight transport companies in

Germany. The findings show that freight organizations perceive automated driving as an attractive feature of future trucks, but that it is less prioritized than other factors, such as long driving range and low operating costs, which may be correlated features. Furthermore, driverless and automated trucks are generally seen as more attractive than manually driven trucks, but the results reveal no difference in preference between automated and driverless trucks.

Talebian and Mishra (2022) used choice modeling based on a stated preference survey from 400 trucking companies in the US to analyze characteristics of companies that may be early adopters of driverless trucks. They found that smaller companies transporting perishable goods are more likely to be early adopters, and that across all firm sizes, the expected substantial upfront investment required for adopting driverless trucks is a barrier. Amiri et al. (2022) analyzed the readiness among logistics companies to adopt autonomous vehicles, including driverless trucks, in Ontario's Niagara Region using structural equation modeling with survey data from 101 companies. The study highlights the role of third-party logistics reliance and firm characteristics (e.g. the amount of transportation assets, and e-commerce reliance.) Firms that rely more on third-party logistics providers or have higher reliance on e-commerce show higher interest in driverless trucks. The study also reveals some skepticism regarding the timeline for driverless trucks becoming widespread, with many firms anticipating that adoption may not occur soon.

2.2.3 Impacts on transport costs

While there are multiple industry and consultancy reports that make claims about how driverless trucks will affect the cost of road freight transport operations, there are only a few examples in the academic literature analyzing this topic. These analyses are often based on a total cost of ownership (TCO) approach.

Wadud (2017) analyzed how automated driving may affect TCO for trucks of various sizes in the UK. Results for the scenario intended to represent the "most likely" assumptions show TCO reductions in the range of 15%-22% with larger relative TCO reductions for smaller truck types and vice versa. Ghandriz et al. (2020) used an optimization approach to minimize TCO for D-ETs and D-ICETs for different types of transport routes by selecting vehicle and infrastructure configurations. Their findings suggest that TCO can be reduced by 33-41% for D-ETs compared to M-ETs and 22%-33% for D-ICETs compared to M-ICETs depending on trip distance. Another TCO study was performed by Lee et al. (S. Lee et al., 2023) and used baseline costs from a South Korean logistics company to estimate potential cost impacts of replacing trucks of various sizes (1-tonne to 25-tonne GVW) with driverless trucks under various sets of assumptions about costs. They found cost reductions between 25% to 65% depending on truck type and assumption set.

A central factor in the identified cost reduction in these TCO studies is reduced labor costs due to the elimination of the in-vehicle driver. However, all studies stress that not all labor costs related to the transport operations will likely be eliminated because non-driving tasks typically performed by the driver will still need to be completed. Additionally, there might be a need for human remote operators to monitor or actively support the trucks during parts of the transport mission (Ma et al., 2024; Zhao et al., 2020). To assess the effect of the driver cost reduction rate, Jungblut et al. (2024) made a simplified cost assessment of driverless semi-trucks by calculating the payback-period for driverless trucks compared to a manually driven truck, considering an ADS cost of 40k EUR and various driver cost saving ratios. If 100% of driver costs are eliminated, break-even occurs after 120kkm (roughly one year of typical operations), while 20% driver cost reduction corresponds to a payback time of about five years.

Ma et al. (2024) used discrete event simulation on realistic data for driverless truck disengagement rates to study remote operator utilization and truck wait times for remote support. The results show that in the scenarios with low disengagement rates, one operator can handle up to 20 trucks with minimal truck wait times and still only be utilized 20% of the time. For high disengagement rates, one operator may manage eight trucks and be utilized 50% of the time, but truck wait times longer than one minute were observed.

Tjaden and Flämig (2024) used activity based costing to compare costs in an urban parcel delivery case between a conventional manual electric truck (with the driver performing the package delivery) and a driverless electric truck carrying two delivery robots performing the deliveries. In their base setting, they estimate the automated case to be almost 60% more expensive, primarily due to additional labor costs for loading the parcels into the delivery robots and high capital costs for the delivery robots.

Another potentially important cost-related aspect is the impact on overall energy efficiency or fuel economy of driverless trucks. Bray and Cebon (2022a) analyzed how fuel savings due to a reduced driving speed during highway driving may reduce operating costs for driverless trucks. They conclude that for a driverless tractor-trailer truck with 50% mass-payload, a reduction in driving speed from 90km/h to 70km/h would reduce fuel consumption by 26%. If accounting for a cargo value of time at 1GBP/h/tonne, reducing speed would result in a total cost reduction of 4%. Overall, the cost saving potential is highly sensitive to truck size, cargo value of time, and drive cycle.

Jungblut et al. (2023) combined three factors: reduced energy consumption due to smoother driving, speed reductions and additional energy demand to power the ADS. They found that a speed reduction during highway driving from 90km/h to 70km/h may reduce total energy demand by around 15% when accounting for less recuperated braking energy. Smoother driving likely has a smaller effect, reducing energy demand by 1-3% at highway driving

speeds, while the ADS would increase total energy consumption by about 1%. They further speculate that other energy savings such as not requiring a driver's cabin, may save additional energy, estimated to potentially reduce specific energy consumption (kWh/km) by up to 0.1 for electric trucks (Sigle & Hahn, 2023).

2.2.4 Impacts on logistics and freight transport systems

Flämig (2016) provides an overview of possible developments and impacts of automated trucks and highlights that introducing driverless trucks would require changes to logistic activities upstream and downstream of driving as well as adaptations of non-driving related driver tasks, for instance, administrative tasks such as document handling. This aligns with the stakeholder interview literature, which also highlights such aspects as important challenges. One area of importance is the loading and unloading of goods from driverless trucks for which one option may be to use automated forklifts or other automated solutions. While this is a feasible approach, there are challenges in its integration within the logistics processes, as suggested by Agrawal et al. (2023) who performed a requirement analysis for a case involving driverless trucks carrying industrial components from a production plant to a distribution center.

Another operational issue is how the driver's sensing and decision-making functions for fault detection and short-term maintenance planning should be handled for driverless trucks. Rylander and Englund (2024) developed a conceptual system for how fault handling can be managed through a remote control center supported by an intelligent decision support system, in which faults are detected, diagnosed, and addressed using onboard sensors, real-time data monitoring, and remote human expertise to replace the role of the driver. Tao et al. (2022) proposed a risk-based decision-making approach for driverless trucks to optimize maintenance decisions during freight missions by integrating the economic risks of availability loss and maintenance costs.

Driverless trucks are expected to affect logistics and transport planning in various ways. Dong et al. (2021) reviewed the impacts of emerging digital technologies on freight transport and concluded that the introduction of driverless trucks would come with structural and logical changes to logistic operations making these operations more complex and difficult to manage. The paper further highlights that initial adoption of driverless trucks is likely to happen for applications in confined areas. Also, the paper suggests that driverless trucks are one of several potentially important emerging technologies that may generate substantial change within the freight transport sector.

Tavasszy (2020) also assessed a range of technologies within automation and digitalization and their potential to reshape logistics, with driverless trucks being one of the technologies in focus. The paper stresses the need to improve

freight modeling to better understand these novel technologies by developing appropriate representations of new structural elements and addressing their relations with adjacent systems such as climate, finance, energy, information technology, and communication systems. Also, the paper emphasizes a general need to improve models of logistics decision-making, which has been a prioritized area of freight modeling during recent years (De Jong et al., 2021) and to improve the understanding and modeling of the change dynamics of freight systems including adoption and adaptation to emerging technologies.

Chen and Lu (2020) studied how logistics managers may adjust shipment sizes if driverless trucks would require increased upfront costs but, on the other hand, reduce transport lead times compared to manually driven trucks. The analysis is based on the economic order quantity model and concludes that driverless trucks would increase the optimal shipment size compared to manually driven trucks under these circumstances. They further speculate that driverless trucks may lead to a centralization of inventory locations in supply chain networks.

Bray and Cebon (2022b) investigated how driverless trucks may change decisions on truck size and the use of multi-stop tours in an urban distribution network and a regional delivery network. In their urban case, with smaller loads and more dispersed deliveries, cost savings may be achieved by using smaller vehicles and fewer stops per trip. For regional distribution, the cost reduction potential is smaller. However, in both settings, the major factor contributing to cost reductions is reduced driver costs rather than the changes in operations. Pudasaini and Shahandashti (2021) speculate that driverless trucks may also affect the requirements for, and operations of, logistical facilities such as warehouse and fulfillment centers. Inefficiencies and capacity constraints in such facilities might be a bottleneck for realizing high utilization of driverless trucks if increases in the inflow and outflow of goods cannot be managed. On the other hand, driverless trucks may catalyze the adoption of automation and digital warehousing solutions.

A broader perspective on how logistics and freight transport networks may change is provided by Monios and Bergqvist (2019). They anticipate that the introduction of driverless trucks capable of highway driving in combination with truck electrification may change the structure and transport geography of road freight. Transport networks may develop into a model similar to multimodal transport networks with driverless trucks performing long-haulage between fixed logistics terminals located close to the highway network, and M-ETs performing the pre-and post-haulage for these transport flows. These terminals serve as hubs in the network and facilitate trailer swapping between trucks as well as charging infrastructure for electric trucks.

Similar concepts have been assessed from an operational perspective. Lee et al. (2023) analyzed the “transfer hub model” where driverless semi-trucks operate long-haul middle-mile routes in a network of transfer hubs in which trailers

can be swapped from driverless trucks to manually driven trucks which pull the trailer the first and last legs from and to the trailer's origin and destination. The study focused on the optimization of routing and scheduling in such networks but includes a case study based on real world data for a US-wide transport network. The result shows that the transfer-hub model reduces the total transport costs for the considered network. With an assumed 25% reduction in per-km operating cost for driverless trucks, the attainable system-level cost savings were up to 28% due to substantial reductions in empty distance driven and increased truck utilization.

The potential impact of driverless trucks on the relationship between road transport and other modes has only been addressed to a limited extent, and primarily in a North American context. Bao and Mundy (2018) examined how driverless trucks and platooning could decrease relative costs for road transport compared to rail and thereby result in modal shifts from rail to road. Their findings suggest that road transport could be cut by approximately 40% which increases the transport distance at which road transport is cost-competitive against rail transport. The resulting decrease in rail demand is estimated by using two methods, an elasticity-based method suggesting an effect of 45% and a cost-curve-based method which suggests an effect of around 20%.

Huang and Kockelman (2020) studied the impacts of driverless trucks on freight transport patterns in the US by using a multiregional input-output model with a random-utility-based mode-choice model including rail and road transport (the "RUMBRIO" model). Given an assumed reduction in road freight costs of 25% per tonne-kilometer (tkm), road-tkm increase by 11%, partly due to a mode shift from rail and partly by changes in origin choice. The use of trucks increases for all but one commodity type and all distances except for those over 4800km. In a follow up study Huang et al. (2024) enhanced the RUBMRIO model by incorporating more detailed commodity pricing impacts and using updated demand data and assuming a 40% reduction in operating costs for driverless trucks, instead of 25% as in Huang and Kockelman (2020). Furthermore, the potential for reducing travel times with driverless trucks was considered. However, the results show similar increases in truck ton-miles, with the largest effect occurring for trips between 800 and 1200 km. Rail continues to play a role in longer-distance freight, but the study finds that driverless trucks dominate for distances over 400km, with rail's share becoming less significant for shorter trips.

Chowdhury et al. (2024) used a regional macroscopic trip-based model to analyze the transport network effects of various scenarios for a driverless truck introduction in the Greater Toronto and Hamilton Area. The scenarios varied by market penetration rate, whether driverless trucks were only allowed on highways, and the driving behavior of driverless trucks. All scenarios show an increase in total vehicle kilometers travelled (VKT) for trucks with increased

congestion and travel times with partial adoption of driverless trucks, but reduced travel times with full adoption.

Research on driverless trucks with a focus on electric trucks is rare (Hasiri & Kermanshah, 2024) with the previously mentioned study by Ghandriz et al. (2020) being one example focusing on truck-level economics. On the system-level, Hong et al. (2023) studied impacts of driverless ETs on transport costs and the electricity grid by using a US national-level transport and grid model that simultaneously solves cost-minimizing dispatching of truck fleets and the operation of the electricity systems through dispatching of power plants. They found that increasing the penetration of electric driverless trucks from 1% to 99% leads to a fleet size reduction of 20%, due to higher vehicle utilization enabled by automated driving. Additionally, a cost reduction of 18% for transport and charging related costs was achieved, and peak grid load decreased by 14% due to charging planning being more flexible with driverless trucks.

2.2.5 Wider impacts and sustainability implications

Questions related to how driverless trucks will affect energy and greenhouse gas emissions on a transport-system level are addressed by Wadud et al. (2016) and Taiebat et al. (2018). Wadud et al. (2016) examined impacts on energy and carbon emissions from automated vehicles at a transport system level by combining vehicle and network-level effects. The study included both passenger vehicles and trucks. The analysis shows that the total road transport energy demand depends on the combined impacts of automated vehicles on energy intensity and travel demand, and a key finding is that it is plausible for increasing transport demand to outweigh energy intensity reductions, resulting in a net increase in total energy demand and emissions. Taiebat et al. (2018) performed a literature review and analysis of energy, environmental, and sustainability impacts of automated vehicles. They highlight the complexity arising from interactions of impacts on vehicle-, transport system-, and societal levels and stress the need for holistic energy and sustainability assessments of automated vehicles on the societal level.

Driverless trucks may also have implications for both the physical road infrastructure and the digital infrastructure needed to support vehicle-to-vehicle communication, vehicle-to-infrastructure communication, and communication between vehicles and remote operators. Kulmala et al. (2019) assessed potential infrastructure adaptations and associated costs that might be required to support driverless vehicle operations. They found that road markings and signage will need to be standardized and kept in high condition to be reliably detectable by ADS, lane widths and shoulders may need adjustment, and parking, curbside management, and the allocation of lanes will be influenced. They also discuss the increasing need for road operators and

authorities to ensure digital infrastructure for supporting driverless vehicle operations.

There have been a few efforts to study the wider economic and socio-economic impacts of driverless trucks. Lunkeit et al. (2019) developed a system dynamics model to simulate the impacts of an introduction of driverless trucks in Germany on an aggregated system level. They found that road transport demand could grow significantly due to reduced truck operating costs, leading to a net increase in total fuel consumption and total transport system costs. Pernestål et al. (2020) used systems thinking to link vehicle-level effects and transport system effects, suggesting that an introduction of driverless trucks could shift the system equilibrium to a state of higher VKT and that there may be a potential to reduce the truck fleet size (for a given transport demand) due to increased truck utilization.

Fagnant and Kockelman (2015) estimated potential societal benefits of automated vehicles for passenger and freight applications in the US. The analysis covers aspects such as improved road safety, traffic flow, and fuel economy and estimates substantial societal benefits, although separate calculations and results for trucks are not provided. Clements and Kockelman (2017) assess the impacts of automated vehicles across multiple industrial sectors in the US and concluded that the road freight transport sector would experience the largest economic gain from automated vehicles, mainly due to productivity gains. Andersson and Ivehammar (2019) applied a cost-benefit analysis framework to estimate societal costs and benefits in speculative scenarios for the use and impacts of automated trucks in Sweden in 2025 and 2040. Their indicative calculations show that automated trucks could generate substantial net benefits to society, which for 2040 are estimated in the range of 1B€, with almost 90% of the total benefits being driver cost reductions. An important finding is that most benefits accrue to the buyers and producers of road freight transport services and not to other actors in society.

Another topic is he examination of macroeconomic effects and labor impacts, such as potential driver job loss. Waschik et al. (2021) used a dynamic computable general equilibrium model to study the macroeconomic impact of a gradual introduction of driverless trucks for long-haulage road transport in the US. They found that driverless trucks could improve productivity in the road freight sector, which would increase GDP, capital, employment, and wages. Another important finding is that although the net hiring of truck drivers will decrease when driverless trucks are introduced, it does not lead to layoffs in the road transport sector in the slow- and medium- adoption scenarios (assuming current turnover rates), while there are lay-offs in the fast adoption scenario. Several other reports have studied the impact of driverless trucks on the labor market and potential job loss among truck drivers, in particular for the US. The conclusions differ significantly between publications, with some estimating that millions of jobs may be lost in the freight sector if a

rapid introduction of driverless trucks takes place (Center for Global Policy Solutions, 2017; Groshen et al., 2018). Others claim that the effect is much smaller, although some types of road transport jobs, primarily long-haul drivers could be severely affected (Gittleman & Monaco, 2020; Viscelli, 2018). The discrepancy between the studies is mainly due to different assumptions about what types of driver jobs (and tasks) could be replaced by driverless trucks, how demand for road transport will develop, and how fast driverless trucks will be introduced. A study by the International Transport Forum (2017) highlights that the current and anticipated future shortage of truck drivers needs to be accounted for in the analysis but still estimates that more than two million truck drivers may be displaced in the EU and the US combined if driverless trucks are introduced rapidly.

2.2.6 Summary

The literature reveals that freight transport professionals hold varied perceptions of driverless trucks and recognize both potential benefits and challenges. On one hand, experts anticipate that driverless trucks could lead to substantial cost savings, alleviate driver shortages, and increase vehicle utilization. Long-haul highway transport between logistics terminals is seen as a plausible initial application due to its relatively controlled environment. However, concerns are expressed regarding operational aspects such as cybersecurity risks, the absence of an onboard driver to handle non-driving tasks such as load handling and administrative duties, and complexities in delivery processes. Additionally, skepticism exists about the feasibility of fully automating all transport, potential issues with international cross-border transport and the long-term cost performance. The industry also anticipates shifts in business models, with increased collaboration both vertically and horizontally, potentially leading to consolidation in the logistics sector. These varying perceptions highlight the deep uncertainty and complexity surrounding the future of driverless trucks.

Research on the market adoption of driverless trucks is limited and indicates uncertainty in potential uptake patterns. Studies suggest that adoption rates could vary widely over the next decades, influenced by factors such as technological advancements, cost barriers, and company priorities. Unsurprisingly, freight companies often prioritize factors like long driving range and low operating costs over automated driving features as such and many companies expect that substantial upfront investments may hinder adoption. There is skepticism about the timeline for widespread availability, with many logistics firms not believing that adoption will occur soon. This gap in understanding complicates efforts to predict when and how driverless trucks will penetrate the market.

Driverless trucks have the potential to significantly reduce transport costs, primarily through decreased driver-related costs, with studies indicating

possible total cost reductions between 15% and more than 40%. However, these estimates are subject to considerable uncertainty. Not all labor costs may be eliminated, as tasks like loading, unloading, and maintenance still require human involvement. The need for remote operators to monitor or assist driverless trucks could further offset labor savings. Energy efficiency improvements are another potential benefit, as optimized driving behaviors and reduced speeds might lower energy consumption.

A future adoption of driverless trucks might bring substantial changes to logistics and freight transport systems. Some of the discussed changes in the literature are related to driverless truck enablers: adaptations required to support driverless truck operations or to compensate for their constraints or limitations. The lack of an onboard driver necessitates reconfiguring logistics processes, such as handling loading and unloading without driver assistance and managing vehicle maintenance remotely which may involve automation or additional personnel. Moreover, structural shifts in logistics networks may emerge, including the development of hub-to-hub operating models where driverless trucks handle long-haul segments between transfer points. The different cost structure for driverless trucks compared to manually driven trucks may affect shipment sizes, vehicle types, and delivery routes. Driverless trucks could also influence modal shifts by altering the competitive balance between road and other transport modes, potentially increasing road freight volumes. These potential changes introduce substantial uncertainties for long-term forecasting of key transport system outcomes such as modal splits, road freight volumes, and transport patterns.

3 Theoretical background

This chapter provides a background to key theoretical perspectives and frameworks that are either directly applied in the thesis papers or are important for the overarching research design and problem formulation of the thesis.

First, Section 3.1 outlines basic concepts and frameworks for understanding freight transport as a system and presents a multi-layer market-based framework for modeling freight transport decisions, which is applied in Paper D and Paper F. Section 3.2 then briefly introduces decision making under deep uncertainty, which serves a foundation for the methodological approaches used in Paper E, Paper F, and Paper G. Section 3.3 summarizes transition theory, which relates to Paper B. Finally, Section 3.4 presents a research framework integrating the thesis papers and research objectives with the theoretical perspectives.

3.1 A systems perspective on freight transport

Systems can be defined and understood in a variety of ways (Backlund, 2000), but for the purpose of this thesis, a definition based on Meadows (2008) suffices as an overarching definition of systems as a general concept:

> *A system is a set of elements that is coherently organized and interconnected in a pattern or structure that produces a characteristic set of behaviors, classified as its purpose.*

This definition highlights three central aspects of systems: elements, interconnections, and a purpose (Arnold & Wade, 2015). Central to Meadows' understanding of systems is that the purpose of a system is often the key factor determining the system's behavior, but it might not be obvious or explicit.

Freight transport can be understood as a complex sociotechnical system that is shaped by decisions by multiple types of actors who operate with different

objectives and on varying timescales and interact through a range of mechanisms (Auvinen & Tuominen, 2014; Holguín-Veras et al., 2011; L. Tavasszy & de Jong, 2014). Moreover, freight transport systems co-evolve with other societal systems, such as the energy system, industrial systems, and the economy at large (E. Rosenberg et al., 2023; Saidi & Hammami, 2017).

3.1.1 Freight transport fundamentals

The demand for freight transport is a derived demand resulting from logistical activities required to move materials and goods from production sites to consumption sites (Rodrigue, 2020; L. A. Tavasszy et al., 2012; L. Tavasszy & de Jong, 2014). Although transport fills an important function in supply chains, transport decisions can be of peripheral focus and are often implicitly determined by product and market related decisions (Lundberg, 2006; Reis & Macário, 2019; L. Tavasszy et al., 2020; L. Tavasszy & de Jong, 2014).

Freight demand is largely driven by factors external to the transport system, such as the technological and spatial structure of the economy and industry, trade patterns, supply chain structures and strategies, land use systems, and cultural and behavioral aspects (van de Riet et al., 2007). However, freight transport demand is sensitive to generalized transport costs through a number of mechanisms, for instance, by changing sourcing locations or even changing the demand for the end product (de Jong et al., 2010).

Globally, the demand for freight transport is expected to continue to increase due to economic growth and associated trade during the coming decades. For instance, the two scenarios in the latest ITF Transport Outlook (International Transport Forum, 2023), which differ in the extent of implemented climate policy measures, predict substantial increases in global tkm by 2050 compared to 2019 levels (see Figure 4). However, the trajectory for GHG emissions differs substantially between the scenarios with a net increase in emissions projected for the Current Ambition scenario and a net reduction of almost 80% in the High Ambition scenario. The ITF scenarios highlight a central challenge: facilitating growing transport demand while rapidly decarbonizing.

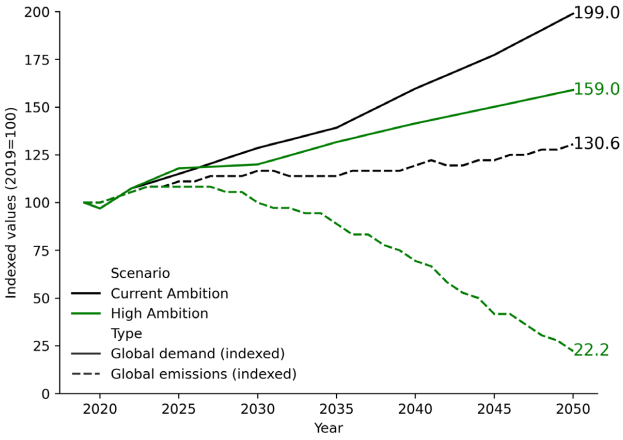


Figure 4 Change in global freight demand and direct CO_2 emissions from freight for ITFs scenarios Current Ambition and High Ambition from ITF Transport Outlook 2023 (International Transport Forum, 2023). The data is indexed to 2020 = 100. Figure by the author based on data from International Transport Forum (2023).

In Sweden, freight transport activity has remained relatively stable over the last two decades at around 100B tkm per year. The most recent STA reference forecast predicts growth in freight demand from 2019 to 2045 of around 21% in terms of tonnes, 23% in tkm and 69% in cargo value while GDP is expected to grow by around 55% during the same period (Wikström, 2024). By mode, the forecast predicts road freight transport on Swedish territory to increase by 42%, rail transport to increase by 32% while sea transport is expected to decrease by 13%. The trend of freight demand (in tonnes or tkm) growing slower than GDP is not specific to Sweden, but has been observed generally in the EU during the last decade (Manzanedo et al., 2024).

Freight transport can be categorized into four main transport modes: air, road rail, and sea. The different transport modes have different cost structures. Two basic cost components can be distinguished: fixed costs, which are independent of transport distance (e.g. loading/unloading of vehicles and terminal handling, etc.) and distance dependent costs (e.g. driver and energy cost). Compared to rail and sea transport, road freight has low fixed costs but high distance dependent costs (Rodrigue, 2020).

Often, several transport modes, or different vehicles of the same mode, are combined for transporting a shipment from the shipper (production site) to its receiver (consumption site). Such a combination is called a *transport chain*. A distinction is made between PC-flows (production to consumption zone flows) and OD-flows (origin to destination flows in a transport network). A PC flow is the flow of goods from a production to a consumption site, i.e. between the

origin and destination of a shipment but may consist of one or several legs, where each leg is an OD flow. A transport chain can be either unimodal (using a single transport mode, although possibly combining several vehicle types), or multimodal (composed of several legs combining two or more transport modes).

The example in Figure 5 illustrates four alternative transport chains.

(1) Direct road (full truckload): a shipment is transported directly from its origin to its destination, and the shipment size is well-matched with the vehicle capacity.

(2) Road (less-than-truckload): several smaller shipments from nearby origin locations to similar destination locations are transported by a single truck. This is done to achieve unit cost reductions by having multiple transport buyers sharing capacity and costs. The shipments are picked up in sequence in the origin region and then delivered sequentially in the destination region, which adds costs and time compared to a direct road transport chain.

(3) Consolidation-Distribution: many smaller shipments (parcels) are transported by utilizing a network of terminals that facilitates consolidation in the origin region and distribution within the destination region. Between the terminals, line-haul transport with large trucks is performed, while the first- and last mile tours typically are performed by smaller trucks.

(4) Multimodal: rail or sea is used for the main haul and road transport is used for the first and last mile to/from the origin/destination and the terminal providing access to rail or sea transport. Intermodal transport is a sub-set of multimodal transport in which the goods are transported in a standardized load carrier (e.g. container) and no handling of the goods is done when transferring between modes or vehicles.

Transfers (between vehicle types as in (3) or between modes as in (4)) add costs and time and there is a trade-off between the added costs and time for transfers and the reduced distance costs that may be obtained by switching to another mode or vehicle type. Some production or consumption zones may have direct access to the rail or sea networks without the need for intermediate road transport.

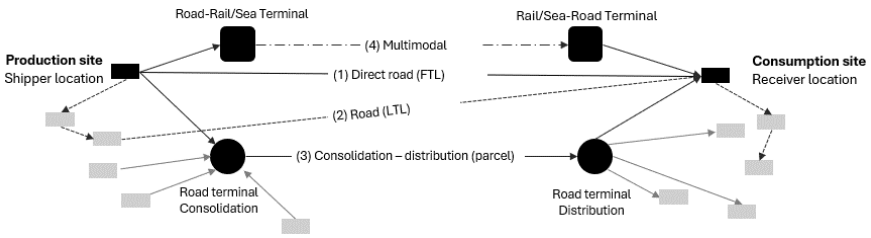


Figure 5 Schematic illustration of different transport chains. Adapted from Engholm (2021).

From the perspective of driverless trucks, it is noteworthy that in all the above examples, the truck driver may fulfil several important roles in addition to driving the truck. At the origin this may include performing or supervising loading, securing the load, managing shipping documents, etc. During transport, the driver may need to perform vehicle inspections, customs handling, etc. At a terminal or destination location the driver may supervise or perform unloading, load returns or handle delivery confirmations. For (2), the non-driving tasks are particularly important since many origins and destinations are served during the same trip, often under volatile conditions (e.g. stops being added or removed, time slots changed, etc.) and the shippers and receivers may not have available staff or equipment for performing these tasks.

3.1.2 Freight transport as a multi-layer system

One influential approach to describe the relationships between various decisions and sub-systems of the freight transport system conceptualizes it as a set of connected system layers that interact via market mechanisms (L. Tavasszy & de Jong, 2014). An early such framework was developed by Wandel et al. (1992) and it distinguishes three system layers: material flows (the movement of goods in the logistic system), transport flows (the movement of load units and vehicles in the transport system), and the infrastructures within which vehicle movements take place. In short, the framework states that the demand for material flows generates a demand for load unit movements which in turn generates a demand for vehicle movements. The demand for load unit movement is satisfied by transport services, and the demand for vehicle movement is satisfied by traffic services.

A similar multi-level framework is provided by Tavasszy (2008) (see Figure 6) which consists of five layers: production and consumption zones, trade patterns linking production and consumption, logistic services organizing the movement of materials and products, transport services providing the transport within logistic networks, and network services providing the infrastructure capacity for performing transport. The layers affect each other, and thus, changes within one layer may affect decisions and activities in the adjacent layers and propagate further in the system. For instance, changes to the cost, quality or availability of transportation services may influence decisions in the logistic system. Various system responses to the introduction of new technology happens on various time scales where some decisions and their implementation take multiple years or decades (e.g. relocation of industry) while some are more immediate (e.g. adjusting the routing and scheduling of vehicles) (Lundberg, 2006; Stenemo et al., 2021; L. Tavasszy & de Jong, 2014).

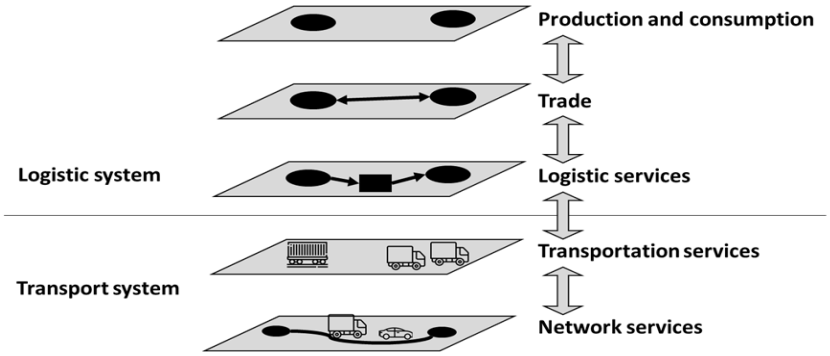


Figure 6 System layers in the logistic and freight transport system. Adapted by the author based on Tavasszy (2008)

The system layers interact via markets. For instance, the transport market matches the demand for material flows with the supply of load unit movements offered by transport service providers. The notion of a single market, "*the* freight transport market", is used in this thesis, but it is a simplifying abstraction that highlights the market-based interactions between transport buyers and transport service providers.

However, for this thesis, a simplified framework will suffice, see Figure 7. This simplified framework includes two layers: the demand for material flows and the supply of freight transport services, complemented by the transport market, which mediates matching of demand and supply. The framework also illustrates a simple way to represent the impact of emerging truck technology adoption, such as driverless trucks, namely as a factor influencing road transport supply, which then has repercussions throughout the system, generating changes in system outcomes.

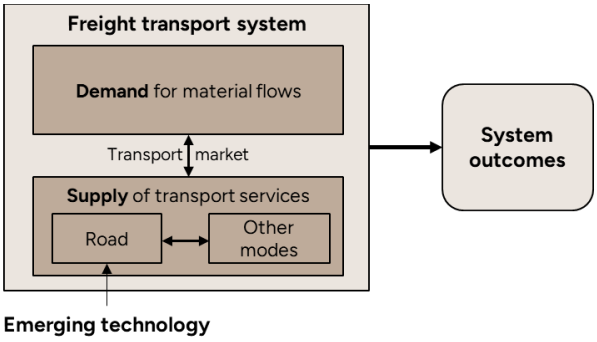


Figure 7 Simplified conceptual model of the freight transport system.

3.1.3 Understanding decisions on the freight market

Transport buyers represent the demand side of the transport market and select and purchase transport services on the transport market. The transport buyer can be either the company shipping the goods or the one receiving the goods, and they set the requirements for the transport service based on the specific logistical context (Lammgård et al., 2013). A key decision is the selection of transport chain, which is often made simultaneously with the selection of shipment size and can be understood as a cooperative process between the shipper and carrier converging towards the solution that best serves the transport buyer's interests (Holguín-Veras et al., 2011, 2021).

The selection of transport chain (or mode) is often the focus of transport policy interventions seeking to achieve a modal shift from road to rail and/or sea (Johansson et al., 2024; Nassar et al., 2023; Trosvik et al., 2023). However, in practice, this decision is often taken without much contemplation and consideration (L. Tavasszy & de Jong, 2014), or it may not be much of an choice since there is only one mode that is a feasible alternative (Rich et al., 2011; Sundin, 2017).

There are multiple factors affecting the decision (or lack thereof) for selection of transport chain, which is shown in the framework by Landborn et al. (1981)[8]. The framework describes this decision process as being affected by three main aspects: restrictions, choice and inertia, see Figure 8.

- Restrictions are physical prerequisites (e.g. access to harbors or the rail network) or dimensioning requirements which may exclude certain transport options. Restrictions thus determine what transport options are feasible for the specific flow.
- Choice is when the transport buyer chooses between several feasible transport options offering different levels of cost, quality, transport times, etc. However, many times, the restrictions leave only one feasible option (Rich et al., 2011), and competition between modes only occurs when there is an overlap in availability, service quality, and geographical coverage.
- Inertia are biases or lock-ins leading to sub-optimal transport decisions or delaying the choice of the optimal solution for the transport buyer. It can also be present for transport service providers, e.g. by not developing their service or adopting new technology.

[8] I was not able to obtain a copy of the publication by Landborn et al. (1981) and thus rely on the extensive summary of the framework provided in Lundberg (2006).

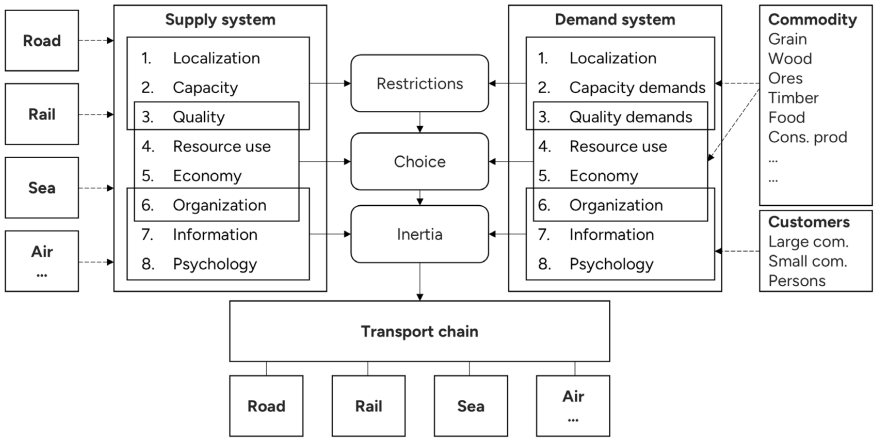


Figure 8 Conceptual model of the choice situation at the transport market. Re-drawn and adapted by the author based on Lundberg (2006).

Previous research has evaluated factors that influence decisions on the transport market, and in particular, the choice of transport service and transport chain. A review of empirical studies examining European freight transport buyers' selection of transport services concludes that cost is the primary factor considered (Flodén et al., 2017). However, this consideration is only made after ensuring that the service meets qualifying requirements regarding transport quality. An empirical study of the US identifies transport price and reliability as the two most important factors determining mode choice (Holguín-Veras et al., 2021).

One important topic is how decisions on the transport market can be predicted for a given (hypothetical) situation. Models of these decisions can be either aggregated (e.g. modeling the mode split within a region or nation as a continuous function of transport mode characteristics) or disaggregated by modeling the decisions for a single transport buyer or a single freight flow, see Holguín Veras et al. (2021) for an overview of model types.

A common approach for disaggregated modeling is to assume that transport buyers operate based on principles of inventory theory (Combes, 2014; Holguín-Veras et al., 2021). The general principle is that transport buyers seek to minimize total logistic costs through decisions on shipment size, the use of consolidation centers and terminals, transport chain choice and vehicle choice (de Jong & Ben-Akiva, 2007).

This approach is the basis of the logistics model (Logmod) used within the Samgods model framework (de Jong & Baak, 2020) which is used in Paper D and Paper F. Here, a simplified version of the formulation provided by Ben-Akiva and de Jong (2013), which is consistent with the one used in Samgods, is introduced, see *Eq. 1*.

$$G = O + T + D + Y + I + K + Z \quad (Eq. 1)$$

G is the total logistic cost for a specific transport solution serving a given transport flow. *O* is order costs, *T* is transport costs, *D* is cost for deterioration and damage of the goods during transit, *Y* is capital costs of goods during transit, *I* is inventory storage costs, *K* is capital costs of inventory and Z is stockout costs. Transport costs can be further broken down into link costs (costs for driving and operating vehicles on transport network links) and node costs (costs for loading and unloading goods into vehicles or containers, and costs for transfers between different types of vehicles or transport modes).

This theory and modeling framework is well established within the freight modeling literature and is useful for modeling the decisions made on the freight transport market for a given set of conditions on the supply- and demand side (L. Tavasszy & de Jong, 2014). By comparing different scenarios with varying assumptions for the characteristics of the supply and/or demand side it can be used to study the effects of changes in the freight transport system (e.g. by comparing scenarios "before and after" an introduction of driverless trucks).

3.2 Deep uncertainty

An important perspective in this thesis is that the development and impacts of driverless trucks can be understood as being characterized by *deep uncertainty*, as defined in Section 1.4.5[9]. This perspective has implications for what can be regarded as appropriate analysis approaches for model-based foresight and policy analysis.

This chapter provides an overview of frameworks for understanding and classifying uncertainty in model-based analysis and policy analysis (Section 3.2.1) and an introduction to decision making under deep uncertainty (DMDU) (3.2.2), which includes approaches used for Paper E, Paper F, and Paper G. The text in this section is partly based on a previous literature review (Engholm & Kristoffersson, 2024).

[9]*Deep uncertainty exists when analysts do not know, or the parties to a decision cannot agree on, (1) the appropriate models to describe the interactions among a system's variables, (2) the probability distributions to represent uncertainty about key variables and parameters in the models, and/or (3) how to value the desirability of outcomes. (Lempert et al., 2003, pp. 25–26)*

3.2.1 Deep uncertainty in model-based analysis

In the simplest sense, uncertainty is a *limited knowledge of future, past, or current events* (Walker et al. 2013). However, obtaining more information about a system does not necessarily reduce the level of uncertainty about it. Rather, more information may sometimes increase uncertainty about the system as more is revealed about its complexities and additional uncertainties are identified (Saltelli & Funtowicz, 2014).

As described in Section 1.4.5, deep uncertainty refers to situations in which it is difficult for decision makers and stakeholders to assign probabilities to events or predict outcomes due to complexity, lack of information or inherent unpredictability of a system. Deep uncertainty thus signifies situations with a high *level* of uncertainty. The literature on model-based foresight and policy analysis under uncertainty has conceptualized uncertainty as a multi-dimensional phenomenon, in which the level of uncertainty is one of these dimensions (Agustdinata, 2008; Funtowicz & Ravetz, 1990; van Asselt, 2000).

For this thesis, a framework by Kwakkel et al. (2010) is used for separating various types, and aspects, of uncertainty in model-based analysis. The framework distinguishes three dimensions of uncertainty: 1) the location, 2) the level, and 3) the nature of uncertainty.

The *location* of uncertainty specifies six areas in an analysis in which uncertainty may be present: system boundaries, the conceptual model, the computer model, inputs (including uncontrollable external factors and controllable variables such as policy levers), parameters, and the processed output data.

The *level* of uncertainty specifies four degrees of uncertainty. Shallow uncertainty is when the likelihood or plausibility of uncertain alternatives can be described with probabilities. Medium uncertainty is when alternatives can be enumerated, and rank-ordered in terms of their likelihood, but how much more or less likely different alternatives cannot be specified. Deep uncertainty is when alternatives can be enumerated using a nominal or categorical scale, but decision makers cannot agree on, or do not know, appropriate probabilities or a rank ordering of alternatives. Finally, recognized ignorance refers to situations when alternatives cannot be enumerated, admitting the possibility of being surprised.

The *nature* of uncertainty distinguishes between three types. Epistemic uncertainty results from a lack of knowledge and can potentially be reduced through further research. Variability refers to uncertainty caused by inherent system variability, particularly in human and natural systems, as well as in social, economic, and technological developments. Ambiguity arises when different actors interpret data and results differently due to variations in frames of reference or values.

A related and complementary perspective is uncertainty within the policy making process, for which additional factors are relevant. In the context of policy making, uncertainty can be understood as a difference between the available knowledge and the required knowledge for making the best policy decision (Marchau et al., 2019b). A framework for classifying uncertainty in policy analysis is provided by Walker (2000), Walker et al (2013a) and Marchau (2019b), illustrated in Figure 9. A central component of the framework is the system model (R) which specifies the system boundaries and the system's internal structure in terms of its elements and their relationships. The system is affected by two types of factors. External forces (X) which are forces acting on the system that the decision makers cannot control, for instance technological or social developments. Policies (P) are the levers that the actors in the policy domain can use aiming to affect the system so that outcomes of interest (O), change in a desirable way based on the relative valuation of the goals, objectives, and preferences (W) of the decision makers and stakeholders.

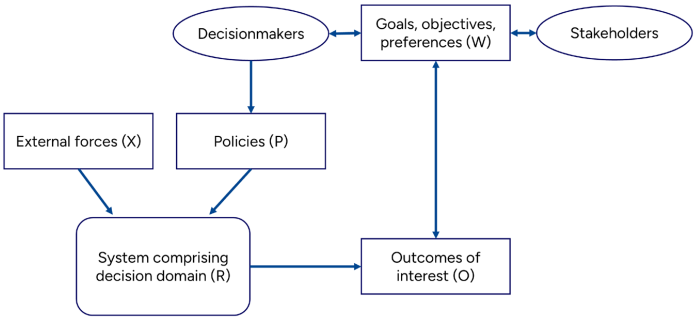


Figure 9 A framework of the policy analysis process, redrawn and modified by the author based on Marchau et al. (2019b, Figure 1.2).

Based on this framework, a few types of uncertainties that may arise can be outlined (Walker, Marchau, et al., 2013b).

- Scenario uncertainty (X). Uncertainties in the development of external factors affecting the system.
- Model uncertainty (R). Uncertainty in the system response to external factors (X and P). The internal causal mechanisms of the system may not be fully understood. Additionally, in complex systems, structural changes may occur when influenced by external factors, and sociotechnical systems may experience endogenous structural changes due to feedback loops, delays, emergent behaviors, or self-organization (Holtz et al., 2015).
- Uncertainty in valuation of outcomes (W). Different stakeholders may have contesting views on the relative importance of outcomes and there

may be disputes on what grounds valuation should be done (e.g., social cost of carbon versus shadow price) or uncertainty due to lack of data. Furthermore, relevant stakeholders might emerge, or the perception of the importance of problems may change over time.

3.2.2 Decision making under deep uncertainty (DMDU)

Situations with deep uncertainty require different approaches for decision making compared to situations with shallow uncertainty for which conventional prediction-based or probabilistic approaches can be used (Dittrich et al., 2016; Lempert et al., 2003; Maier et al., 2016; Malekpour et al., 2016; Marchau et al., 2019b). Generally, approaches for dealing with deep uncertainty account for multiple plausible futures, consider policy robustness, and support adaptive and flexible policy approaches (Lyons & Davidson, 2016; Maier et al., 2016).

DMDU is an umbrella term for a variety of methods and approaches designed to help decision-makers navigate complex and uncertain problems. It is based on three key ideas: exploratory modelling, adaptive planning and decision support (Kwakkel & Haasnoot, 2019). DMDU provides an alternative decision-making approach to the conventional "predict-then-act" method, in which a single or a small set of forecasts serves as the basis for policy design and analysis (Lempert et al., 2003). One way of describing DMDU in contrast to prediction is that it focuses on the decision alternatives and how uncertainty affects their impacts:

> *Which assumptions would I need to believe will hold true to recommend one course of short-term actions over another? (Lempert et al., 2022, p. iv)*

It is beyond the scope of this thesis to provide a comprehensive summary of DMDU methods. Readers interested in a broader overview are referred to Marchau et al. (2019a) which summarizes a number of approaches, including: Assumption-Based Planning, Robust Decision Making (RDM), Many-Objective Robust Decision Making (MORDM), Dynamic Adaptive Planning, Dynamic Adaptive Policy Pathways, Info-Gap Decision Theory, Engineering Options Analysis, Decision Scaling, Scenario Discovery, Adaptation Tipping Points, Many-Objective Robust Optimization.

While DMDU has gained popularity in other fields such as climate adaptation and water management, their use in transport remains limited (Stanton & Roelich, 2021). Examples of transport-related DMDU studies include conceptual assessments for transport policy and urban planning applications (Jittrapirom et al., 2018; Marchau et al., 2010), strategic port- and airport planning (Halim et al., 2016; Kwakkel et al., 2012) and exploratory modeling and analysis for regional transport demand models (Milkovits et al., 2019). In Paper G one specific DMDU-method, MORDM, is applied. In Paper E, and

Paper F, no specific DMDU framework is used, but a variety of EMA techniques commonly employed within DMDU are applied. These methods are further explained in Chapter 4.

3.3 A brief summary of sociotechnical transitions

One approach for understanding transformative change in sociotechnical systems is the concept of sociotechnical transitions and transition theory (Sovacool & Hess, 2017). A large body of literature has emerged within the field of transitions research (Markard et al., 2012) which studies how transitions of societal systems can be understood and managed. Often, these studies are done from the perspective of sustainability, seeking to understand *sustainability transitions*: shifts to new kinds of sociotechnical systems to tackle societal sustainability problems (Köhler et al., 2019).

A central idea in this research domain is that grand societal challenges, such as achieving sustainable transport, are *wicked problems* (Rittel & Webber, 1973) resulting from system failures that cannot be solved with conventional policy measures within the current system structure (Rotmans & Loorbach, 2009). Instead, tackling these problems requires a *transition*, which is a change from one *sociotechnical regime* to another (Auvinen & Tuominen, 2014; Geels & Schot, 2007; Rotmans & Loorbach, 2009).

A (sociotechnical) regime is the set of dominating technologies, practices and rules that enable a system to provide its functions to society, such as the provision of freight transport (Cherp et al., 2018). A transition requires extensive changes across a wide range of factors including political, social, technological, economic and organizational aspects (Markard et al., 2012). Regimes tends to be stable, resilient, and resist pressures to change but may innovate to adapt or expand within its logic (Cherp et al., 2018).

Emerging technologies challenging the dominating technologies or practices in the regime tend to be developed in a *technological niches,* or *technological innovation system*. These are partly separated from the regime, aiming to ultimately diffuse the emerging technology within the regime. However, the resilience of regimes makes it hard for novel technologies or innovations that challenge the existing regime to enter it (Cherp et al., 2018). Transitions create opportunities for these technologies and innovations (niches) to enter the regime (Geels & Schot, 2007).

It has been argued that using a variety of perspectives and methods is beneficial for gaining a comprehensive understanding of transitions. Cherp et al. (2018) propose that national energy transitions can be conceptualized as a co-evolution of three types of systems: energy flows and energy markets, energy technologies, and energy-related policies. Each of these three types of sub-systems tend to be in focus for different research fields that offer distinctly different, but complementary, perspectives (Cherp et al., 2018). For this thesis,

the framework is adapted to the study of (national) freight transport systems with the corresponding three types of systems being: transport flows and markets, transport technologies, and transport-related policies. The research perspectives outlined by Cherp et al. (2018) can then be summarized as:

- *Techno economic perspective* focuses on transport flows and transport markets applying theories and methods from engineering and economics. Typical methods are quantitative system models to project future scenarios and assess the impact of policies or new technologies. A limitation with this perspective is the lack of ability to explain some types of dynamic system behaviors such as momentum, path-dependence and innovation processes, which instead is a focus of the sociotechnical perspective.
- *Sociotechnical perspective* focuses on emerging technologies and the interplay between technology and society related to technological change. In contrast to the techno-economic perspective, in which technology is analyzed from its instrumental capabilities, the sociotechnical perspective considers technology as embedded within society through interaction with knowledge practices, infrastructure and other technological artefacts. One prominent approach is the *Technological Innovation Systems* (TIS) framework (further explained in Section 4.2). However, the socio-technical perspective tends to focus on emerging technologies being significantly different from the regime and thus struggles to explain change from incremental innovations and long-term gradual changes in demand patterns.
- *Political perspective* focuses on the state and policies affecting the transport system. The political perspective examines political actions, interests, and institutions, shapes transitions using theories from political science and policy studies. A general shortcoming of this perspective is the limited consideration to material factors influencing the dynamics of the system and change thereof.

3.4 Research framework

Based on the theoretical background provided in Sections 3.1-3.3, an overarching research framework for the thesis is devised. This framework, illustrated by Figure 10, combines several theoretical concepts and frameworks and is partly inspired by an existing framework for transport foresight (Auvinen & Tuominen, 2014) which in turn is based on a sociotechnical system perspective and the structure of the multi-level perspective approach (Geels, 2002). The framework is primarily meant to conceptualize the thesis scope and how the research objectives and papers relate to each other and, by extension, to the aim and research problems addressed.

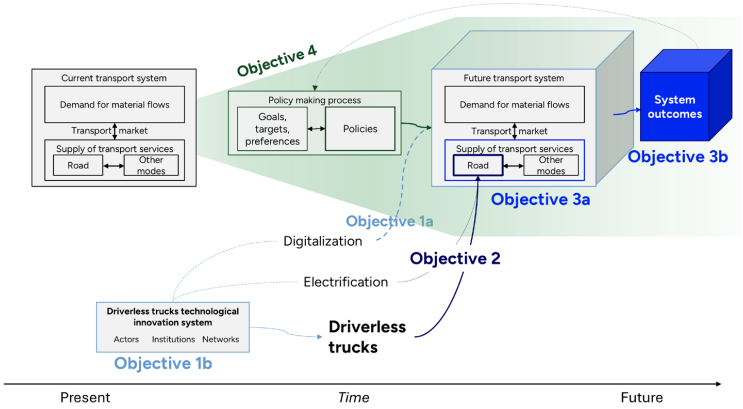


Figure 10 Conceptual research framework and mapping of the thesis objectives.

The freight transport system is conceptualized as a *sociotechnical regime* while driverless trucks are conceptualized as, currently, being a *technological niche*. The research framework assumes the possibility of a *transition* of the current regime to a future decarbonized freight transport system. This transition is largely driven by landscape pressures to decarbonize, for instance those manifested through EU-level and national legislation. The (potential) decarbonization transition is regarded as important context for the development of driverless trucks as it both influences the direction of search for the innovation system, but also since it will affect the characteristics of the future freight transport regime in which a large-scale driverless truck adoption may take place. In addition to driverless trucks, the framework also highlights electrification and digitalization as two key technological domains influencing the future freight transport system and driverless trucks.

The underlying logic in the freight transport system regime is represented using the simplified transport market framework introduced earlier in Figure 7. This conceptualization is used to describe and model how changes to the supply or demand for transport services affects freight- and logistics decisions, including the choice of transport chain and vehicle type, assuming a preference for solutions minimizing logistic costs (including transport, inventory, warehousing). In other words, this links how changes in supply of road freight transport due to driverless trucks affects system outcomes, i.e. their system-level impacts. It is assumed that this underlying market-based logic will be a valid representation and model of the decision logic also in a future freight transport regime, although possibly with significantly different prerequisites.

A central component of the research framework is the notion of deep uncertainty in model-based analysis for the development and impacts of driverless trucks. This is illustrated by the green cone, while the cubic shapes of

the future freight transport system and its system outcomes represent their uncertain future states, which can be considered a space of plausible development pathways, system states, and outcomes. Acknowledging the presence of deep uncertainty affects the choice of approach and methods for studying the system-level impacts of driverless trucks for research purposes and has implications for the decision-making approaches of various stakeholders, including policy makers and planners, represented by the policy making process in the center of the cone.

Within this research framework, the thesis objectives and their relationships can be outlined as follows. Objective 1a focuses on how the freight transport regime might develop in the coming decades, with an emphasis on digitalization. Objective 1b analyzes the technological niche of driverless trucks, its relationship to other technologies, and its prospects for entering the regime. Objective 2 models and examines how driverless trucks may influence the supply of road freight transport if integrated into the regime, i.e., if adopted and used for the provision of road freight transport services. Objective 3a addresses how to model driverless trucks and its effects on transport supply while Objective 3b seeks to provide an outcome space illustrating how the Swedish freight transport system might respond to the introduction of driverless trucks under various scenarios spanning uncertain developments. Objective 4 evaluates methods for model-based analysis under the deep uncertainty surrounding the development and impacts of driverless trucks.

An alternative interpretation of the thesis is as a study on driverless trucks in the sustainability transition of freight transport, mapped to the transition research perspectives by Cherp et al. (2018) introduced in Section 3.3:

- *The techno-economic perspective* is employed in Papers C, D, E, F, and G through analyses of the impacts of driverless trucks from a market-based perspective on the freight transport system.
- *The sociotechnical perspective* is applied in Paper B by using the TIS framework to study the emerging innovation system of driverless trucks in Sweden. To some extent, the scenario development study in Paper A also addresses this perspective by considering digitalization technologies in their sociotechnical context.
- *The political perspective* is not the primary focus of any paper, although political factors are central to the analyses in Papers A and B. Paper G examines transport climate policy in the context of driverless trucks but does so from a techno-economic rather than a political perspective.

4 Introduction to key methods and models

This chapter introduces key methods and models used in the thesis. A variety of methods and models are used, as is shown by Table 2. The thesis combines papers focusing on qualitative data and analysis (Paper A, Paper B) and papers with model-based analysis (Paper C, Paper D, Paper E, Paper F, Paper G). For the model-based papers, some use conventional scenario-based approaches and others apply DMDU or EMA-based approaches.

Table 2 Overview of methods and models used in the thesis papers.

Thesis paper	*Type of method(s)*	*Specific methods or models used*
Paper A	Scenario development	Intuitive Logics scenario development method based on expert workshops
Paper B	Qualitative innovation system analysis	Semi-structured interviews; analysis using the TIS framework
Paper C	Truck cost performance modeling and analysis	Comprehensive literature review and TCO modeling using ASEK cost model
Paper D	National freight modeling using the ADA framework	Samgods v1.2; scenario-based analysis
Paper E	EMA applied to truck cost performance modeling	Dedicated truck operations and cost performance simulation model; computational experiments (EMA)
Paper F	EMA applied to national freight modeling using the ADA framework	EMA-Logmod-AE; computational experiments (EMA)
Paper G	DMDU using transport climate policy model	STA's Scenario tool; MORDM

The remainder of this chapter is organized as follows: Section 4.1 gives a brief overview of scenario planning and specifically the Intuitive Logics (IL) approach which is used in Paper A. Section 4.2 summarizes the technological innovation system framework used in Paper B. Section 4.3 gives an overview to the models of truck cost performance analysis used for Paper C and Paper E. Section 4.4 presents the Swedish national freight transport model Samgods used for Paper D, and an extended version thereof used for Paper F. Finally, Section 4.5 introduces EMA, used in Paper E and Paper F, as well as Many Objective Robust Decision Making (MORDM), applied in Paper G.

4.1 Scenario development and the intuitive logics approach

Scenarios can be used for various purposes. Börjeson et al. (2006) distinguish three main scenario types: predictive scenarios (*what will happen*), typically in the form of forecasts of the most likely development; explorative scenarios (*what can happen)* which are used to identify a range of plausible futures in the face of uncertainty; normative scenarios (*how can a specific target be reached*) used within backcasting. In the context of transport planning, scenario planning has historically played a peripheral role but has more recently become increasingly integrated into actual planning and policy-making as a tool for handling uncertainty and stress-testing policies against multiple plausible futures (Lyons et al., 2021).

In many model-based fields, including transport modeling and EMA, the term *scenario* is typically used as *a description of the conditions under which the system or policy to be designed, tested, or evaluated is assumed to perform* (Walker, Marchau, et al., 2013a). This definition of scenario is applied in Paper C, Paper D, Paper E, and Paper F.

However, in the field of future studies, the concept scenario is typically used with a more restrictive meaning which is more relevant for the scenarios developed in Paper A. In future studies, scenarios usually have the following characteristics (Spaniol & Rowland, 2019): they are future-oriented, they deal with external context, they are plausible, they have a narrative description, they constitute a systematized set, the scenarios in the set are comparatively different.

A distinction can also be made between *scenario development* and *scenario planning* where scenario development refers to the creation of narratives about potential future states, while scenario planning involves additional steps where organizations use the scenarios to take action and to guide strategy development (Bishop et al., 2007). Considering this distinction, Paper A constitutes a scenario development study.

There is a range of techniques and methods for scenario development, surveyed in detail by Bishop et al. (2007). In Paper A, the Intuitive Logics (IL) approach is used. IL is used to generate explorative scenarios and is considered

as one of the most prominent scenario development method (Curry, 2022; Spaniol & Rowland, 2019). IL is a *dimension of uncertainty scenario technique* in which scenarios are specified as states within a set of uncertain dimensions (Bishop et al., 2007). Often, two uncertainties are in focus which are combined to form a 2x2 matrix where each quadrant of the matrix constitutes a scenario. The IL process comprises the following steps (Spaniol & Rowland, 2019):

1. Scope the study including specifying the issues of concern, the audience and user needs for the scenarios, and set the time horizon for the scenarios.
2. Identify driving forces and trends and rate them in terms of their degree of uncertainty and potential impact.
3. Frame scenarios by creating the scenario matrix by combining strategic uncertainties, describe scenarios in quadrants, map causal elements among system elements.
4. Refine scenarios by creating scenario narratives and assess strategic implications.

A key data source for the scenario development process is typically qualitative expert judgement data. This is also the case for Paper A, where such data was collected through a series of workshops.

4.2 The technological innovation systems (TIS) framework

The TIS framework is an analytical construct for analyzing and assessing the innovation process of technological systems, and focuses on conditions that foster technological innovations, such as learning processes, knowledge networks, and institutional support (Cherp et al., 2018). In Paper B, the Swedish driverless truck innovation system is analyzed based on the TIS analysis framework by Bergek et al. (2008). In this framework, a TIS is conceptualized as consisting of three types of structural components: actors, networks, and institutions that need to successfully perform a set of functions (i.e. key processes) to achieve the system goal. The TIS framework combines a structural perspective (mapping the actors, regulations, and other system components) with a functional perspective (understanding how key processes and activities for the success of the innovation system are performed) to enable an understanding of the dynamics of an innovation system and to critically assess its performance and identify areas for improvements (Hekkert et al., 2007).

4.3 Truck cost performance modeling and analysis

The basis for the cost calculations in Paper C and Paper E is the Total Cost of Ownership (TCO) framework. A TCO analysis intends to capture all costs for owning and operating a vehicle during the vehicle lifecycle and is commonly

used to compare the cost performance between alternative vehicle technologies (Ghandriz et al., 2020).

In Paper C, TCO is assessed for a set of generic type trucks corresponding to typical truck configurations for four different truck sizes (16-ton, 24-ton, 40-ton, and 64-ton maximum gross vehicle weight). The calculations are based on the TCO framework and cost data for Swedish manually driven trucks by ASEK 6.1 (Trafikverket, 2018), as shown in Figure 11. The calculations consider internalized costs from the perspective of the vehicle owner and exclude externalities or cargo-related costs.

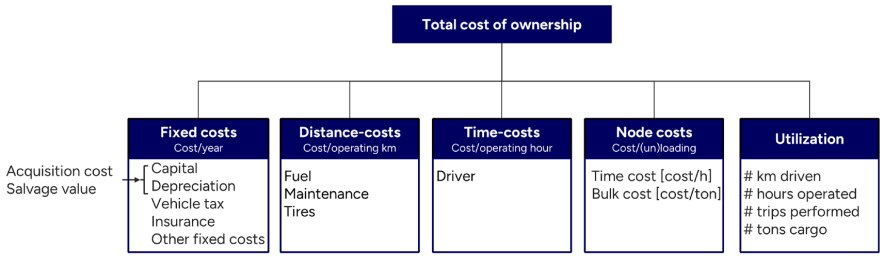


Figure 11 The TCO framework used in Paper C.

Potential impacts of driverless trucks on each cost component are reviewed, and TCO parameters are adjusted accordingly. To account for potential differences in truck utilization, the comparison is made in TCO normalized by total transport activity (tkm). A benefit of using the ASEK TCO model is that it is used to derive inputs to the Samgods model. Therefore, the results from Paper C can be used to derive cost input parameters for driverless trucks in Paper D.

Paper E has a different scope compared to Paper C, as it focuses on electrification and D-ETs. The analysis in Paper E is based on computational experiments and the vehicle utilization and energy consumption are simulated for individual transport missions over the truck life. However, the cost performance calculations are performed using a similar type of cost framework but extended to also consider electrification related costs such as charging and battery costs, as shown in Figure 12.

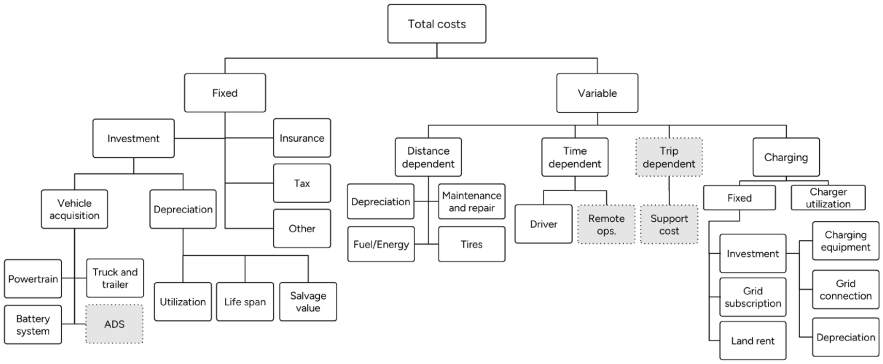


Figure 12 Truck cost framework used for Paper E. Highlighted boxes are cost elements specific to driverless trucks.

The approach for Paper E is to explore how the cost performance for M-ETs and D-ETs relative to M-ICETs varies with different assumptions of techno-economic factors such as technology costs and performance (e.g. cost for ADS, battery cost, degree of remote operations required) and energy costs, the characteristics of the transport application, and charging strategy.

To facilitate the analysis, a novel model for simulating truck operations and associated costs over the vehicle's life from a first buyer perspective was developed. Figure 13 provides an overview of this model. Each instance runs the corresponding experiment for an M-ICET, an M-ET and a D-ET which allows for direct comparison of their cost performance, defined as the difference in tkm-normalized truck lifecycle costs compared to the corresponding value for M-ICET.

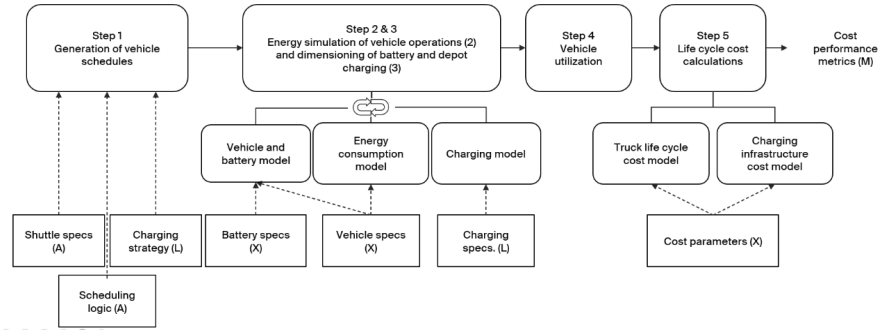


Figure 13 Overview of the simulation model used for Paper E.

A key difference compared to many previous cost performance studies is that a weight-dependent energy model is used, and that battery degradation is modeled, albeit simplistically. This allows analysis of how charging strategy and battery sizing decisions are interdependent, how they impact the energy

feasibility for various types of transport flows as well as the battery life and thus total vehicle utilization. Furthermore, the model endogenously dimensions the battery based on the energy requirements of the specific experiment, see Figure 14. This allows for analysis of the operational feasibility of ETs in terms of the resulting cargo payload capacity instead of the energy feasibility given a pre-determined battery capacity.

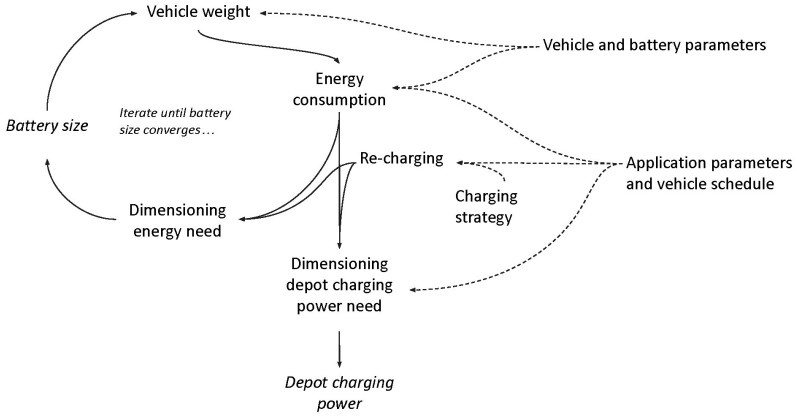


Figure 14 Relationship between charging strategy, battery size and energy consumption in the model used for Paper E.

4.4 National freight transport modeling using the aggregate-disaggregate-aggregate (ADA) framework

The basis for the modeling in Paper D and Paper F is the STA's national freight model Samgods (Trafikverket, 2024c). In Paper D, Samgods v.1.2 is used to analyze various scenarios of driverless trucks in Sweden. Driverless trucks are represented by adjusting parameters and input data of the existing vehicle types in the model which by default represent M-ICETs. In Paper F, the Samgods model is extended to enable the modeling of scenarios with a combination of M-ICETs, M-ETs and D-ETs by adding additional truck types to the model. Additionally, several input models were developed to facilitate EMA through automatic generation of scenarios and corresponding input data.

The Samgods model, as used for Paper D, is introduced in Section 4.4.1. The modifications made for Paper F are presented in Section 4.4.2.

4.4.1 The Samgods model: a Swedish national ADA model

Samgods is based on the aggregate-disaggregate-aggregate (ADA) freight transport modeling framework (Ben-Akiva & de Jong, 2013) and is primarily used to provide decision support for national transport planning, including the production of the Swedish reference forecast. Samgods has also been used in

various research studies, for example, to examine the economics of electric road systems (Börjesson et al., 2021) and for prospective policy analysis (Johansson et al., 2024; Vierth et al., 2024). ADA refers to the aggregation level of three central components in the modeling framework, as summarized below (De Jong et al., 2021):

- An aggregate model of production to consumption (PC) flows. This describes the demand for flow of goods between production zones and consumption zones, typically expressed in annual volumes (e.g. ton/year). PC flows are separated into different commodity types and can be modeled using multiple approaches and data, typically involving a combination of economic statistics like production and consumption data, input-output tables, and trade statistics.
- A disaggregate model of logistics decisions based on a normative principle of transport buyers seeking to minimize total logistics costs. This is a model of decisions on the freight transport market, as described in Section 3.1.4 and follows the cost formulation outlined in Eq.1. To enable modeling decisions on a firm level, the PC flows are disaggregated by splitting each PC flow into different categories based on the combination of size of the shipping and receiving firm using a simplified synthetic population of firm types. These disaggregated flows are denoted sub-cell flows. The output of the logistics model is the cost-minimizing logistic solutions per sub-cell flow including the choice of transport chains and vehicle types, shipment size and frequency, and transshipment locations.
- Aggregated network model. The solutions for the sub-cell flows are then aggregated up to OD flows which can then be assigned to the transport networks using standard network assignment techniques.

Samgods includes road, rail, sea, and air traffic and intends to cover all Swedish freight flows including domestic flows (starting and ending in Sweden), import and export flows and transit flows. The focus is on long-distance inter-municipality transport with distribution transport within municipalities modeled in a simplified way.

Freight demand is represented using exogenous PC-flow matrices of annual ton-flows for sixteen commodity types between 590 production and consumption zones. Out of these PC zones, 291 are domestic zones which correspond to the Swedish municipalities, while the other 299 represent international regions.

The core of the model is the logistics module, Logmod, (de Jong & Baak, 2020), the disaggregate model of logistics decisions described previously. Logmod uses exogenously defined transport chain options when searching for the cost minimizing solution. This is done in two iterative steps: BuildChain identifies the most cost-effective solution for each transport chain option and PC flow, ChainChoice then identifies the most cost-effective logistical solution, including choice of shipment size, given the available transport chain options.

Normally, Samgods contains 6 truck types, 9 train types and 22 ship types with different characteristics and cost structures. The six truck types are used to model trucks of different configurations and maximum gross combination weight (max GCW) which differ in terms of their loading capacity and cost structure. Samgods comes with two default scenarios: one base year scenario which typically is one or a few years behind the Samgods release, for which the model has been calibrated against real world transport statistics (Edwards, 2023), and one future scenario which corresponds to the STA reference forecast. For Paper D, the base year scenario for 2017 used in Samgods v. 1.2 (Trafikverket, 2020b) was used as the baseline scenario. In Paper F, a scenario for 2045 corresponding to the 2024 edition of the STA reference forecast (Wikström, 2024) was used as baseline.

4.4.2 EMA-Logmod-AE: extending Samgods to enable EMA for driverless and electric trucks

For Paper F, Logmod is extended with two additional groups of trucks, each with six truck types, in addition to the existing group with six truck types. This extended version of Logmod is denoted *Logmod-AE* (where AE denotes Automated Electric) and contains a total of 18 truck types. This allows to operate three groups of trucks with different characteristics in parallel. Therefore, it supports analysis of how different truck technologies may compete and complement each other without compromising on the granularity in representation of truck sizes. For Paper F, the three groups of trucks are used to represent M-ICETs (operated using diesel constituting100% biofuels), M-ETs D-ETs.

To enable computational experiments, several input models were developed to automatically generate scenario-specific Logmod-AE input files based on the scenario parameters. Also, scripts for setting up, performing and storing output for sets of scenarios were developed. This integrated system is referred to as EMA-Logmod-AE, an overview of which is provided in Figure 15.

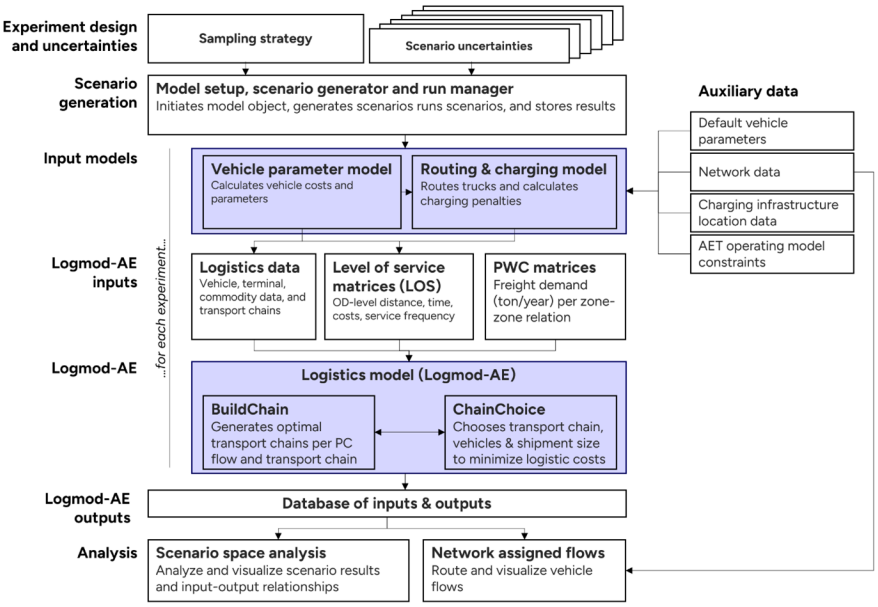


Figure 15 Overview of the EMA-Logmod-AE system developed for Paper F. Here AET = D-ET. The diagram is inspired by Johansson et al. (2024).

Three main aspects are considered for the modeling of M-ETs and D-ETs in Logmod-AE, as summarized below and with an overview of the calculations in Figure 16:

- Changes to the truck parameters in terms of weight, payload capacity and operating costs compared to M-ICETs.
- Route-specific energy feasibility and costs for charging in relation to the use of charging infrastructure and charging time.
- For D-ETs, route-specific operational feasibility given constraints imposed by technological or operating-model related factors.

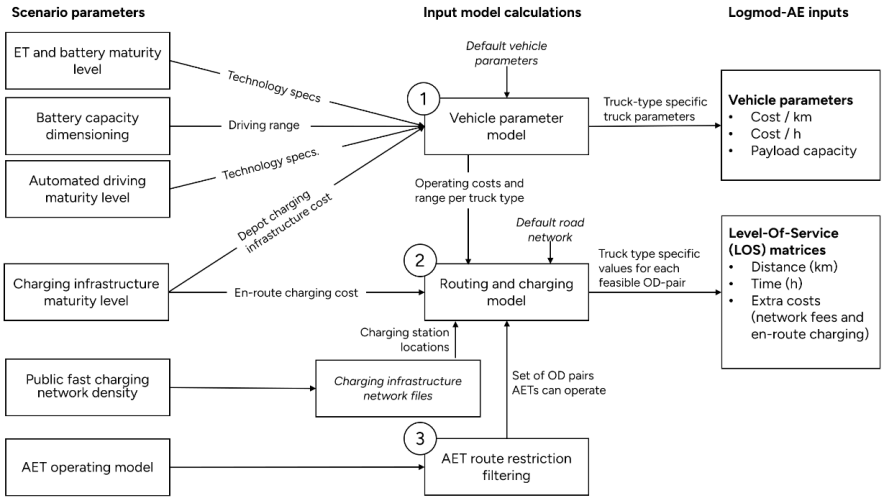


Figure 16 Overview of calculations for generating scenario-specific data for M-ETs and D-ETs in EMA-Logmod-AE. Here, AET = D-ET.

The combination of truck-type specific costs with route-specific costs and constraints provides a rich representation of the relative cost performance between the three groups of trucks. This representation is illustrated by Figure 17, which shows the total trip cost per OD flow over trip distance for the 60ton truck type for M-ICET, M-ET and D-ET for a scenario with low maturity of electrification and automation with small batteries (left panel), and high maturity of electrification and automation, and large batteries (right panel).

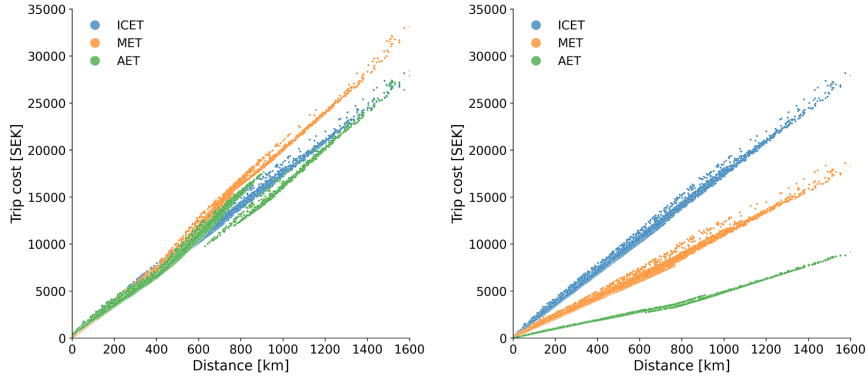


Figure 17 Examples of OD trip costs for 60t M-ICET (blue), M-ET (orange), and D-ET (green, denoted AET in Paper F). The left panel is a scenario with low maturity electric truck and battery technology, a battery sized for 4.5 hours of driving and low maturity for automated driving technology. The right panel denotes a scenario with high maturity for electric truck and battery technology, a battery sized for 9 hours of driving and high maturity for automated driving technology.

4.5 Exploratory Modelling and Analysis (EMA) and Many Objective Robust Decision Making (MORDM)

Paper E and Paper F both apply EMA but follow no specific DMDU framework while Paper G applies the MORDM framework. EMA is introduced in Section 4.5.1, MORDM in Section 4.5.2. followed by an overview of their applications to these three papers in Section 4.5.3.

4.5.1 EMA

Exploratory modeling is the use of computational experiments to explore how a system would behave under a large set of uncertain assumptions about the system and its inputs (Bankes, 1993). This approach is useful when dealing with complex or uncertain systems for which a single model, or a single set of assumptions for uncertainties affecting the system, cannot be derived (Kwakkel & Pruyt, 2013).

The concept of exploratory modelling was originally outlined by Bankes (1993) who distinguished exploratory modelling from predictive modelling. The goal of predictive modelling is to produce an accurate prediction of the system's behavior by incorporating known facts into a single unifying surrogate model of the system. In contrast, EMA is used to generate one or more plausible models that are consistent with available data and knowledge, scenarios, and policy options for the system of interest. The space spanned by scenarios, and policies is then sampled or searched, and inferences are made about how scenarios, policies and alternative model formulations affect system outcomes of interest. In exploratory modeling, a single instance drawn from the experiment set is not a prediction but rather a what-if experiment that reveals how the real-world system would behave if the assumptions about the various uncertainties formalized in the specific model and scenarios were correct.

EMA covers several methods and techniques, such as sampling techniques, sensitivity analysis, uncertainty analysis, scenario discovery, and (multi-objective) optimization, which can be applied to various types of system analysis and decision problems. EMA is closely related to the field of sensitivity and uncertainty analysis (Tarantola et al., 2024) and uses many techniques from that domain.

EMA problems can be structured using the XLRM framework (Lempert et al., 2003), see Figure 18.

- X: external uncertain factors. A specific set of parameter values of the external factors constitute a *scenario*
- L: policy levers are the decision variables the considered actor has available to influence the system. A specific set of parameter values of policy levers constitutes a *policy*.

- R: relationships in the system which collectively constitute a model of the system.
- M: performance metrics, the system outcomes of interest.

A unique combination of a scenario and policy comprises an *experiment*.

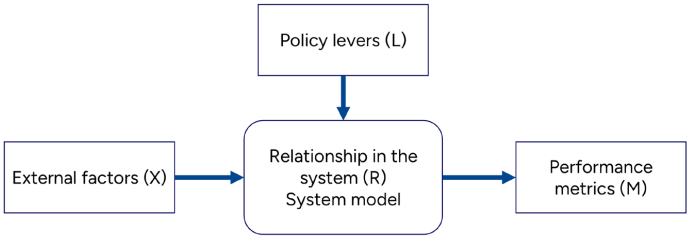


Figure 18 The XLRM framework (Lempert et al., 2003), figure based on Kwakkel (2017).

With this framework, EMA can be understood as an approach for analyzing how regions of the uncertainty space spanned by X, R, and the decision space spanned by L map to the outcome space, spanned by M. In some cases, it is also of interest to study uncertain parameters or relationships within the system model (R). Then, the analysis includes an additional parameter set for the uncertain model parameters. In cases when it is not clear what type of model is appropriate for representing the system, the analysis can be performed over multiple sets of models.

Kwakkel (2023) provides a useful distinction between two types of EMA techniques: *open exploration* and *directed search*. Open exploration is when sampling is used to explore the uncertainty space, and/or the decision space and may be used for various purposes, some examples are:

- Spanning the outcome space to understand what ranges of outcomes are plausible given a plausible set of uncertain input combinations, and policy options.
- Analyzing policy robustness by testing policies over scenarios spanning uncertain scenario parameters and quantifying their robustness using robustness metrics (Lempert et al., 2003).
- Assessing the influence of various uncertainties on model outputs by perform sensitivity or uncertainty analysis (Pianosi et al., 2016).
- Analyze under what circumstances a policy fails in achieving its intended outcomes. This can be done using scenario discovery methods to identify regions of the uncertainty space with high density of cases where the policy fails (Bryant & Lempert, 2010a).
- Identifying small sets of policy-relevant scenarios. This can be done by combining scenario discovery and search techniques for identifying diverse scenario sets (Carlsen, Eriksson, et al., 2016; Carlsen, Lempert, et al., 2016; Eker & Kwakkel, 2018)

Directed search which is the application of optimization to search the uncertainty space or decision space to identify sets of policies or scenarios with specific characteristics. For instance, the uncertainty space can be searched to identify the worst (or best) case scenario for a given policy (Halim et al., 2016). More common however, is to apply directed search to the decision space to identify candidate policies.

The application of directed search for policy identification in EMA often deals with complex, multi-objective policy problems. Such policy problems can be formulated mathematically as multi-objective optimization problems whose solution set is a set of non-dominated solutions, i.e. solutions that cannot be outperformed by any other solution across all objectives, representing a set of trade-off solutions among the objectives (Coello et al., 2007). Due to the complex search space and non-linearity of the problems typically being studied using EMA, approaches for solving such problems tend to be based on meta-heuristics such as multi-objective evolutionary algorithms (MOEA) (Eker & Kwakkel, 2018; Kasprzyk et al., 2013a).

Open exploration and directed search are often complementary and used in combination, as is the case for MORDM, which is further described in the next section.

4.5.2 MORDM

MORDM is used for generating robust policies for complex and uncertain policy issues with multiple conflicting objectives (Kasprzyk et al., 2013a). It is an extension of Robust Decision Making (RDM) for which the basic idea is to identify policies that perform well across a wide range of future conditions (Lempert et al., 2003). RDM was originally devised as a method to evaluate the robustness of pre-generated candidate policies. MORDM extends RDM by also facilitating the generation of candidate policies through directed search for multi-objective problems.

Typically, MORDM involves the following four steps, which are iterative and intended to be performed collaboratively with problem stakeholders and policy makers (Eker & Kwakkel, 2018; Kasprzyk et al., 2013a).

1. Problem formulation. The policy problem is conceptualized, for instance using the XLRM framework (Figure 18).
2. Generation of candidate policies. A multi-objective optimization problem is formulated based on the policy problem. The policy levers (L) constitute the decision variables and (a set of) the outcome metrics (M) comprise the objective functions. A MOEA is then used to identify a set of Pareto optimal solutions which constitute the candidate policies.
3. Robustness analysis. The candidate policies are subjected to uncertainty analysis, where their performance is evaluated over a large set of scenarios representing plausible developments of deeply uncertain

factors. The candidate policies' robustness is then evaluated by using one or several robustness metrics (Kwakkel et al., 2016; McPhail et al., 2018).

4. Vulnerability analysis. While the previous steps aim to generate candidate policies that perform well in the reference scenario and are robust, these policies may still fail under specific scenarios. Vulnerability analysis identifies such scenarios and their characteristics, often using scenario discovery methods such as the Patient Rule Induction Method (PRIM) (Bryant & Lempert, 2010b; Friedman & Fisher, 1999)

Different methods and tools can be used for each step. Choosing the most appropriate ones, and determining how best to apply them, requires careful consideration based on the specific nature of the policy problem, data, and model (Kasprzyk et al., 2013b; Kwakkel & Haasnoot, 2019).

It has been proposed that MORDM can serve as a learning process for policy makers helping refine their understanding of the policy problem and the system under study, generating better-performing policy alternatives, and fostering a deeper understanding of trade-offs (Kasprzyk et al., 2013a). The use of MOEA for generating policy alternatives alleviates some challenges and biases decision makers may face in situations involving many possible policy alternatives, multiple objectives to consider simultaneously, and deep uncertainty about the future (Kasprzyk et al., 2013a; Singh et al., 2015).

4.5.3 Application of EMA and MORDM in this thesis

Paper E and Paper F both apply EMA in the form of open exploration while Paper G applies MORDM. This section provides an overview of how EMA and MORDM are applied in these papers (see Figure 19), with more details available in the thesis paper summaries in Chapter 5.

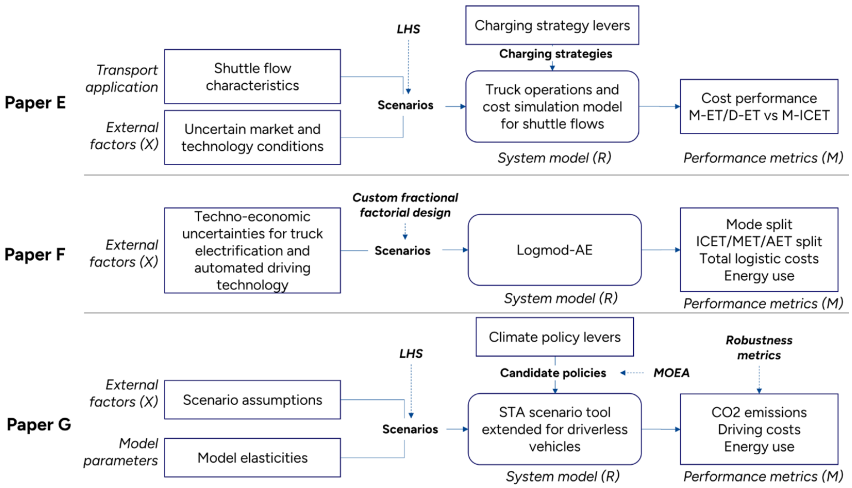


Figure 19 Overview of the approach for Paper E, Paper F, and Paper G using the XLRM framework (Figure 18).

In Paper E, computational experiments are used to explore the influence of uncertain market and technology conditions and the choice of charging strategy on the cost performance for M-ETs and D-ETs compared to M-ICETs for shuttle flows with different characteristics. The experiments thus cover three sets of input parameters: the shuttle characteristics, uncertain market and technology conditions (X) and a set of charging strategies (L). Scenarios are generated from the joint set of A, X parameters using Latin Hypercube Sampling (LHS) (Helton et al., 2006) and a full factorial is performed over the scenarios and a set of pre-specified charging strategies. The cost performance in each experiment (scenario-charging strategy combination) is analyzed using the truck operations and cost simulation model for shuttle flows presented in Section 4.2. Analysis is then performed using various data visualizations, and dimensional stacking is used to identify regions of the input parameter space with high concentrations of experiments where D-ETs (and M-ETs) are cost-competitive.

In Paper F, computational experiments are used to explore how techno-economic uncertainties for electrification and driverless trucks (X) may affect a set of system-level outcomes (M) for the Swedish freight transport system in 2045, which is modeled using Logmod-AE as the system model (R) (see Section 4.4.2). The uncertainties are modeled as categorical uncertainties, and the scenario set is designed using a custom fractional factorial design over the parameter categories.

Paper G applies MORDM to identify robust climate policies. The problem formulation includes a set of external uncertainties for the development of the transport system (X), a set of climate policy levers (L), uncertain model

elasticities (P), and a set of outcome metrics (M). A transport climate policy analysis tool developed by the STA, extended for modeling driverless vehicles in Paper G, is used as the system model (R). Candidate policies are created using a MOEA, and scenarios are sampled using LHS from the joint set of X, P parameters. The policy robustness is assessed using a set of robustness metrics and a variety of visualization techniques and statistical tests. Scenario discovery with PRIM (Bryant & Lempert, 2010b; Friedman & Fisher, 1999) is used for the vulnerability analysis.

5 Summary of thesis papers

This chapter provides a summary of the seven thesis papers in terms of their motivation and aim, method, and findings. Note that a synthesis of the findings, their contribution to the thesis objectives and relation to the literature is not provided here, but in Chapter 6. The summaries for Paper B, Paper C, and Paper D are partly based on Engholm (2021).

5.1 Paper A: How Will Digitalization Change Road Freight Transport? Scenarios Tested in Sweden

5.1.1 Motivation and aim

Digitalization, including automation, digitized information flows, data sharing and artificial intelligence, is expected to influence freight transport substantially during the coming decades. However, there is significant uncertainty about how these technologies will develop, be deployed and how they will interact with other developments affecting freight transport. The aim with this paper is to explore how digitalization may impact the road freight transport sector in Sweden by 2040 through a scenario development study. The paper analyzes the impacts of digitalization on transport efficiency, logistics, and the business ecosystem within the freight transport sector.

5.1.2 Method

Given the uncertainty surrounding the future of digitalization and the development of the transport sector, the study uses the Intuitive Logics scenario development method (see Section 4.1) to explore multiple potential futures. The scenario development was done in a workshop setting involving 52 experts from 33 different organizations from a wide range of domains related to the Swedish freight transport sector such as logistic service providers

and carriers, vehicle manufacturers, academia and research institutes, shippers, real estate, and consulting.

Three workshops were held. After each workshop, the collected data was analyzed by the paper authors and three additional participants, and the results were fed into subsequent workshops. During the first workshop, participants identified and described developments likely to influence the sector until 2040, rating them by uncertainty and potential impact. Strategic uncertainties were then selected, and a 2x2 scenario grid was constructed. Certain developments, trends regarded as largely scenario-independent and less uncertain, were summarized and used as a foundation for all scenarios. The second workshop refined the strategic uncertainties and elaborated drafts for the four scenarios. The analysis group then further developed the scenarios and prepared written scenario narratives. During the third workshop, the participants assessed and critiqued the scenarios and then analyzed potential impacts related to transport demand, transport efficiency, vehicle kilometers, logistics, business models and ecosystems. The analysis group finalized the scenarios and refined the impact analysis by assessing implications on the different stages of a generic supply chain model for each scenario.

5.1.3 Results

The certain development of trends that are expected to materialize independently of the four scenarios is summarized in Figure 20.

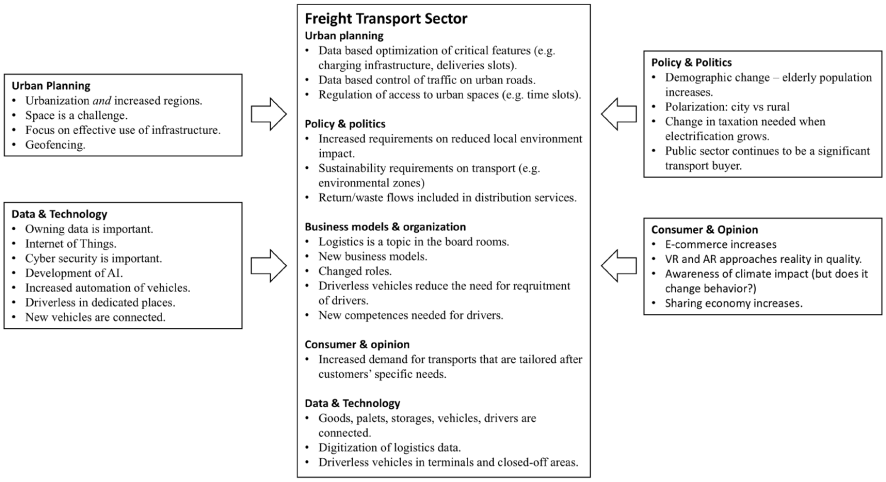


Figure 20 The certain, scenario-independent, development identified during the expert workshops for Paper A.

Figure 21 illustrates the two strategic uncertainties that constituted the scenario axes and spanned the four scenarios:

- Sustainability paradigm: whether climate issues become the top sustainability priority or remain one of many societal goals.
- Business logic for data sharing: whether traditional hierarchical industrial organization persists or shifts to a network-based, data-sharing ecosystem.

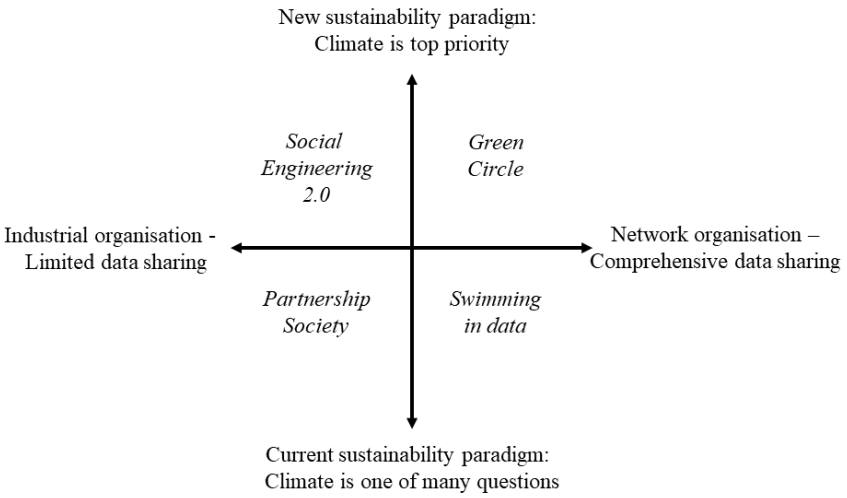


Figure 21 Scenario axes and the developed scenarios in Paper A.

The four scenarios that were developed are summarized below:

Social Engineering 2.0: is driven by transformative climate policies and strict regulations, such as high taxes on fossil fuels and raw materials, and mandatory recycling. Data sharing is limited, partly due to a general focus on data integrity, leading to sets of actors having created alliances in which data is shared. The impacts include increased use of biomaterials, a widespread electrification of road transport, rise in recycling flows, and improved transport efficiency through close collaborations within supply chains, but limited data sharing restricts horizontal optimization.

Green Circle: climate action becomes the most prioritized societal goal, leading to a circular economy with strong sustainability regulations. There is extensive data sharing that supports transparency and efficiency and enables a circular economy through coordination and minimal transaction costs. There is a significant increase in recycling and peer-to-peer transport flows, improved efficiency in product transport, but challenges in optimizing bulky goods transport.

Partnership Society: is characterized by established large companies dominating the business sector through strategic partnerships and data sharing within these networks. However, data sharing is limited between these networks. Climate actions are slow due to economic priorities, but data-driven services enhance overall supply chain efficiency which results in slightly improved transport efficiency. In total, there is a substantial increase in transport demand.

Swimming in Data: comprehensive data sharing drives innovation and efficiency. Public and private sectors collaborate in network-based settings which has sparked a wealth of new services and improved use of infrastructure. There is a major increase in transport demand, largely driven by consumer patterns and e-commerce, with delivery services being personalized offering high flexibility and short delivery times. These demands result in limited transport efficiency, in particular for urban and last-mile transport.

5.2 Paper B: The emerging technological innovation system of driverless trucks

5.2.1 Motivation and aim

Sweden is home to two leading truck manufacturers with stated ambitions to develop and offer driverless trucks to the market. In addition, there are several Swedish startup companies developing freight technology solutions for scaling the deployment of D-ETs, as well as technology companies developing automated driving technology. Thus, there are arguably ongoing efforts within Sweden to develop, deploy and utilize driverless trucks, and a greater understanding of the structure and dynamics of this development may be valuable to better understand the prospects for a driverless truck introduction in Sweden.

In paper B, an analysis of the Swedish innovation system for driverless trucks is presented. The motivation is that understanding the innovation system's capabilities, shortcomings, and the motivations for pursuing driverless trucks can provide insights into what factors may shape the development and what uncertainties are present.

The aim of the paper is to outline the structural components and analyze how the TIS functions are performed. There is also an ambition to perform initial assessment of the TIS capabilities and shortcomings, primarily through identifying critical uncertainties about the future development of driverless trucks perceived by the involved actors.

5.2.2 Method

The analysis followed the TIS framework by Bergek et al. (2008). Figure 22 provides an overview of how this framework is applied to the driverless truck TIS.

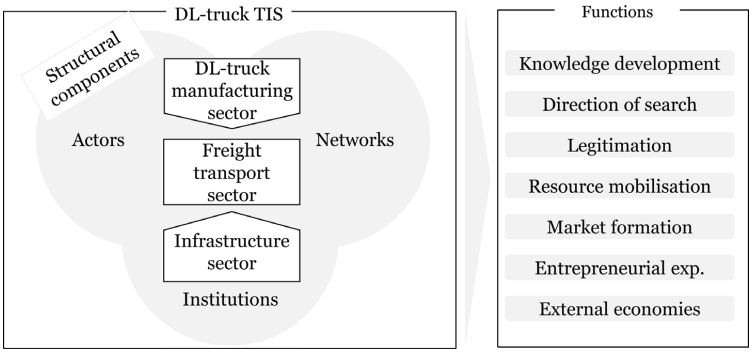


Figure 22 The driverless truck TIS conceptualization used in Paper B.

The method is summarized in Figure 23. A total of 20 expert interviews were conducted with 23 respondents from 16 different organizations during the autumn and winter of 2018. The interviewees represented various types of actors from both the public and private sectors, such as public authorities, truck OEMs, road carriers, transport buyers, and others (see Table 1 in Paper B). Most of the data collection and initial processing were conducted by two of the paper co-authors as a part of their master thesis project. The interview data collection and data processing followed standard practices for semi-structured interviews, summarized in Paper B and presented in detail in Björkman and Joelsson (2019).

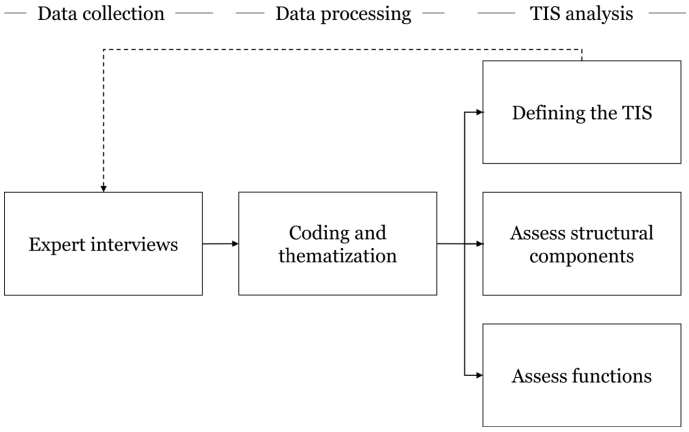


Figure 23 Overview of the methodological procedure in Paper B.

5.2.3 Results

The formation of a technological innovation system of driverless trucks in Sweden (in this section referred to as "the TIS") is in an early formative phase with relatively loose structures. The following general observations can be made.

- Significant uncertainties exist regarding the timeline, infrastructure requirements, and regulatory landscape needed to enable widespread driverless truck deployment in Sweden.
- There is a general view among TIS actors that driverless trucks are a significant opportunity for the Swedish automotive sector and industry at large, and there is a political ambition for Sweden to become both a leading supplier and user of driverless trucks.
- The development of driverless trucks is influenced by incumbent firms in the vehicle industry but actors from the telecom sector, energy sector, and emerging truck technology companies are also entering the area and shaping the development. Developing driverless trucks is not an innovation process taking place only within the existing truck and road transport value chains but is characterized by a higher degree of cooperation by a broader set of actors.
- The current relatively rigid institutions for truck development and production and road freight transport operations will require significant alignment to adapt to driverless truck development and operations

The analysis indicates that the TIS is predominantly supply-driven rather than demand-driven. Although several actors highlight the potential economic benefits for the driverless truck users and their customers (e.g. transport buyers, logistics service providers, and road carriers) such as reduced transport costs, this demand is in general not clearly articulated by potential users and customers of driverless trucks but rather by other actors in the TIS. However, interviewees from the logistics sector highlight that driver shortage is a critical industry challenge which driverless trucks could alleviate.

Several interviewees believe that driverless trucks likely will be offered "as a service" which could put pressure on truck OEMs and road carriers, and lead to that the traditional customer-supplier relationship between road carriers and truck OEMs will change. Driverless trucks are, therefore, from this perspective, both an opportunity and a threat for several of the existing actors in the truck and road freight transport value chains while it could also provide significant business opportunities for new entrants.

Driverless trucks require advancements in business model design, closer collaboration with transport buyers, institutional and regulatory alignments, and potentially also investments in new (digital) infrastructure that requires new ways of working and collaborating in areas that traditionally have not

been key considerations for truck OEMs. Several publicly financed innovation networks for automated driving and freight transport have emerged and provided arenas for cross-organizational collaboration on areas that are not easily managed within each organization's existing R&D process. Several interviewees view driverless trucks as a component of a broader digital transformation of the entire freight transport and logistics sector. However, the needs and requirements for a (new) digital infrastructure to support driverless truck operations and a broader digitalization of freight transport is an area of large uncertainty.

5.3 Paper C: Cost analysis of driverless truck operations

5.3.1 Motivation and aim

One important input for analyzing how driverless trucks may be diffused and to model system-level impacts is plausible estimates on the costs for owning and operating them. The aim of this paper is to provide a transparent analysis of potential total cost of ownership (TCO) implications for driverless trucks by reviewing published cost impact estimates and systematizing them into a simple, comprehensive TCO framework. This represents one of the first comprehensive TCO analyses of driverless trucks in the academic literature and the first study to derive such estimates within a Swedish context.

5.3.2 Method

The analysis is performed by using a total cost of ownership (TCO) approach, as presented in Section 4.2. Three driverless truck scenarios are considered, reflecting pessimistic, intermediate (base), and optimistic assumptions on cost impacts as reported in the literature.

The focus is primarily on calculating the relative difference in TCO for driverless trucks compared to a baseline for manually driven trucks. The methodology follows four main steps:

1. Review the literature and map the reported cost impacts of driverless trucks on transport costs to the components of the TCO model.
2. Set assumptions for the three cost scenarios for driverless trucks per TCO component (see Table 2 in Paper C) based on a literature review.
3. Calculate the TCO components for the manually driven truck baseline and the three driverless truck cost scenarios.
4. Convert the results from step 3 into TCO per tkm to enable comparisons between the truck types.

For step 4, an aggregated approach is used to represent a typical truck within each truck type, assuming an average mission profile (e.g., trip distance, mix of commodities) representative of long-distance transport. For this purpose, official statistics for inter-regional truck trips are used. The transport mission

profile per truck type is assumed unchanged between the manually driven truck baseline and the driverless truck scenarios but the number of trips per year may vary due to changes in truck utilization rates. This “average truck” approach provides useful indications for the overall cost development for different truck types, but since truck transport applications are heterogeneous, this method is not suitable to evaluate the cost impacts of driverless trucks for specific truck owners or transport applications.

5.3.3 Results

The TCO results suggest that driverless trucks would, in a mature stage, enable substantial reductions in road freight transport costs. Figure 24 shows the TCO in the driverless truck scenarios normalized to the manually driven truck baseline per truck type. The cost reduction is 12%-23% in the pessimistic scenario, 29%-45% in the base scenario, and 43%-58% in the optimistic scenario depending on the truck type. There is a relationship between truck size and relative cost savings, with smaller truck types having larger relative cost savings while the relative cost savings are smaller for larger truck types.

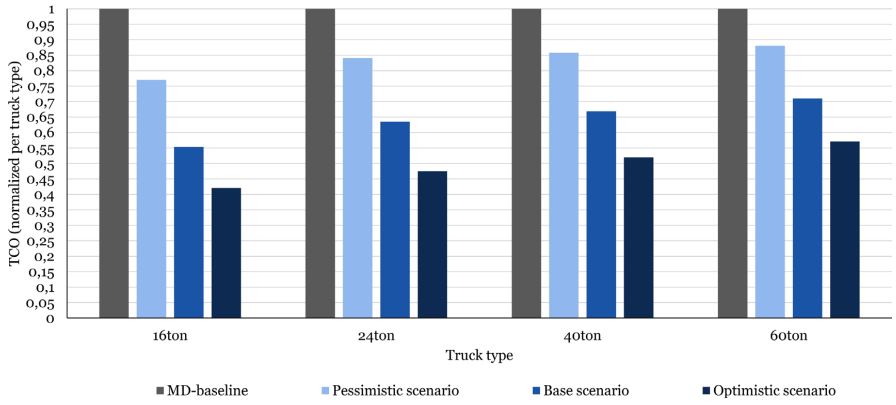


Figure 24 Total cost of ownership per tkm normalized to the manually driven truck baseline per truck type. MD=manually driven truck.

The obtained cost reductions are primarily a result of eliminating in-vehicle driver costs. This is shown in Figure 25 which provides a breakdown of cost changes for the 40-ton truck type. Driver costs are modeled as a time-dependent cost and are independent of the truck size which explains why smaller truck types foresee larger relative cost reductions than larger trucks. However, the cost for remote operations is an important and highly uncertain cost component for which little guidance is provided in the literature.

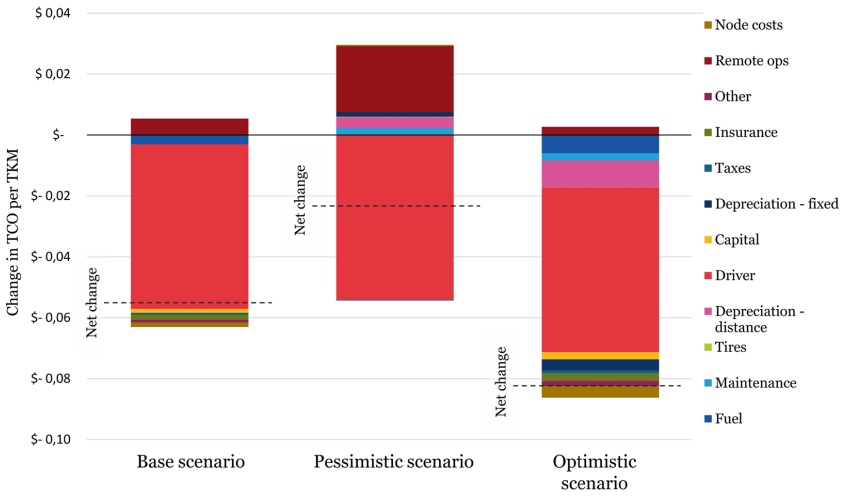


Figure 25 Breakdown of cost changes for individual cost components for the 40-ton truck type compared to the manually driven truck baseline.

If in-vehicle driver costs can be eliminated, the reduction in TCO is relatively robust to increased truck acquisition costs due to costs for the automated driving system. For instance, the truck acquisition cost would have to increase by 230% compared to the manually driven truck baseline to fully offset the TCO savings in the driverless truck base scenario for the 40-ton truck, all else equal. Similarly, it would require that remote operations are used during 60% of the driverless truck's operating time, with the assumed 2x cost for a remote operator per hour compared to a truck driver, to fully offset the TCO savings for the 40-ton truck type in the DT base scenario, all else equal.

Figure 26 shows the cost structure for the 40-ton truck. In all scenarios, the relative share of node costs (costs for loading and unloading) increase compared to the manually driven truck baseline. This is a result of the assumptions of fixed labor and process costs for loading and unloading but highlights that node costs may constitute a significant share of road transport costs also in a driverless future. Another key effect on the cost structure is that time costs are significantly reduced (by around 75% in the driverless truck base scenario) while distance costs are only marginally reduced (around 5% in the driverless truck base scenario) and even slightly increased in the pessimistic driverless truck scenario.

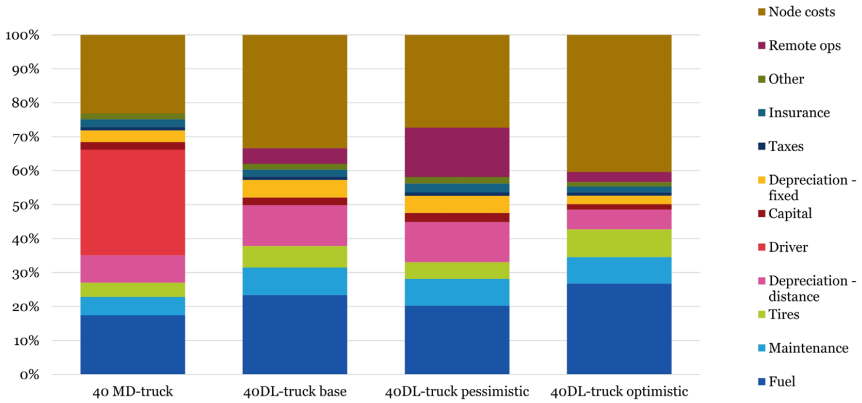


Figure 26 Cost structures for the 40-ton truck type in the various scenarios. Note MD=manually driven truck, DL-truck=driverless truck.

5.4 Paper D: Impacts of large-scale driverless truck adoption on the freight transport system

5.4.1 Motivation and aim

Quantitative estimates of potential system-level impacts of driverless trucks are valuable for informing transport policy and long-term infrastructure planning. For instance, modal shifts from road to rail and sea are often highlighted as key for improving the sustainability of the freight sector, but the cost performance improvements from driverless truck adoption could significantly influence the efficiency of policies seeking to achieve such as shift. Previous research on system-level effects of driverless trucks is scarce and often lacks detailed insights into actual transport flows, transport chains, and geographical effects.

The aim of this paper is to analyze the potential impacts of large-scale adoption of driverless trucks on the national freight transport system in Sweden. The study extends the application of the Swedish national freight transport model, Samgods, to incorporate driverless trucks and evaluates two scenarios: full adoption of driverless trucks across the entire road network, and a mixed scenario where driverless trucks operate on major roads between logistics hubs, but manually driven trucks are required to perform the first- and last leg of the transport chain. Also, the influence of assumptions on driverless truck operating costs are considered as part of the scenario analysis. The study quantifies changes in transport patterns, system costs, and modal shifts resulting from driverless trucks in these various scenarios. It also showcases a first approach for how driverless trucks can be represented in national-level freight models based on the ADA framework.

5.4.2 Method

In Paper D, the Swedish national freight transport model Samgods v.1.2 (see Section 4.4) is used to analyze the impacts of driverless trucks on the transport system.

The scenario analysis revolves around two main uncertainties: the driverless truck operating model and the operating costs for driverless trucks.

Two main scenarios are considered, representing two types of operating models, see Figure 27. The All trucks driverless scenario (*AllDL*) represents a future where driverless trucks are a direct substitute to manually driven trucks and there is full penetration of driverless trucks across the whole truck fleet. In this scenario, there are no restrictions to the driverless trucks' ODD, corresponding to SAE level 5.

The Hub-to-hub scenario (*H2H*) represents a situation where driverless trucks have a restricted ODD and can operate between fixed logistics hubs in the transport network in the form of terminals (road terminals, rail terminals, harbors, airports) but cannot be used for the first or last transport leg in a transport chain. Therefore, for a shipment to be transported by driverless trucks, it first must be transported to a terminal (e.g. by a manually driven truck) where it can be transferred to a driverless truck. In the H2H scenarios, an additional 40-ton driverless truck type is introduced as a complement to the existing manually driven truck fleet.

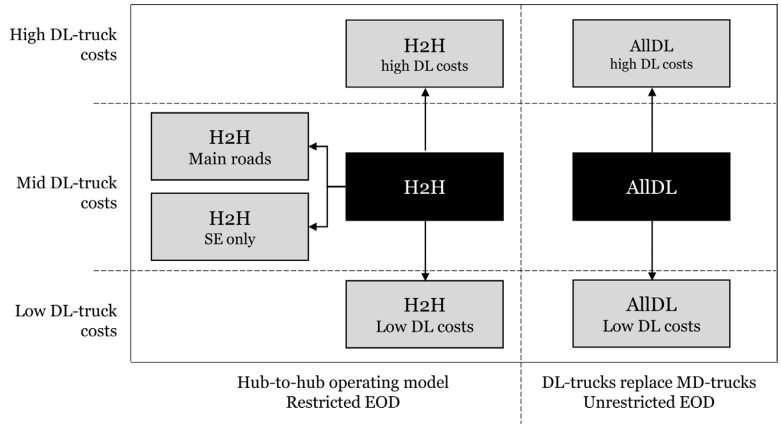


Figure 27 Overview of the scenarios analyzed in Paper D. Here, EOD=ODD.

To explore the influence of assumptions on driverless truck operating costs, scenarios in which the driverless truck cost inputs vary between a low-, mid-, and high level are considered. These cost levels map directly to the optimistic, base, and pessimistic cost scenarios in Paper C.

For H2H, also variations of the driverless truck network are analyzed: 1) driverless trucks can only operate the international E-road network and access closely located hubs (*H2H main roads*) 2) driverless truck can only operate on Swedish territory (*H2H SE only).*

5.4.3 Results

The key findings on an aggregate system level are summarized below, with additional data in Table 3.

- Driverless trucks generate mode shifts from rail and sea to road in both the *AllDL* scenarios and the *H2H* scenarios. The increase in road tkm is approximately twice as large in *AllDL* as in *H2H*. In the H2H scenarios, there is also a shift from manually driven trucks to driverless trucks although manually driven trucks still perform the majority of domestic road tkm.
- There is a larger relative increase in road VKT than road tkm. This is mainly because the average truck size decreases slightly in the driverless truck scenarios. One reason is that the increased use of road transport for import and export flows means that there is a larger number of small trucks (up to 40-tons) operated in Sweden since 60-ton trucks are not allowed outside of Sweden and Finland.
- Total logistics costs decrease substantially in all scenarios except *H2H SE only*.
- Restricting driverless trucks to the international E-road network *(H2H main roads*) does not significantly reduce their competitiveness compared to the full *H2H* scenario. However, limiting their operation to Sweden alone (*H2H SE only*) significantly reduces their aggregate impact.

Table 3 Summary of results from Paper D. All values are domestic values (transport on Swedish territory) except for total logistics costs which also include international costs.

Scenario	Road tkm [M tkm]	Road share of total tkm	Road VKT [M VKT]	DL-truck use (H2H) [M tkm]	DL-truck share of road tkm (H2H)	Total logistics costs, all flows [M€]
Baseline	50 200	48%	2 700	N/A	N/A	22 400
AllDL	61 100	60%	3 700	N/A	N/A	20 700
	+22%	+12pp	+35%			-8%
AllDL high DL costs	52 900	51%	3 000	N/A	N/A	21 900
	+5%	+3pp	+9%			-2%
AllDL low DL costs	67 800	66%	5 300	N/A	N/A	19 800
	+35%	+18pp	+58%			-12%
H2H	55 700	54%	3 100	15 900	28%	21 200
	+11%	+6pp	+15%	N/A	N/A	-5%
H2H high DL costs	51 200	50%	2 800	5 800	11%	22 100
	+2%	+2pp	+4%	N/A	N/A	-2%
H2H low DL costs	60 600	58%	3 500	22 500	37%	20 600
	+21%	+10pp	+29%	N/A	N/A	-8%
H2H Main roads	56 100	54%	3 000	14 600	26%	21 200
	+12%	+6pp	+13%	N/A	N/A	-5%
H2H SE only	51 600	50%	2 800	2 900	6%	22 600
	+3%	+2pp	+4%	N/A	N/A	+1%

Figure 28 shows changes in annual transport flows in tons per mode for *AllDL* and *H2H* compared to the baseline. The most substantial increases in road transport occur along major road corridors. Rail and sea import/export flows are significantly replaced by road which generates substantial increases in road ferry transport connecting Sweden to neighboring countries.

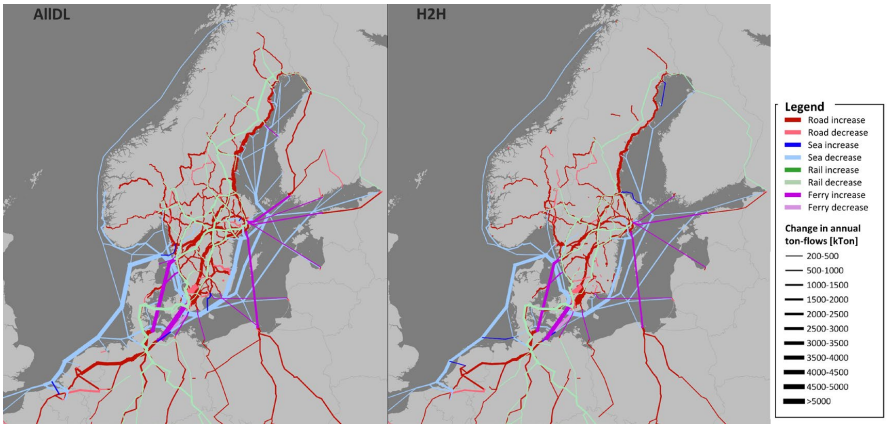


Figure 28 Changes in ton flows in AllDL (left) and H2H (right) compared to the baseline.

Figure 29 shows that the increase in road tkm for various commodity types differs between *AllDL* and *H2H*. For some commodity types, e.g. “coal, petroleum, gas”, “non-metallic mineral products” and “food and beverages” there are only a small differences between *H2H* and *AllDL* while for some other commodity types, e.g. “metal ores, etc.”, “timber” and “refined petroleum products” there are substantial differences.

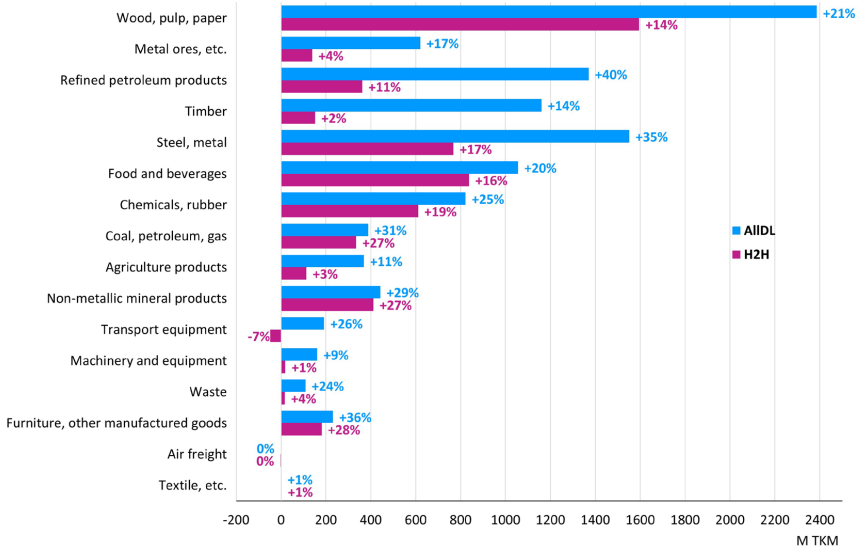


Figure 29 Change in domestic road tkm per commodity type in AllDL and H2H compared to the baseline without driverless trucks. Percentages indicate the relative change in road tkm, compared to the total increase of 22% in AllDL and 11% in H2H.

5.5 Paper E: Exploring cost performance tradeoffs and uncertainties for electric- and autonomous electric trucks

5.5.1 Motivation and aim

This study compares the cost performance of M-ETs and D-ETs against M-ICETs, considering various market and technology conditions, charging strategies, and transport applications, with a focus on heavy-duty tractor-trailer trucks operating full truckload shuttle flows in Sweden. An overview of the shuttle flow operations and conceptual energy consumption profiles for the three charging strategy types considered is shown in Figure 30.

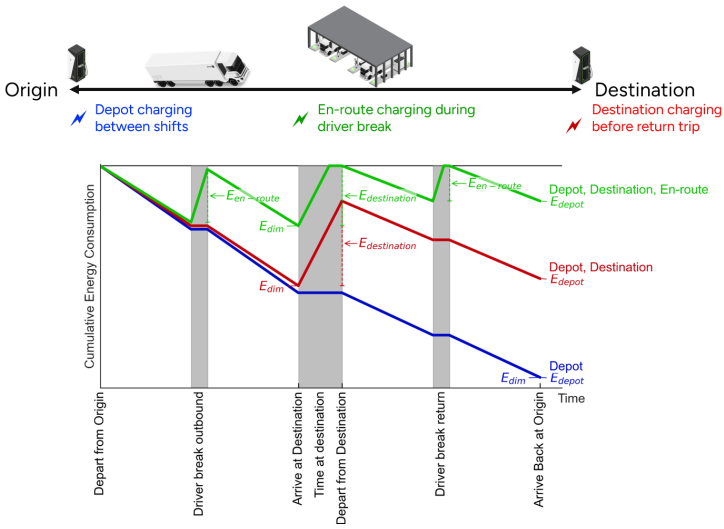


Figure 30 Overview of the shuttle flow application and conceptual diagram of energy consumption for various charging strategy types in Paper E.

The study also demonstrates methods for computational experiments for cost performance analysis and tools for mapping the high-dimensional input space to outcomes, which allows for insights into the boundary conditions for cost-effective deployment of M-ETs and D-ETs.

5.5.2 Method

The analysis is made using computational experiments with a novel cost simulation model of truck operations and cost performance (see Section 4.3) to analyze the feasibility and cost performance of M-ETs and D-ETs, compared to M-ICETs. The experiments include 1200 scenarios, representing different transport applications and techno-economic conditions. Sixteen charging strategies, combining depot, destination, and en route charging, separated into three charging strategy types (CSTs) are evaluated. In total, 19200 experiments

are performed based on a full factorial of the scenarios and charging strategies. A summary of the methodology is presented in Figure 31

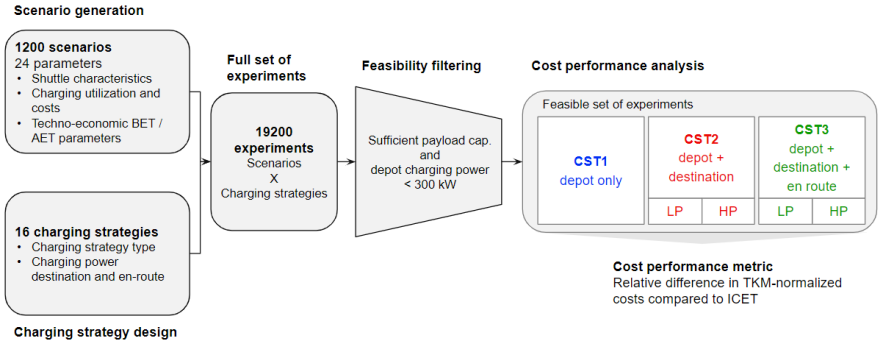


Figure 31 Overview of the methodological process for Paper E. LP=low power (<600kW), HP=high power (≥600kW). BET=M-ET, AET=D-ET.

The scenarios are designed by sampling 24 scenario parameters using LHS. Four of the scenario parameters are used to model different variants of the shuttle flow, such as its distance and the cargo weight. The remaining scenario parameters are used to model plausible techno-economic and market conditions for M-ETs and D-ETs in short- to mid-term.

5.5.3 Results

The results indicate that both M-ETs and D-ETs are cost-competitive compared to M-ICETs in a substantial share of the investigated scenarios, in many cases regardless of which charging strategy is used. For most scenarios, at least one charging strategy enables the shuttle flow to be served by an M-ET or D-ET without exceeding vehicle weight constraints. In many scenarios where the payload and distance are low to moderate, it is possible to rely only on depot charging between shifts. The results suggest that shuttle flows, which have highly predictable energy demands and allow for a high utilization of the battery, constitute a highly electrifiable transport application.

For most scenarios, the charging strategies with the best cost performance are either using only depot charging between the round-trips (CST1) or a combination of depot and destination charging before commencing the return trip (CST2), see Figure 32. Using en route charging between the origin and destination (CST3) in addition to depot and destination charging is only the best choice in a small share of cases, namely for highly energy demanding shuttles with long driving distance and heavy cargo.

For M-ETs and D-ETs to be cost competitive relative to M-ICETs, high asset utilization is critical for offsetting the additional investment costs in vehicles and chargers, see Figure 33.



Figure 32. Share of scenarios for which the corresponding charging strategy type (CST) has the best D-ET cost performance out of the set of feasible scenarios for various combinations of shuttle distance and payload.

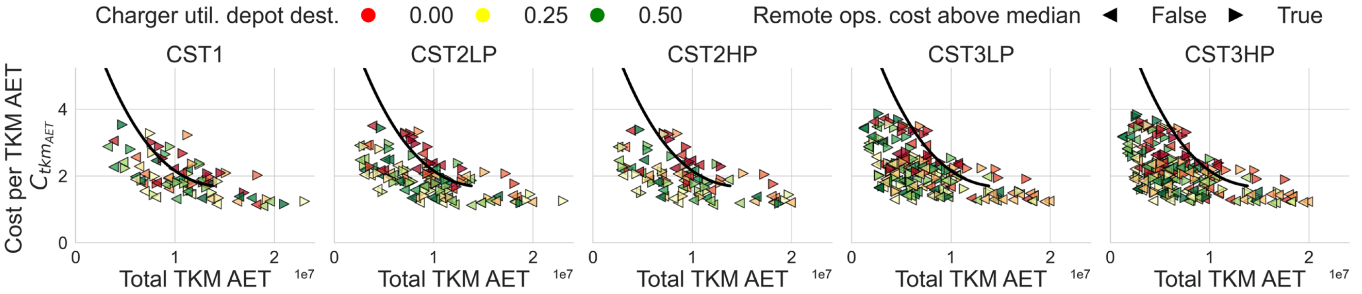


Figure 33 Relationship between total truck cost normalized per tkm (y) and total tkm performed over the truck life (x) for D-ETs (AET=D-ET). Data is from a random sample of 100 scenarios, each marker represents one experiment. The black line is a trend line for the M-ICET results. The color coding denotes the value for the assumed utilization of depot and destination charging infrastructure by other trucks, used as a proxy for charging infrastructure utilization, while the marker direction shows whether the remote operations cost, used as a proxy for operator utilization, is below or above the median value over scenarios.

Dimensional stacking is used to identify sub-spaces of the scenario parameter space in which there are high concentrations of cases in which D-ETs are feasible and cost competitive compared to M-ICETs, see Figure 34. The analysis suggests that when using depot charging only (CST1), energy related factors have the largest influence. When also destination and high-power en route charging is used (CST3HP), energy constraints are less of an issue, and instead, remote operations costs, the cost of the ADS, the charging infrastructure utilization and diesel price (used as a proxy for the electricity-diesel price differential) are most important. It can be noted that even in cases when the remote operations cost is in the highest tercile (300-400 SEK/h) and ADS cost is also in its highest tercile (roughly 1.2-1.7 MSEK), D-ET may still be cost competitive in a rather large share of scenarios.

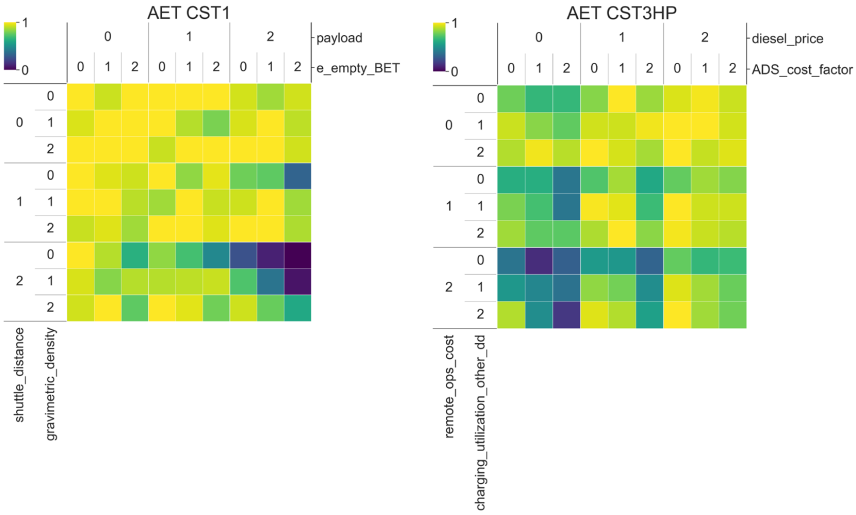


Figure 34 Dimensional stacking for concentration of scenarios where D-ETs are cost-competitive relative to the D-ET using depot charging only (CST1, left) and a combination of depot, destination and en route charging (CST3HP, right).

5.6 Paper F: Impacts of electric and driverless heavy-duty trucks on the future decarbonized freight transport system

5.6.1 Motivation and aim

The development and deployment of truck electrification and automated driving technology are uncertain. This is a challenge for conventional long-term transport planning relying on forecasting the expected future development of the transport sector and formalizing the expected development in transport forecasting models.

Paper F presents an exploratory analysis of the Swedish freight transport system in 2045, focusing on the impacts of techno-economic uncertainties surrounding electric trucks (ETs) and automated driving technology. The study aims to provide insights into how these uncertainties could shape a future decarbonized road freight transport system, assuming decarbonization primarily occurs through electrification and biofuels.

The main objectives of this paper are to:

1. Develop an improved representation of M-ETs and D-ETs for use in national freight transport models.
2. Develop and demonstrate a tractable approach for performing EMA to explore road freight transport uncertainty using national freight transport models.
3. Provide a plausible outcome space around the current STA reference forecast for 2045, considering uncertainties for the development of M-ETs and D-ETs and their potential impact on: mode split and road freight transport demand; total logistics costs (system costs); demand for biofuels and electricity by the road freight sector; and the use of M-ICETs, M-ETs, and D-ETs

5.6.2 Method

EMA is used to derive an outcome space for the Swedish freight transport system in 2045 using the EMA-Logmod-AE model system, see Section 4.4.2. The outcome space is spanned by scenarios reflecting uncertainties in the development of electrification and automated driving technology, using the current STA reference forecast for 2045 as a baseline.

Six uncertainties are considered, hereafter denoted *scenario parameters*. Four scenario parameters relate to the development and state of ETs (affecting both M-ETs and D-ETs) and two relate to the development and state of automated driving technology and operations (affecting D-ETs). All uncertainties are modeled as categorical parameters.

- *ET and battery maturity level*: what will be the level of technical performance and costs for typical M-ETs and D-ETs operating in 2045?
- *Battery capacity dimensioning*: how will the battery capacity be dimensioned for typical M-ETs and D-ETs operating in 2045?
- *Charging infrastructure maturity level*: what will be the level of technical performance and costs for typical charging infrastructure technology used in 2045?
- *Public fast charging network coverage*: how dense will the network of public fast charging network be in 2045?
- *Automated driving technology maturity*: what will be the level of technical performance and costs for typical automated driving system technology for D-ETs operating in 2045?

- *Driverless truck operating model*: what will be the dominating operating model in terms of what type of freight flows D-ETs operate in 2045 considering technical, regulatory and operational factors? Four operating models in which D-ETs are restricted to certain types of OD-flows are considered (see Figure 35), a fifth unconstrained case corresponding to automation at SAE level 5 (equivalent to the AllDL scenario in Paper D), and a case without any driverless trucks.

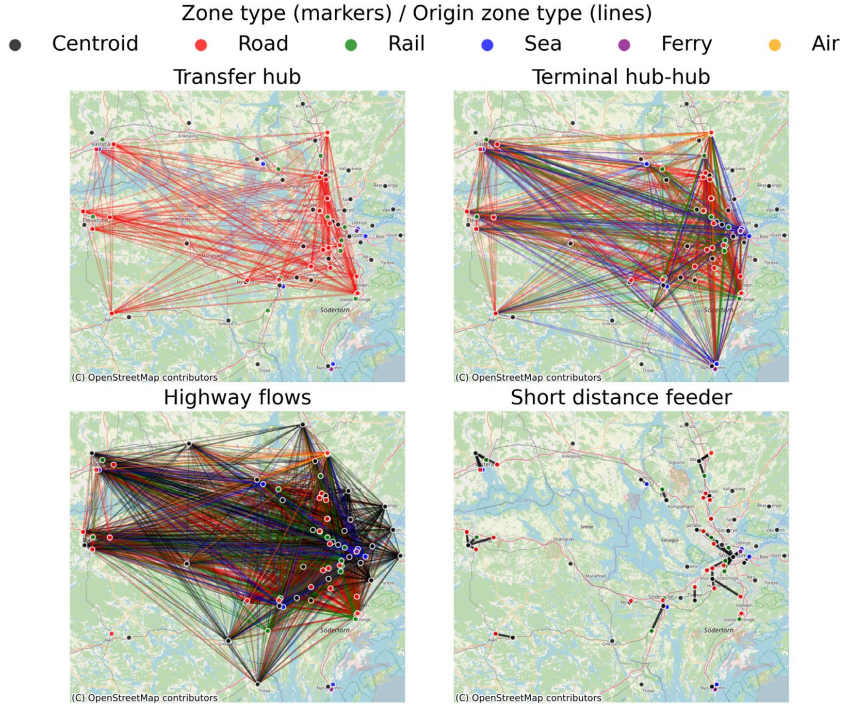


Figure 35 Visualization of OD-pairs allowed for D-ET operations for the various D-ET operating models starting and ending in the greater Stockholm area. Line color represents the origin node type, i.e. a red line indicates that the origin is a road terminal. Transfer hub allows D-ET to drive only between road terminals, Terminal hub-hub allows D-ET to drive between all terminals but not centroids, Highway flows allow D-ETs to drive on the highway network and to reach any node <20km from it, Short distance feeder allows D-ET to drive to/from the zone centroid and terminals within the same zone.

The scenario set is designed using a hybrid fractional factorial design consisting of two complementary scenario sets. Together, these two sets comprise 306 scenarios. In addition, a scenario corresponding to the STA reference forecast for 2045 (denoted the reference forecast scenario) is run. The scenario parameter values, and the two scenario sub-sets are shown in Table 4.

Table 4 overview of scenario parameters, categorical values and which values are considered in each of the two scenario sets.

Scenario parameter	**Categorical values**	**Scenario sub-set 1** *vertex-M-ET-scenarios*	**Scenario sub-set 2** *mid-M-ET-scenarios*
ET and battery maturity level	low	low	
	mid		mid
	high	high	
Charging infrastructure maturity level	low	low	
	mid		mid
	high	high	
Public fast charging network density	low	low	
	mid		mid
	high	high	
Battery capacity dimensioning	half shift	half shift	
	full shift	full shift	full shift
Automated driving maturity level	low	low	low
	mid	mid	mid
	high	high	high
D-ET operating model	No D-ET	No D-ET	No D-ET
	Short distance feeder	Short distance feeder	Short distance feeder
	Highway flows	Highway flows	Highway flows
	Transfer hub	Transfer hub	Transfer hub
	Terminal hub-hub	Terminal hub-hub	Terminal hub-hub
	Level 5	Level 5	Level 5

5.6.3 Results

The scenarios exhibit substantial variation in system-level outcomes. Figure 36 provides an overview, displaying the pairwise relationships between scenario parameters and key outcomes, with the dashed line indicating the corresponding value in the STA reference forecast.

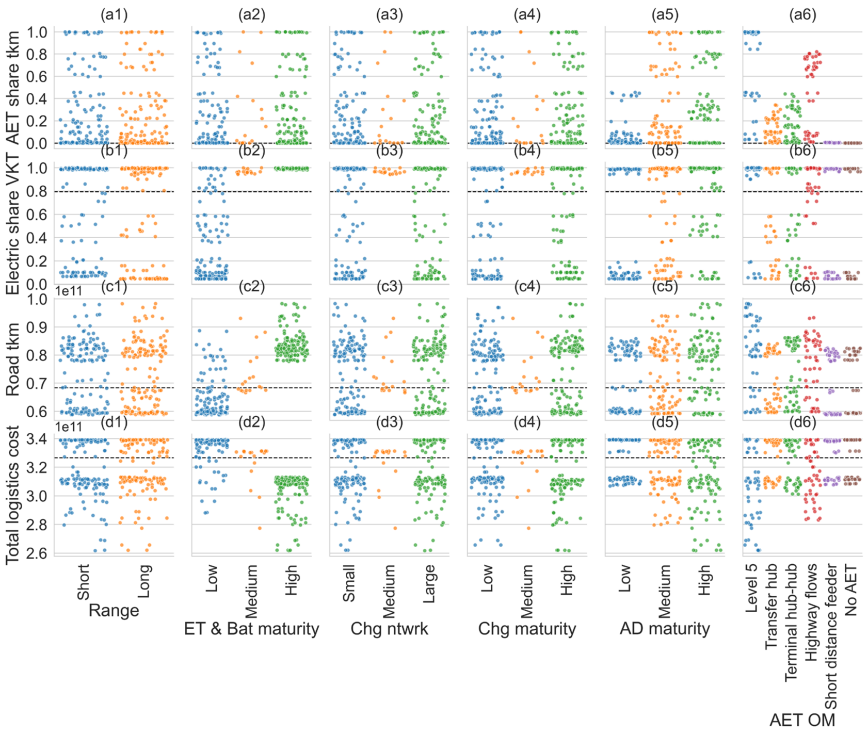


Figure 36 Pairwise relationship between scenario parameters (x-axis) and a selection of key outcomes (y-axis). The dashed horizontal lines represent the reference forecast scenario value. Here, AET = D-ET.

Electrification rates (share of domestic road transport VKT by M-ETs or D-ETs) vary between 0-100%, with many scenarios concentrated at either end of the range, although a majority on the higher end. This separation occurs because in many scenarios, either M-ICETs or M-ETs/D-ETs are the most cost-effective option, causing Logmod's cost minimizing logic to assign most flows to one truck type. The *ET and battery maturity level* largely determines the electrification rate with the high maturity level yielding nearly 100% electrification, regardless of other factors, while scenarios with low maturity exhibit more variation.

The D-ET share of road tkm also varies substantially, showing differences across D-ET operating models. The large variations in road tkm imply significant differences in mode share between scenarios. The road mode share ranges from just above 45% to 85% while rail and sea shares vary from 6-33%

and 8-22%, respectively. In the reference forecast scenario, the mode split is 56% road, 24% rail, and 19% sea.

Figure 37 shows that both D-ET operating models and automated driving maturity levels heavily influence system outcomes. At low maturity, D-ETs have limited impacts regardless of the operating model. As maturity increases, the impacts grow, and differences between operating models become more pronounced.

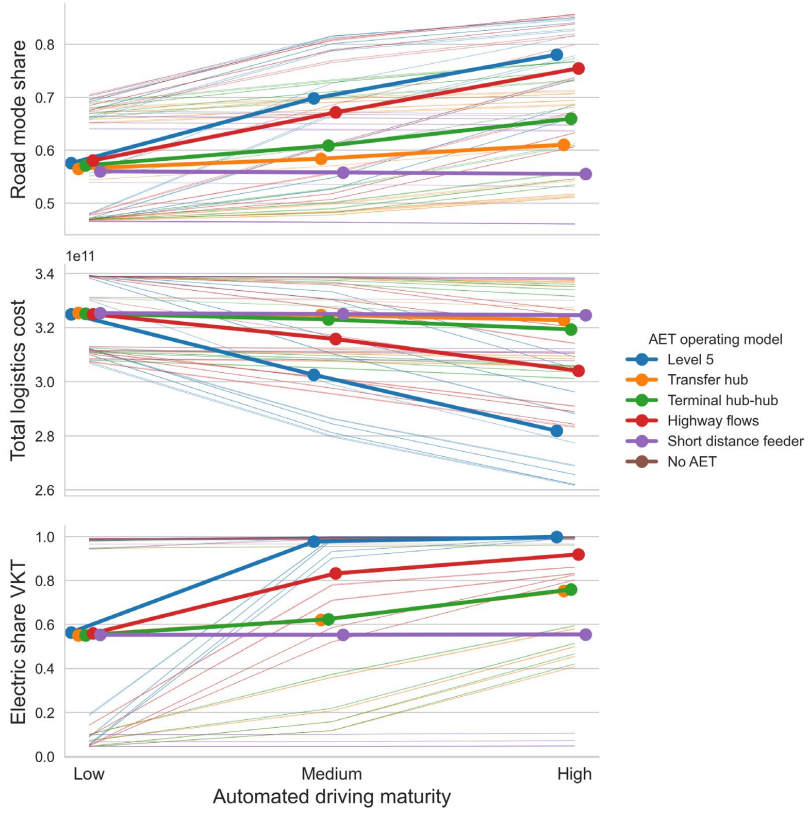


Figure 37 Impacts of automated driving maturity level and D-ET operating model on road mode share, total logistics cost, and electrification rate. The thick lines and corresponding markers show averages over scenarios per D-ET operating model, and the thin lines show results for individual scenarios.

Compared to the reference forecast, the outcome space spans a broad range. For instance, 38% of scenarios exhibit at least 20% higher road tkm, while 52% show at least 20% lower rail tkm. Even in the absence of D-ETs, substantial variability exists, with road transport demand ranging from 59 to 82 billion tkm. When compared to official statistics since 2000 and recent STA forecasts

for 2040 and 2045, as shown by Figure 38, the results illustrate the considerable impact of assumptions about emerging technologies on future projections.

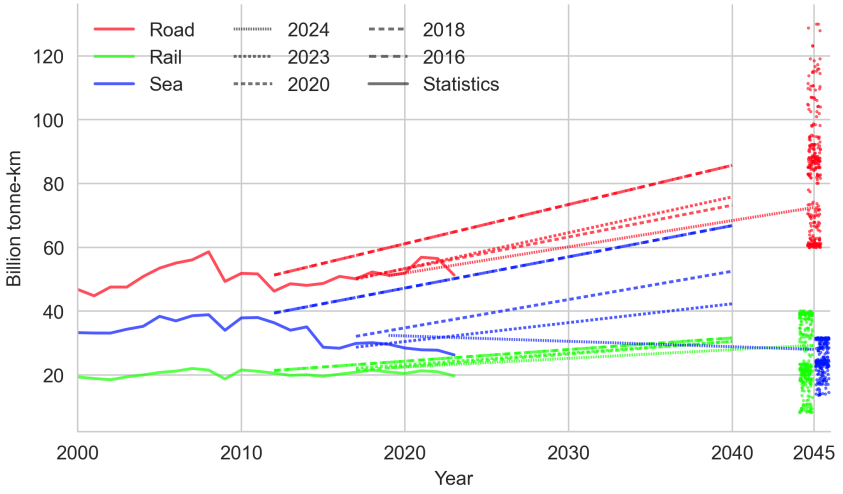


Figure 38 Freight transport activity by mode (road, rail, sea) for historical data (full lines), STA reference forecasts (dashed lines) and results for 2045 from the computational experiments.

5.7 Paper G: Using exploratory modelling and analysis to assess the robustness of national transport climate strategies: a case study of Sweden

5.7.1 Motivation and aim

This paper explores the use of MORDM to improve the generation and evaluation of national transport climate policies. The analysis focuses on policies aimed at achieving a 90% reduction in direct CO_2 emissions from domestic transport by 2040 compared to 2010 levels. The objective is to assess whether MORDM can generate more diverse and robust policy options compared to a set of policies identified by the STA in a previous analysis (Trafikverket, 2020c), and to evaluate potential benefits and challenges with using MORDM.

5.7.2 Method

MORDM is applied to the Scenario tool (Trafikverket, 2020a), a simplified policy analysis tool used by the STA. The Scenario tool covers domestic road transport and models both car traffic and truck traffic. To incorporate the

impacts of automation, the model is extended to include representations for shared autonomous vehicles (SAVs) for passenger transport and driverless trucks, accounting for uncertainties related to their adoption, diffusion, and effects on energy efficiency and transport demand.

The problem formulation is summarized in Figure 39 and includes a set of external uncertainties (X), a set of climate policy levers (L), uncertain model parameters (P), a set of outcome metrics (M), and the Scenario tool (R).

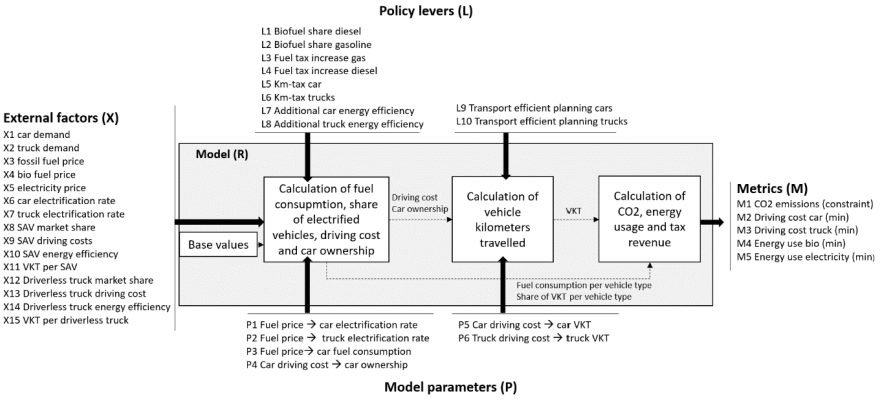


Figure 39 Problem formulation for Paper G using the XLRM framework.

The method follows the four steps of the MORDM process (Section 4.5.2), implemented as follows:

- Problem formulation and candidate policy generation. A MOEA (ε-NSGAII) is used to solve a multi-objective optimization problem seeking to minimize driving costs for cars and trucks respectively and the use of biofuels and electricity while meeting the 2040 climate target. Two sets of candidate policies are considered: one set for which policy levers for biofuel admixture, km-tax, fuel tax and measures for transport efficiency are considered (denoted *All levers*) and one set without the transport efficiency measures (denoted *No transport efficiency*). A set of 200 candidate policies are generated, 100 All levers policies, and 100 No transport efficiency policies.
- Robustness analysis. The candidate policies are evaluated across 2,100 scenarios that represent various deeply uncertain factors related to transport demand, energy and fuel prices, the market uptake of electric vehicles and the diffusion and impacts of SAVs and driverless trucks. In addition to the candidate policies generated with the MOEA (denoted the MORDM-policies), the eight policies previously identified by the STA are included. Three robustness metrics measuring various aspects of robustness, are calculated for each policy: $RM_{90\%dev}$ measures the relative deviation of the 90th percentile of an outcome metric distribution compared to its value in the reference scenario; $RM_{MeanStdev}$ is the product of the mean and standard deviation of an

outcome metric across scenarios indicating overall performance consistency; and RM_{CO2sat} is the share of scenarios in which a policy meets the climate target.

- Vulnerability analysis. Scenario discovery using PRIM (Bryant & Lempert, 2010b; Friedman & Fisher, 1999) is performed to identify key vulnerabilities for not reaching the climate target for each policy.

5.7.3 Results

The results highlight trade-offs among some of the optimization objectives. Specifically, there are trade-offs for driving costs versus biofuels and electricity use. Policies that minimize driving costs yield higher transport demand and thus require higher usage of biofuels and electricity. Conversely, policies that prioritize minimizing biofuel and electricity use tend to increase driving costs to curb energy demand. There is also, as expected, a tradeoff between the use of biofuels and electricity as they are partly complementary means of reducing emissions.

In terms of robustness, the MORDM policies do not distinctly outperform STA policies. However, there are some differences in robustness between the *All levers* policies and *No transport efficiency* policies.

The analysis also explored the relationship between reference scenario performance and robustness. If reference scenario performance and robustness are positively correlated, it might be sufficient to focus on achieving good reference scenario performance to also achieve robust policies. This relationship is different depending on which robustness metric is considered. For most outcome metrics in $RM_{MeanStdev}$, there is a positive correlation between reference scenario performance and robustness, indicating that policies performing well in the reference scenario are also robust according to this metric. For $RM_{90\%dev}$, there is a negative correlation for driving costs, suggesting a trade-off between robustness and reference scenario performance for driving costs: policies that minimize driving costs in the reference scenario tend to be less robust when faced with deep uncertainties.

The vulnerability analysis identifies the rate of vehicle electrification, especially for cars, as the primary factor influencing whether the climate target is reached, as is illustrated by Figure 40. Driverless vehicles have only a marginal impact of whether the CO_2 target is met or not, compared to the other factors considered.

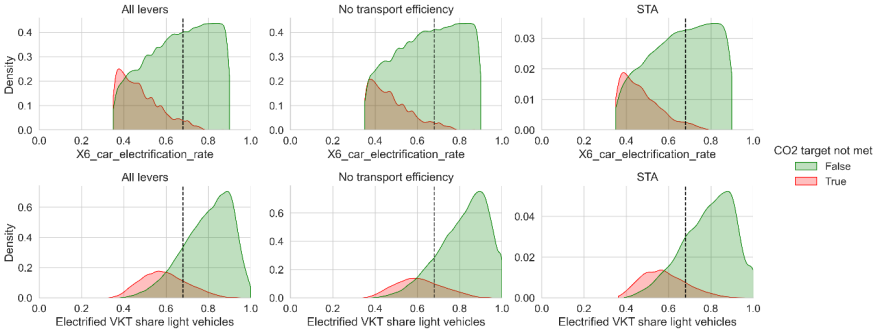


Figure 40 Distribution of experiments over the scenario parameter car electrification rate (top row) and over the share of electrified VKT for cars after accounting for policy effects (bottom row) by Policy type separated by whether the climate target is met or not. The dashed vertical line is located at 68% which is the car electrification rate in the reference scenario under which the policies are designed.

In total, the results suggest that for this case study, the value of MORDM is primarily in providing a framework for systematic evaluation of policy robustness and for considering deep uncertainties. MORDM policies did not distinctly outperform STA policies which already covered a large share of the solution space.

6 Key contributions

This chapter presents key contributions and findings for each of the thesis objectives and their relation to the literature.

6.1 Objective 1: scenarios for future freight and the innovation system of driverless trucks

Objective 1a: Develop and analyze scenarios for the future freight transport system in Sweden.

Objective 1b: Analyze the Swedish innovation system of driverless trucks and identify key uncertainties for its development.

The key contributions to Objective 1 are:

1. Digitalization scenarios (6.1.1).
2. Analysis of the Swedish driverless truck innovation system (6.1.2).

6.1.1 Digitalization scenarios

Paper A presents four distinct scenarios for the Swedish freight transport system in 2040, based on two critical uncertainties: *the sustainability paradigm* (whether climate issues become the top sustainability priority or remain one of many societal goals) and the *business logic for data sharing* (whether traditional hierarchical industrial organization persists or shifts to a network-based, data-sharing ecosystem). In addition to these two uncertainties, several other trends shaping development of the Swedish freight transport system, both internal and external to the freight transport system, are identified.

The scenarios revealed that a significant degree of digitalization is expected across all futures, but societal developments heavily influence the specific

problems that digitalization addresses, and thus the development is different in character. Common features across scenarios include an increase in total road freight VKT compared to today and substantial increases in recycling flows in three out of four scenarios.

The scenario development process identified the development of automated driving technology, including driverless trucks, as part of the certain development, i.e. trends that are expected to materialize in all scenarios. The experts anticipate that trucks will become increasingly automated, with driverless trucks primarily introduced in dedicated areas with relatively simple operating environments, such as terminals and other restricted areas. Overall, the scenarios suggest that some development and adoption of driverless trucks is likely to materialize, but the context for their adoption in terms of transport climate policies, the distribution of various types of transport flows, and the organization of transport services, may vary significantly.

Both Paper A and Paper B suggest that actors within the freight transport sector regard driverless trucks as one component of a broader, ongoing digital transformation of the entire freight transport and logistics sector. This perception aligns well with the literature of emerging freight transport technologies (Dong et al., 2021; L. A. Tavasszy, 2020). The view that driverless trucks are likely to be developed and deployed to some degree during the coming decades has also been identified in other studies synthesizing Swedish expert opinions (Melander et al., 2019).

6.1.2 Analysis of the Swedish driverless truck innovation system

There is a general view among actors within the TIS that driverless trucks represent a significant opportunity for the Swedish automotive sector and the industry at large which is supported by an implicit political ambition for Sweden to become a leading supplier and user of driverless trucks. The development is influenced by incumbent firms in the Swedish vehicle industry, but actors from the telecom sector, energy sector, and emerging freight technology companies are also entering the area and shaping the development.

Also, many actors discuss the need for connectivity and digital infrastructure to support driverless truck operations and that their relation to the broader digitalization of freight transport is an area of great uncertainty. In particular with regards to which actor(s) will lead this development. This relates to the finding in Paper A that the logic underlying digitalization of road freight is a central factor for how the freight sector develops.

The innovation process of driverless trucks is characterized by a higher degree of cooperation among a broader set of actors compared to many previous truck technology innovations. Driverless trucks require significant advancements in areas not traditionally in focus for truck OEMs, such as business model design, closer collaboration with transport buyers, institutional and regulatory

alignments, and investments in digital infrastructure. These issues have sparked a higher degree of collaboration with other companies and public bodies.

The TIS analysis indicates that the development of driverless trucks is primarily driven by the supply side, i.e., OEMs and technology companies. One reason for the more cautious position by actors on the demand side might be that driverless trucks could disrupt the value chain of road freight transport and marginalize the role of current road carriers. It is expected that driverless trucks will most likely be offered "as a service," which could challenge existing business models for truck OEMs and road carriers, changing the logic for transport and logistic service providers.

Several findings about the Swedish driverless truck TIS in Paper B support findings in the literature on the impacts of driverless trucks on business models and value chains. The driverless truck TIS is indeed characterized by close partnerships by truck OEMs, technology suppliers, and logistic services providers (Fritschy & Spinler, 2019; Lind & Melander, 2021), at least during the initial stage of the TIS which was observed in Paper B.

Similarly, the actors in the TIS believe that driverless trucks will challenge existing business models for truck OEMs and road carriers (Fritschy & Spinler, 2019; Monios & Bergqvist, 2019) and that the driver shortage is a key challenge that driverless trucks could alleviate (Müller & Voigtländer, 2019).

By performing the analysis on the level of the innovation system and not focusing solely on a specific type of actor, which has been the focus in previous empirical studies on perceptions on driverless trucks, Paper B lifts several perspectives which are not covered in other literature. For instance, the slightly different positions towards driverless trucks between the supply and demand side and the dynamics created by new types of actors entering the area. This observation is in line with the prediction about shifting value creation and capabilities related to driverless trucks by Monios and Bergqvist (2020).

6.2 Objective 2: cost impacts of driverless trucks

Objective 2: Model and analyze impacts of driverless trucks on the production cost of road freight transport

The key contributions to Objective 2 are:

1. Analysis of driverless truck cost impacts (6.2.1).
2. Insights into electrification effects (6.2.2).

6.2.1 Analysis of driverless truck cost impacts

The results from Paper C and Paper E suggest that driverless trucks would be cost competitive compared to M-ICETs in all (Paper C), or most of (Paper E), the investigated scenarios.

The reduction in TCO for D-ICETs compared to M-ICETs identified in Paper C is between 12%-23% in the pessimistic scenario, 29%-45% in the base scenario, and 43%-58% in the optimistic scenario depending, where smaller truck types have larger relative cost savings and larger truck types have smaller relative cost savings. In Paper E, the experiment-average result in terms of total cost per tkm is roughly 20% lower for D-ETs compared to M-ICETs.

Driverless trucks will change the cost structure of truck operations by substantially reducing marginal time-dependent operating cost while the distance-dependent marginal costs will decrease to a smaller extent or possibly increase slightly (compared to its ICET or M-ET counterpart). Depending on the ADS cost, fixed costs may increase.

The identified TCO reductions are primarily driven by net reductions in labor cost for the truck operations. Both Paper C and Paper E find that the potential for reduction in labor cost for truck operations is a key sensitivity for the level of TCO reduction and a significant uncertainty. One critical factor is remote operations, for which the literature provides little guidance on central aspects such as: how many trucks one operator can handle on average, under what conditions remote operations of different types will be required, the labor costs per operator, technology costs, and how the need for remote operations will develop over time with increasing maturity of automated driving systems. However, a recent paper simulating remote operator utilization, published after writing Paper C and Paper E, suggests that the remote cost operator ranges considered in these papers (5-40% of hourly driver cost in Paper C, 100-400 SEK/h in Paper E) seem reasonable, and may even be conservative (Ma et al., 2024).

Furthermore, both Paper C and Paper E suggest that if in-vehicle driver costs can be eliminated, the cost performance of driverless trucks compared to M-ICETs is relatively robust to increased truck acquisition costs resulting from technology costs for the automated driving system and remote operations costs. For instance, in Paper C, the truck acquisition cost would need to

increase by 230% compared to the manually driven truck baseline to fully offset the TCO savings in the driverless truck base scenario for the 40-ton truck, all else equal. Similarly, a remote operator dedicated to the truck for 60% of the truck's operating time (at an assumed 2x cost per hour compared to a truck driver) to fully offset the TCO reductions, all else equal. In Paper E, ADS costs exceeding one million SEK and remote costs above 300 SEK/h could still result in cost-competitive D-ET operations for shuttle flows if other conditions are favorable.

Compared to similar studies by Wadud (2017) and Ghandriz et al. (2020), Paper C estimates slightly larger reductions in TCO (in the driverless truck base scenario). The differences seem to stem from that these other studies assume higher costs for human labor for supporting driverless truck operations (e.g. through remote support and loading/unloading), and higher technology costs than in Paper C. A more recent paper using a similar methodology identified more substantial cost savings for driverless trucks (S. Lee et al., 2023) compared to Paper C. The relative cost savings estimated in Paper C and Paper E are likely transferable to markets with similar cost structures for road transport as in Sweden. However, in markets with significantly different cost levels or structures, the impacts of automated driving on TCO may vary substantially, as shown in studies on driverless commercial vehicles for passenger transport (Becker et al., 2020).

The potential cost reductions and impacts on the cost structure of driverless trucks may have several types of implications for transport planning and truck operations. For instance, the results from Paper C have been used for studies of operational adaptations for driverless such as reduced operating speeds to reduce energy consumption (Bray & Cebon, 2022a), and impacts on vehicle size selection and routing decisions (Bray & Cebon, 2022b). Paper C has also been used to guide cost parameter assumption a study on transfer hub model operations planning (C. Lee et al., 2023).

6.2.2 Insights into electrification effects

Paper E provides additional findings related to driverless trucks in an electrification context. One aspect is the potential indirect benefits from reducing truck tare weight if not having a driver's cabin. Such a reduction may partly or fully offset the additional battery weight, which results in fewer cases in which the battery weight causes payload capacity reductions. Also, the reduction in tare weight and auxiliary power demand would reduce energy consumption which means that a smaller battery would be required for a given truck, ceteris paribus. These factors have a minor effect on direct operating costs but may allow for more energy demanding shuttles to be energy-feasible for electric truck operations and may allow for a larger share of energy to be provided through depot charging rather than more costly public fast charging, thus improving the overall economics.

Paper E is one of the first papers that provides a cost performance modeling framework that accounts for the impacts of long-term use patterns, battery sizing and charging strategy on battery degradation and the consequential impact on total truck costs. The results from Paper E aligns with other studies also identifying shuttle flows as a highly electrifiable transport application both in terms of energy- and economic feasibility, and that reliance on depot charging is, in general, a beneficial charging strategy for such flows (Bergqvist et al., 2023; Karlsson & Grauers, 2023b).

6.3 Objective 3: Modeling and analysis of system-level impacts of driverless trucks

Objective 3a: Develop appropriate model representations of driverless trucks for use in national freight transport modeling.

Objective 3b: Provide a plausible outcome space of system-level impacts of driverless trucks using national freight transport modeling.

The key contributions to Objective 3 are:

1. Development of model representations for driverless trucks (6.3.1).
2. System-level impact analysis (6.3.2).
3. Insights into operating models and electrification (6.3.3).

6.3.1 Model representations for driverless trucks

Paper D and Paper F present and apply model representations of driverless trucks for several types of scenarios and operating models within the Swedish national freight transport model Samgods (Logmod). The papers are, to the author's best knowledge, the first studies that have modeled driverless trucks within the ADA freight model framework, which, is also used for national models in other countries including Denmark and Norway (De Jong et al., 2021)

Driverless trucks are modeled as distinct truck types with their operating characteristics and parameters adjusted to represent various operating models. Paper D provides an initial modeling effort for two types of operating models, while Paper F extends the work by covering a larger set of operating models, modeling a broader set of driverless truck aspects (e.g. truck weight and payload capacity, operating time) and providing an enhanced representation of electric trucks. This enhanced modeling is useful for both M-ETs and D-ETs.

The following factors are considered when representing driverless trucks in the papers:

- Cost parameter adjustments (Paper D and Paper F): The truck-type specific cost parameters are adjusted to represent estimated costs for

driverless trucks, based on findings for Objective 2. In Paper D, three different cost levels are used, based on the pessimistic, intermediate, and optimistic cost scenarios in Paper C. For Paper F, a broader set of cost parameter values resulting from combined effects of automated driving technology and electrification uncertainties, combining insights from Paper C, Paper E and other literature.

- ODD-related restrictions (Paper D and Paper F): Several techniques are used to model ODD-related limitations and to represent various driverless truck operating models. This includes manipulation of truck-type specific access to the road network and zones (centroids, terminals) and the specification of transport chain options for the driverless truck types.
- Differences in time-related operational constraints (Paper F): in Paper F, it is accounted for that driverless trucks may not be constrained by driver-related driving time regulations and scheduling constraints, potentially reducing transit time for long-haul flows. This is modeled as a route-specific adjustment to operating time, resulting in D-ETs having higher average operating speeds over long flows where driver breaks or multiple shifts are required for manually driven trucks.

For the modeling of D-ETs in Paper F, additional considerations are made to represent electric truck aspects, such as route-specific assessments of energy feasibility and charging time and cost penalties, accounting for different charging infrastructure scenarios.

An important difference between Paper D and Paper F is the extended set of truck types used in Paper F. While adding more truck types is a conceptually simple change, it is a substantial modeling improvement since this setup allows studying how different types of trucks complement each other and compete under various technology and market assumptions that may affect the truck technology types differently.

However, there are many potential improvement areas. Some examples include improving the cost functions of driverless truck operations to model different ways of handling non-driving driver tasks, route- and ADS dependent modeling of the need for remote operations, and more sophisticated charging logics for driverless trucks.

6.3.2 System-level impact analysis

Paper D and Paper F contribute to the identification of a plausible outcome space of system-level impacts of driverless trucks through model results for a total of 315 scenarios of driverless trucks in the Swedish freight transport system. These scenarios vary in terms of the assumed cost performance level and operating model for driverless trucks, as well as varying assumptions for different aspects of truck electrification in Paper F.

A general finding is that the introduction of driverless trucks may result in substantial impacts by reducing the cost of road transport relative to other

transport modes. This change would affect freight transport decisions and generate system-level impacts by increasing the road mode share and reducing total logistics costs. However, the magnitude of these effects varies depending on the assumed cost reduction for road freight transport and the driverless truck operating model being used. The variation in system-level impacts between scenarios is large, which suggests that uncertainty in techno-economic assumptions for future driverless trucks propagates to substantial uncertainty about their system-level impacts.

An overview of key results is presented in Table 5. Results for Paper F are the min and max values obtained for each operating model in the set of *mid-M-ET-scenarios*, see Table 4. Calculations of *Δ road tkm* and *Δ total logistic* cost are made using the baseline scenario for Paper D as reference, while for Paper F, the reference is the *mid-M-ET-scenario* without any driverless trucks. However, comparisons between results from Paper D and Paper F should still be made with the caveat that they use different baseline scenarios (2017 data with only ICETs for Paper D, 2045 forecast with biofuel ICET and METs for Paper F), use different versions of Samgods (Logmod) and use different cost models and assumptions for driverless trucks.

Table 5 Results for selected system-level impacts from Paper D and Paper F. Δ road tkm and Driverless truck share of road tkm consider tkm on Swedish territory. Δ total logistic cost covers total costs for all flows (domestic, export, import, transit). Adjacent scenarios highlighted in grey are the most comparable scenarios between the two papers.

Paper: Operating model	Δ road tkm		Driverless truck share of road tkm		Δ total logistic cost	
	Min	Max	Min	Max	Min	Max
D: AllDL (Level 5)	+5%	+35%	100%	100%	-12%	-2%
F: Level 5	+3%	+56%	38%	99%	-16%	-0%
F: Highway flows	+5%	+66%	7%	82%	-8%	-0%
F: Terminal hub-to-hub	+3%	+31%	2%	42%	-2%	-0%
D: H2H	+2%	+21%	11%	37%	-8%	-2%
F: Transfer hub	+0%	+14%	2%	30%	-1%	0%
F: Short distance feeder	-1%	+0%	0%	0.5%	-0.3%	-0%

There are only a few studies that are similar to Paper D and Paper F in that they model driverless truck impacts on large scale (national level) and for multi-modal systems. The two most similar studies, Huang and Kockelman (2020) and Huang et al. (2024), are both for the US, and consider scenarios corresponding to as the AllDL/Level 5 scenario in Paper D and Paper F

respectively. However, the model and scope of the analysis differ. Samgods/Logmod uses fixed freight demand matrices while the RUMBRIO model, used for the US studies, models origin choice endogenously. Samgods models transport chain choice and includes road, rail, sea and air transport while RUMBRIO only considers road and rail. Due to these differences, direct comparisons of results are challenging but results point in the same direction: increasing road mode share and increasing demand for road transport.

6.3.3 Insights into operating models and electrification

A key finding is that driverless trucks with SAE level 4 automation may generate substantial impact at the system level. For instance, Paper F suggests that the Highway flows operating model can, in cases of a high automated driving maturity level, result in a majority of road tkm being performed by driverless trucks and substantially increase road transport demand. This operating model requires that the driverless trucks perform the first and last mile of the trip so that full PC flows can be covered. Other operating models, which also primarily depend on highway driving but require transfers between trucks or modes, such as H2H (Paper D), Terminal hub-to-hub (Paper F), and Transfer hub (Paper F), are not as attractive but may still generate substantial impacts.

Paper F further shows that, in scenarios where both a high maturity level for electrification technologies (ETs, charging, etc.) and automated driving technology are achieved, cost reductions compound, further reducing road freight costs. On the other hand, if electrification technology maturity is low, scenarios without driverless trucks achieve a very low electrification rate, and M-ICETs using biofuels would be the cost-minimizing choice for most flows. However, in such cases, driverless trucks may increase the electrification rate if the automated driving technology maturity is medium or high. In other words, under these conditions, automated driving compensates for costly electrification and makes D-ETs attractive relative to M-ICETs.

Moreover, the results in Paper F suggest that building a more widespread public charging network (in addition to the smallest considered network of 60 well-chosen locations in Sweden) has no major impact on the electrification rate, whether with or without driverless trucks, for the battery sizes considered (allowing for 4.5h or 9h of driving). This supports other assessments in the literature, which indicate that the number of charging locations needed to serve a large share of demand is fairly limited, but ensuring access to sufficient grid power is a more pressing issue (Lange et al., 2024).

Another contribution is that Paper D is, to the author's best knowledge, the first study to quantify the impacts of the hub-to-hub operating model. The findings suggest that this operating model might be an attractive option, although both the representation and analysis of this operating model in Paper D are associated with several limitations. A subsequent study that considered

the operational aspects of this operating model in detail confirmed this finding (C. Lee et al., 2023). In Paper E, a broader set of operating models was considered, including variations of hub-to-hub operations as well as another type of operating model, the short distance feeder. The latter is especially relevant since it has not previously been studied using large-scale modeling, but this type of application may be an initial application in which driverless trucks will be deployed. Another interesting aspect with this operating model is that it boosts multi-modal transport in contrast to other operating models, although the results from Paper F suggest that this effect is likely marginal.

Another contribution by Paper F is the assessment of driverless trucks from an electrification and decarbonization perspective, which has largely been overlooked in the literature. In addition to the contributions related to driverless trucks, the modeling of M-ETs in Paper F advances the state of knowledge for system-level impacts of electric trucks in general. Previous literature on large-scale electrification impacts has typically used fixed truck OD-matrices and performed analysis of economics, energy feasibility, charging infrastructure network design and charging infrastructure competition (Lange et al., 2024; Rogstadius et al., 2023; Shoman et al., 2023). Paper F complements this research by using a multi-modal model and studying how M-ETs may compete or complement other transport modes.

6.4 Objective 4: Methods for deep uncertainty in freight transport modeling

Objective 4: Assess and demonstrate methods for managing deep uncertainty in model-based foresight and policy analysis for driverless trucks.

The key contributions to Objective 4 are:

1. Application of methods for managing deep uncertainty (6.4.1).
2. Identification of challenges and opportunities with EMA and DMDU (6.4.2).

6.4.1 Application of EMA and DMDU

Transport is one area for which EMA and DMDU can be highly beneficial. Despite their potential benefits, they have only been tested and applied for transport planning to a limited extent so far (Stanton & Roelich, 2021), see Section 3.2.2 for previous examples.

This thesis contributes through initial efforts to apply EMA and DMDU to freight transport in general and the analysis of driverless trucks in particular. Several approaches for managing deep uncertainty have been explored for different types of applications and purposes: vehicle level cost performance modeling using computational experiments (Paper E), national level freight modeling using exploratory modeling and analysis (Paper F), and transport climate policy analysis using the MORDM framework (Paper G). To support the analyses, both the application and modification of existing tools such as EMA-workbench (Kwakkel, 2017) as well as newly developed tools (e.g. EMA-Logmod-AE and the cost performance model in Paper E) have been used.

In Paper E, computational experiments are used to cover three parameter domains 1) heterogeneous characteristics of transport flows 2) deep uncertainties in future techno-economic performance of electric and driverless trucks and their market conditions and 3) the influence of different types of charging strategies. Previous studies making assessments of techno-economic potential for ETs and driverless trucks typically focus on one or two of these dimensions, for instance by making case-studies of specific transport networks (Bergqvist et al., 2023; Karlsson & Grauers, 2023a; Link & Plötz, 2022), performing analytical analysis of charging strategies for a single set of techno-economic truck parameters (Karlsson & Grauers, 2023a) or probabilistic analysis of using type vehicles without detailed representation of transport operations (ITF, 2022; Nykvist & Olsson, 2021; Sripad & Viswanathan, 2019). While these approaches have merits and bring value, there is also a need to understand how the electric truck performance potential varies over all three types of dimensions, which Paper E contributes to.

For Paper F and Paper G, the contributions focus on national-level planning and policymaking and how uncertainties related to driverless trucks can be incorporated into long-term model-based foresight (Paper F) and transport climate policy analysis (Paper G).

Paper F applies exploratory modeling and analysis to national freight transport modeling to explore the influence of technological uncertainty on a decarbonized freight transport system. The EMA-Logmod-AE system (Section 4.4.2) includes model representations of key aspects of truck electrification and automated driving, tools for designing scenario ensembles, and tools for analyzing and visualizing scenario ensemble results. Paper F showcases the practical use of this approach, identifying critical uncertainties and issues for further exploration.

Paper G explores how MORDM may improve the generation and evaluation of national transport climate policies from a robustness perspective. A broader range of uncertain factors are considered compared to the other papers and driverless trucks are considered as one of many uncertain factors. Paper G contributes by evaluating if and how MORDM can improve transport climate policy analysis. It assesses whether MORDM helps generating policies that have better performance and robustness across multiple relevant policy objectives than previously identified policies and to what extent robustness- and vulnerability analyzes brings additional policy insights. The transport policy literature has highlighted the need for handling scenario uncertainty and to explore a larger set of policy levers in transport climate policy analysis (Lyons & Marsden, 2021; Marsden & McDonald, 2019; Witzell, 2021). Through MORDM, Paper G addresses these gaps and evaluates its efficacy.

6.4.2 Identification of challenges and opportunities

Paper E, Paper F, and Paper G collectively demonstrate the feasibility of applying EMA and DMDU for analyzing driverless trucks and that such analyses can provide a richer and more nuanced understanding of the economics and impacts of driverless trucks. However, they also highlight conceptual and practical challenges with applying such methods. Some of the main identified challenges for incorporating DMDU into transport planning and policy analysis in Sweden are:

- Existing practices and inertia. Swedish transport planning is mainly based on a predictive paradigm and primarily bases investment and policy evaluation on a single forecast. Transitioning to DMDU requires altering these practices and the development of new tools.
- Tension between planning approaches. Conventional planning methods, like cost-benefit analysis (CBA), prioritize single-metric evaluations by aggregation of outcomes and standardization using standardized appraisal values and the use of a single reference forecast. DMDU approaches, such as MORDM, seeks to avoid

aggregation over future states of the world and valuation of outcomes and emphasize robustness across multiple futures and metrics.

- Perceived validity and communication. DMDU analyzes a broad range of potential outcomes instead of a single, or small set of, predictions. Policymakers may question the validity and practicality of the results. It may be challenging to effectively communicate DMDU results and their implications. Also, the use of simplified models in DMDU, such as the Scenario tool, contrasts with the conventional more detailed national models like Samgods. The use of simplified models allows for large-scale experiments, but may be perceived as too coarse for policymaking. Bridging this gap was one motivation for exploring the feasibility of performing EMA with Samgods in Paper F.
- Knowledge requirements. Implementing DMDU among analysts requires technical skills in exploratory modeling and scenario analysis, which is partly complementary to the skillset for conventional transport modeling. However, the emergence of open-source software tools for EMA and DMDU drastically reduces technical hurdles. But still, the use of DMDU requires a shift in mindset and conceptual approach for the use of modeling. For policy makers and decision makers receiving DDMU analyses and acting upon them, there is likely also a need to develop skills and experience in how to understand and interpret DMDU results.
- Computational complexity and modeling issues. Mainstream national transport models are generally relatively computationally demanding, and the general trend is toward more detailed models and a larger focus on simulation. This may limit the practical feasibility of applying them for EMA and DMDU which relies on large sets of experiments. One option is to use faster simplified models, or a meta-model of a detailed model, for initial exploration to identify relevant scenarios and policies for further analysis in a more detailed model. Such ideas have been present since at least two decades (de Jong et al., 2004) but seem to have gained limited traction in national planning. However, there are examples of EU-level models with a larger focus on short run times (De Jong et al., 2021) and there are new tools to support the generation of meta-models of conventional transport demand models (Lemp et al., 2021).

7 Discussion and outlook

This chapter discusses the thesis findings, provides an outlook for future work and presents concluding remarks.

First, a discussion on driverless trucks in relation to decarbonization is provided in Section 7.1 followed by reflections on additional policy, planning, and industry implications in Section 7.2. Then, the limitations of the modeling studies and their implications for interpreting the results are discussed in Section 7 .3. The chapter concludes with ideas for future research in Section 7.4 and concluding remarks summarizing the contributions of the thesis in Section 7.5.

7.1 Driverless trucks and decarbonization

An important perspective is how driverless trucks may affect the decarbonization of road transport. This is discussed in Section 7.1.2, prior to which a short introduction to the scope and scale of change required to decarbonize in line with politically decided targets is provided in Section 7.1.1.

7.1.1 The decarbonization transition in perspective

The scale of change required to decarbonize the freight transport sector is massive. The EU-27 road transport GHG emissions amounted to 0.755 Gt CO_{2eq} in 2022 of which heavy trucks and buses accounted for roughly 26% (EU Commission, DG Energy, 2024). Plötz et al. (2023) estimate a remaining carbon budget for the EU road transport sector (including passenger and freight transport) to comply with the pathway for limiting global warming to 1.5 °C being about 9.7 $GtCO_{2eq}$ for 2021 to 2050. At current emission rates, the carbon budget would be consumed by 2033. Assuming a linear reduction trajectory, EU transport emissions need to be reduced by 42% by 2030,

compared to 2020, 84% by 2040 and reach zero direct GHG emissions by 2045 (Plötz et al., 2023).

The European climate law stipulates EUs' climate targets of a 55% reduction of CO_2 emissions in 2030 compared to 1990 levels and achieving zero emissions by 2050. It can be noted that the European Scientific Advisory Board on Climate Change recommends the EU to reduce EU-wide GHG emissions by 90-95% by 2040 relative to 1990 (European Scientific Advisory Board on Climate Change, 2023). The EU recently decided on a set of policies for achieving these targets including new requirements for vehicle manufacturers to reduce the tail-pipe emissions of newly sold heavy duty trucks by 65% in 2030 and 90% by 2040 compared to 2019 levels (Börjesson & Eliasson, 2024).

In a Swedish context, the Swedish climate law imposes achieving net-zero territorial GHG emissions by 2045, which in practice means that the road transport system should be decarbonized by then (Trafikverket, 2024b). The climate law and climate targets are becoming institutionalized within Swedish transport planning. For instance, the most recent reference forecast by STA is based on the assumption that the Swedish transport system will be "near-zero" by 2045 with road traffic, rail traffic, and domestic sea transport fully decarbonized by 2045 and other sea transport and air transport being decarbonized by 2050 (Trafikverket, 2024b). The STA expects that this decarbonization will primarily be achieved through electrification and an increased share of biofuels, which implies that the share of ZETs in truck sales reaches 100% by 2040 (Trafikverket, 2024a) i.e. stricter than the EU requirements.

Pathways for achieving a 100% market share of ZETs in terms of newly sold trucks by 2040 have been assessed to correspond to a share of about 65% by 2030, of which the vast majority are expected to be ETs (Energiforsk, 2022). Between 2020 and 2023, just above 500 ETs were registered in Sweden and ETs constituted 4% of the newly registered heavy-duty trucks in 2023 (Trafikanalys, 2024). Thus, an annual market share growth rate of 45% is required to reach 65% ETs by 2030. Such a rapid growth is arguably challenging and requires numerous substantial developments, including a sufficiently rapid deployment of charging infrastructure (Lange et al., 2024), addressing financing issues (Parviziomran et al., 2024) and an accelerated turnover of the truck fleet (Rogstadius et al., 2024). On the other hand, there are also positive signs such as projected drastic cost reductions for batteries and other enabling technologies (Link et al., 2024), although it is recognized that such cost projections are uncertain and should be interpreted with caution (Yeh & Rubin, 2012).

Decarbonization will also impact freight demand. One reason for the projected decrease in sea transport demand in the STA reference forecast is that the decarbonization of transport is expected to drastically reduce the import of fossil fuels which constitutes a large share of sea transport (Wikström, 2024).

Another demand-side development is the ongoing establishment of decarbonization-related industries in Sweden, such as battery manufacturing and carbon-neutral steel production. These industries will generate additional freight demand with specific characteristics. Other structural changes in transport demand, such as increased production and distribution of biomaterials in Sweden, have been identified as potential results of climate policies, such as increased CO_2 tax (Almström et al., 2024). Moreover, other sustainability related trends such as more circular supply chains may further affect the structure and volume of freight transport demand, as discussed in Paper A.

7.1.2 Decarbonization implications of driverless trucks

McKinnon (2018) outlines five types of mechanisms that can be used to decarbonize freight transport

1. Reducing the demand for freight movement
2. Shifting freight to lower-carbon transport modes
3. Improving asset utilization
4. Increasing energy efficiency
5. Switching to lower-carbon energy

Based on the findings in this thesis and related literature, the impact of driverless trucks on each of these mechanisms is here briefly discussed. This discussion is not an attempt at a systematic assessment of decarbonization impacts, but rather intends to make the point that the effect of driverless trucks on decarbonization is unclear, that it will be context-dependent, and ultimately the outcome of a nexus of interdependent factors on vehicle- and system level.

1. Reducing the demand for freight movement

Paper C and Paper E suggest that driverless trucks would in the long run possibly result in reductions of generalized freight transport costs which would rather increase freight demand. This effect is also noticed by modeling studies for the US which predict increasing tkm demand (Huang et al., 2024; Huang & Kockelman, 2020) and a study using a computational general equilibrium model that predicts GDP growth due to the introduction of driverless trucks (Waschik et al., 2021) which would likely result in additional freight demand (McKinnon, 2018).

2. Shifting freight to lower-carbon transport modes

Both Paper D and Paper F suggest that driverless trucks would increase road freight demand due to mode shifts from rail and sea to road in many of the analyzed scenarios. The only exception occurs in scenarios where driverless trucks are used solely as feeders to and from multimodal terminals. In such cases, they may contribute to a slight increase in multimodal transport chains and an associated reduction in the road mode share. However, while modal

shifts from road to sea or rail currently reduce carbon emissions, this effect will diminish over time as trucks become less carbon-intensive (Johansson et al., 2024). Modal shifts to road as an effect of a driverless truck introduction is identified by several other modeling studies (Bao & Mundy, 2018; Chowdhury et al., 2024; Huang et al., 2024; Huang & Kockelman, 2020).

3. *Improving asset utilization*

Driverless trucks may enable increased truck utilization by eliminating driver-related constraints such as daily driving- and rest-time regulations. This would allow driverless trucks to be operated for longer hours, effectively increasing the average operational speed for long trips (Paper F). Furthermore, driverless trucks designed without a driver's cabin would free up weight and volume, which could be used to increase cargo capacity or battery capacity (as explored in Paper E and Paper F). These effects would mean that fewer trucks might be required to fulfill a given transport demand, ceteris paribus (Hong et al., 2023; C. Lee et al., 2023). However, it is uncertain to what extent such utilization benefits can be realized in practice. Additionally, from a life-cycle perspective, increased truck use intensity would not necessarily mean that fewer trucks are used overall since fleet turnover rates might increase. Furthermore, trade-offs exist between high truck utilization with frequent high-power charging and achieving long battery life, as battery life is, generally, negatively correlated with charging power (Paper E).

4. *Increasing energy efficiency*

Driverless trucks could potentially drive more energy-efficiently than human drivers (Paper C). Additionally, weight reductions from not including a driver's cabin would reduce specific energy consumption, all else equal, although this effect is small, likely less than 0.1 kWh/km (Paper E). It has also been speculated that driverless trucks might, in some situations, drive slower to reduce energy use, such as during highway driving at night for goods with a low cargo value of time (Bray & Cebon, 2022a; Kristoffersson & Pernestål Brenden, 2018). Such strategies would however counteract truck utilization benefits. The ADS has a non-negligible energy consumption, but without a driver's cabin there is no need for heating and air conditioning which may substantially decrease auxiliary power consumption (Jungblut et al., 2023), possibly in the order of 0.1 kWh/km (Sigle & Hahn, 2023). However, it seems plausible that increases in road transport demand would largely, or fully, offset improvements in energy efficiency in many implementation scenarios (Paper D, Paper F).

5. *Switching to lower-carbon energy*

It is unclear whether driverless trucks would increase the share of ZETs, or more precisely, the share of vehicle kilometers performed by ZETs. This is a largely unexplored area in the literature, but in general, there is nothing fundamental linking automated driving to a particular propulsion technology

or energy carrier. However, there are some economic and operational aspects that may be of importance. Firstly, the reduced tare weight and energy consumption of driverless trucks might allow for a larger share of transport flows to be energy-feasible for electrification than would otherwise be the case (Paper E). This effect is likely small in relation to other factors affecting the feasibility for electrification but may have a marginal effect. Secondly, the removal of an in-vehicle driver eliminates the indirect driver cost during charging and may allow for more flexible charging planning which may increase the attractiveness of electric trucks and the utilization of D-ETs (Hong et al., 2023). Thirdly, automated driving may offset the added costs of battery electric trucks, i.e. a D-ET might have a lower TCO than a M-ICET in cases where a M-ET would not (Paper F). Also, automated driving might result in larger relative cost reductions for D-ETs than for D-ICETs (Ghandriz et al., 2020). However, this effect may diminish over time as M-ET are expected to achieve lower TCO than M-ICETs (Basma et al., 2021; Link et al., 2024; Rogstadius et al., 2023).

Another related effect is that there is possibly a causal link in the other direction, namely, that a transition to ETs would boost the incentives for adopting driverless trucks since driver costs as a share of total truck TCO may increase substantially for electric trucks (Rogstadius, 2022).

7.2 Implications for policy, planning and industry

The findings in this thesis suggest that if a large-scale introduction of driverless trucks takes place, there could be substantial changes in policy-relevant system outcomes. These prospects have implications for both transport planning and policy as well as the broader freight transport sector and industry.

7.2.1 Driverless trucks may be poorly understood if modeled as "cheap trucks"

The introduction of driverless trucks will not necessarily take place as a direct substitution of manually driven trucks. It is plausible that the initial deployment of driverless trucks will be for trucks with limited ODDs operating e a sub-set of the road network and specific types of transport chains (e.g. in providing the middle-leg in a transfer hub model, or a leg in a multimodal chain). This could change the structure of road transport networks and entail a different logic for road transport operations. Therefore, driverless trucks using such operating models will not directly substitute manually driven trucks but rather represent an additional "road sub-mode" that is complementary to manually driven trucks for some transport applications and competes with manually driven trucks for others. For planning and impact assessment purposes, it is important not to default to conceptualizing and modeling driverless trucks simply as a direct substitute for manually driven trucks or focus only on aspects such as relative costs and their penetration rate.

7.2.2 Level 5 automation likely not required for substantial system-level impacts to occur

A general finding in the thesis is that the "everything somewhere" approach to driverless truck development, i.e. driverless trucks with limited ODDs operating only some types of transport flows, can offer attractive economics and be competitive for a large share of transport flows. While the system-level impacts of driverless trucks using such operating models are not as substantial as in scenarios for SAE level 5 trucks, the timeline for their deployment may be much shorter. From a planning perspective, this is an important insight, as it could justify considering driverless trucks in shorter planning horizons than would otherwise be the case. Moreover, such operating models may result in different spatial patterns of road freight flows.

From an industry perspective, this finding highlights the need to identify and navigate trade-offs when prioritizing development efforts. There may be a balance between enabling operations for a larger set of flows (increasing scope) and improving cost performance for the set of flows that can be operated.

7.2.3 Managing increasing demand for road freight

A fundamental implication of a future introduction of driverless trucks is that it would likely lead to an increased demand for road freight transport. This may compound with effects from truck electrification which may further reduce the cost for road freight as electric truck technology further matures (Link et al., 2024; Rogstadius, 2022). In Paper F, a few scenarios with driverless trucks deployed across the full road network result in road demand increasing by more than 50% compared to the STA reference forecast. However, even without driverless trucks road freight demand may increase by up to 30% due to cost efficient M-ETs. This uncertainty about forecasting results should be considered in long-term infrastructure planning. Furthermore, as road freight becomes decarbonized, the direct climate emissions from road freight will be less of an issue, but other externalities from road freight will still need to be addressed (Johansson et al., 2024).

7.2.4 Climate policy implications

If a large-scale introduction of driverless trucks takes place, it is likely that policies for achieving modal shifts from road to rail and sea become less feasible and more costly since the relative cost performance of road transport increases. Achieve modal shifts from road have been proposed as important measures to reduce freight transport greenhouse gas emissions in European and Swedish transport climate policy (European Commission, 2011; Regeringskansliet, 2018). However, previous research has suggested that even without consideration to driverless trucks, the potential for shifting freight transport from road to sea in Sweden is quite limited and that realizing the

potential would require rather costly policy instruments (Johansson et al., 2024; Rich et al., 2011; Vierth et al., 2020).

Some Scenarios in Paper F saw a decrease in road freight demand due to scenarios in which there is a low maturity level for ET and battery technology making biofuel ICETs a more attractive option. However, such scenarios may turn out not to be feasible due to policies which are not considered as constraints in the modeling. For instance, the recently updated EU CO_2 requirements for heavy duty trucks requires truck manufacturers to reduce tailpipe CO_2 emissions of newly sold trucks by 90% in 2040 compared to 2019 levels (Mulholland, 2024). How trucks operating on 100% biofuels, which do produce tailpipe emissions, will be considered when evaluating CO_2 standards is currently investigated by the European Commission, but it is possible that scenarios relying on ICETs using biofuels would not be feasible under this mandate. If not, scenarios with low ET and battery maturity levels would likely instead either see adoption of ETs in line with the CO_2 mandates but with higher average costs for road transport, or extended use of other zero-emission technologies such as hydrogen fuel cell trucks (Mauler et al., 2022) which are not modeled in this thesis.

7.2.5 Implications for truck electrification and charging

During the last years there has been a growing attention in the literature on the charging infrastructure needs for supporting large-scale electrification of road freight using manually driven trucks. This research includes studies of charging demand and charging localization given current transport patterns (Fu et al., 2024; Lange et al., 2024; Shoman et al., 2023), the interaction and competition between different types of charging infrastructure (Rogstadius et al., 2023), assessments of electric road systems (Ainalis et al., 2022; Börjesson et al., 2021; Deshpande et al., 2023; Rogstadius et al., 2023), and the dynamics of charging supply and electric truck adoption (Raoofi et al., 2024). Another related topic that has been explored is what type of charging strategy is most favorable for different transport applications (Bergqvist et al., 2023; Karlsson & Grauers, 2023a, 2023b; Parviziomran & Bergqvist, 2023), which is also analyzed in Paper E. A common finding in such studies is that charging during planned downtime at the depot, and, if needed, during mandatory driver rest breaks, is generally the most favorable setup. This is since such setups avoid costly non-productive standstill time, has a low planning complexity and minimizes reliance on en route fast charging which is expected to be more costly and lead to more rapid battery deterioration.

For driverless trucks, these findings may not apply. The different cost structure (lower time-dependent marginal operating costs but higher fixed ownership costs) and lack of driver-related operating constraints may have several effects on what charging setup is most attractive. Direct costs for charging standstill time during non-planned downtime is lower for driverless trucks. On the other

hand, driverless trucks will likely have less planned downtime in depots and no driver break-related downtime during operations that can be used for charging without limiting their productivity. In other words, the opportunity cost for charging time is higher for driverless trucks compared to manually driven trucks. This should reasonably point to that higher power charging and charging during other types of downtime, such as during loading and unloading, will be more attractive for driverless trucks. This dynamic has been proposed to affect powertrain choice and favor fuel-cell electric trucks over battery electric trucks using static charging since hydrogen refueling will be faster than charging (Mauler et al., 2022). A related aspect is dynamic charging using electric road systems, that allows trucks to recharge while driving, might be attractive for driverless trucks since it can reduce charging standstill time (Rogstadius et al., 2023). Another perspective is that driverless trucks may allow for more flexible charging scheduling which can be used to distribute charging over the time of day and thus reduce peak charging demand and increase charging infrastructure utilization which would reduce charging costs (Hong et al., 2023).

A final consideration is that if driverless trucks are used primarily for hub-to-hub applications mainly operating on the highway network, the charging demand would have a different spatial characteristic and be more concentrated to these parts of the road network.

7.2.6 Acknowledge uncertainty and its implications for planning and policymaking

Although this thesis indicates that there may be significant economic benefits of driverless trucks for producers and buyers of freight transport services, the policymaking and governance of driverless trucks need to acknowledge and consider the risks for negative externalities and unanticipated effects from the potentially substantial increase in road transport volumes as well as the uncertain sustainability effects.

The thesis builds on a problem formulation where the future freight transport system is regarded as highly uncertain, with driverless trucks being one source of uncertainty. However, the notion that the development of the transport system is deeply uncertain can be contested and there are diverging views among transport scholars. One topic is whether fundamental transport behaviors and preferences can be expected to remain stable over time and thus limiting the degree of uncertainty (Eliasson, 2022) or whether technological and behavioral change may fundamentally change transport (Lyons & Davidson, 2016; Marsden & McDonald, 2019).

In simple terms, current national freight forecasting can be described as a process of predicting changes to input data and model parameters for a future scenario year for the national freight model Samgods (Trafikverket, 2024e). In other words, the fundamental structure of the system is assumed to remain,

but its inputs are adjusted to reflect change over time. Making such forecasts is in no way an easy feat, and it requires compiling a wide range of information from various sources, alignment with forecasts by other public bodies, and making qualified assumptions to manage uncertainty.

It can be noted that driverless trucks, or other factors related to automated driving technology, have not been considered in any STA reference forecast so far. It is not obvious that the assumption that driverless trucks would have no noticeable effect on the transport system during the coming 20 years is the most likely outcome. Moreover, this assumption contrasts by statements by the STAs in other documents, for instance in their roadmap for digitalization of road transport, in which it is stated that the anticipated development is that within 15 years "vehicles with increasingly advanced automated functions are widely available" and that in 20+ years "automated vehicles are widely available" (Trafikverket, 2024d, p. 15). Similar inconsistencies between stated expectations on technological development and forecasting assumptions have been observed for the UK (Marsden & McDonald, 2019). Possibly, there is a reluctance among forecasters to consider uncertain technological change, such as the prospects of driverless trucks, in forecasts (Lyons & Marsden, 2021) and there is therefore a value in exploring alternative analysis frameworks that reduce thresholds for enabling larger scenario sets to be explored.

It is debated whether a predictive based planning approach is appropriate going forward. As previously mentioned, some planning researchers have stressed the need for novel approaches that supports exploration of broader sets of futures and policies (Lyons & Davidson, 2016; Lyons & Marsden, 2021; Marsden & McDonald, 2019; Witzell, 2021). On the other hand, uncertainty does not necessarily make current planning approaches irrelevant. For instance, CBA ranking of infrastructure investments have been shown to be rather robust to uncertainty in forecasted demand and valuations of externalities (Asplund & Eliasson, 2016).

7.3 Limitations in modeling scope

This section presents a discussion of central limitations in the modeling studies in Paper D and Paper F. The focus is on these studies since they provide key policy-relevant quantitative results for the thesis and a qualified and nuanced understanding of these results is crucial for a proper interpretation of the findings of this thesis.

7.3.1 Truck fleet and dispatching

Samgods (Logmod) does not explicitly represent the truck fleet. Thus, it does not allow for representing or analyzing fleet size and fleet composition or considering individual trucks or their dispatching to transport missions. This limitation means that it is not possible to account for a specific penetration rate

of a particular truck type as a constraint in the model. The resulting "market shares" of different truck types from the analyses in Paper D and Paper F should be interpreted as the demand for using these truck types, assuming no supply-side constraints. Instead, insights about fleet effects have to be based on other studies that do model impacts on truck utilization and associated fleet effects(Hong et al., 2023; C. Lee et al., 2023).

7.3.2 Transport demand impact mechanisms

The modeling of system-level impacts in Paper D and Paper F assumes fixed transport demand using exogenous PC flow matrices. This means that freight demand is modeled as insensitive to changes in transport supply. However, both fundamental transport economics and supply-chain theory, as well as empirical studies, suggest that this assumption does not reflect reality.

Modeling with fixed demand means that the impacts of driverless trucks on changes to road freight demand identified in this thesis are underestimated. While it is hard to judge the extent of this effect, some insights are provided by a review of empirically derived road freight elasticities that estimate a tkm price to tkm elasticity of -1.0, out of which roughly -0.4 is attributed to mode shifts and -0.6 due to other changes in transport demand (de Jong et al., 2010). Additional insights are provided by the two American studies analyzing driverless trucks using the RUMBRIO model which show that driverless trucks would both lead to a net increase in freight demand (tkm) and a modal shift from rail to road indicating that these are both potentially important effects (Huang et al., 2024; Huang & Kockelman, 2020).

7.3.3 Road freight adaptations to driverless trucks

The modeling of driverless truck operations in Papers C, Paper D, Paper E, and Paper F focuses on the impacts of automated *driving* but place less emphasis on other non-driving tasks critical to road freight such as loading and unloading, load securing, and similar tasks. These tasks are currently often performed by the driver (Flämig, 2016) and would be handled in different ways through (combinations of) various adaptations such as process redesigns, having tasks performed by other personnel (e.g. by employees at sending/receiving locations or terminals) and automation (Agrawal et al., 2023). Assessing these adaptations is beyond the scope of this thesis, but they are acknowledged as important for enabling large-scale adoption of driverless trucks and might have substantial influence on the economics and service quality of transport services involving driverless trucks.

Other similar aspects that are not explicitly modeled relate to potential changes in the design and locations of logistics infrastructure and facilities to support and integrate driverless truck operations within supply chains. One factor is the need for logistics infrastructure tailored to specific operating models where driverless trucks are limited to operating on specific parts of the road network.

It has been suggested that driverless trucks may operate from "launch and landing zones": dedicated locations strategically located near the road network where driverless trucks operate (Shladover et al., 2024). These zones would act as staging areas where driverless trucks would be prepared for their next trip, and may also be used for pickup and drop-off of trailers when operating in the transfer-hub model (C. Lee et al., 2023; Monios & Bergqvist, 2019). Good network design is likely critical for achieving cost efficient operations of the transfer-hub model at scale (C. Lee et al., 2023).

In both Paper D and Paper F, no changes were modeled for the location and number of transport terminals are modeled, instead Samgods default terminal network was used. Since these locations are not designed or located with any consideration to driverless trucks, this would imply that there might be additional potential for some of the analyzed operating models. Neither the granularity of the network model nor the temporal resolution of Samgods captures detailed operational aspects of such operating models, which are better studied using other methods.

7.3.4 Developments in other transport modes

There is an extensive range of known uncertainties related to the future freight transport system that are not accounted for in this thesis and could interact with the development and impacts of driverless trucks. For instance, the development of alternative ZET technologies such as hydrogen fuel cell trucks might influence the use of driverless trucks (Mauler et al., 2022). Similarly, alternative charging infrastructure solutions, such as electric road systems, could significantly affect road transport patterns and costs (Börjesson et al., 2021; Rogstadius et al., 2023).

Developments in other modes could also have a substantial impact on the relative performance and attractiveness of transport modes. For instance, rail innovations considered in the Shift2Rail research program have been estimated to offset gains in road freight due to driverless trucks, if fully implemented (Kristoffersson et al., 2021). However, while these two factors might balance one another in terms of modal split, both would reduce freight transport costs and thus affect the system in other ways.

It is beyond the scope of this discussion section to provide a comprehensive review of relevant uncertain factors. However, a broader view on uncertainties in technological developments in automation, digitalization, and electrification is provided in one of the additional publications (Stenemo et al., 2021).

7.4 Future research

The work with this thesis has generated many ideas for topics of future research. Here, a selection of such ideas is presented.

7.4.1 Assess model integration for achieving broadened system boundaries

Important limitations of the freight system modeling in this thesis relate to the lack of feedback from changes in transport supply to changes in transport demand, no consideration taken to the energy system, the lack of linkage between charging infrastructure utilization and charging cost, and the lack of a fleet model. To model linkages with other systems- and sub-systems, one potentially promising approach could be to (soft-) link EMA-Logmod-AE with other existing models of these systems. Possible model linkages could be the MOSTACHI[10] model for charging infrastructure time and route-specific charging and battery sizing decisions, the TIMES[11] model for energy systems linkages, and trade and freight demand models sensitive to changes in transport costs. However, it is well-known that such model integration intending to achieve both large-scale, large-scope and high detail is practically and conceptually challenging (L. Tavasszy & de Jong, 2014) and the feasibility of such integrations must first be carefully assessed.

7.4.2 Modeling of broader transition pathways and their dynamics

Paper F presents initial work on modeling various scenarios for driverless and electric trucks in a decarbonized freight transport system. As a continuation of Paper F, the analysis can be extended to a broader set of uncertainties. For instance related to other technological development in the freight transport sector, and uncertainties in freight demand, and energy costs and supply. However, such a study would come with practical challenges in dealing with a broad range of factors consistently as well as conceptual challenges on how to handle correlation structures and feedback between various scenario parameters.

Another limitation is that the system modeling is static and does not model change over time. This limits the understanding of how the introduction of driverless trucks may play out and how their effects might evolve over time. Recent literature has called for modeling that incorporates broader system boundaries, links system levels, and uses simulation-based approaches to better understand the dynamics of adoption and impacts of emerging freight technologies (Browne et al., 2023; Ghisolfi et al., 2022; Nassar et al., 2023; Raoofi et al., 2024; L. A. Tavasszy, 2020).

System dynamics is one approach that has recently been used aiming to incorporate feedback in large-scale freight modeling (Ghisolfi et al., 2024; Raoofi et al., 2024). While system dynamics offers many merits, the structure of system dynamics models provides limited capability of representing

[10] https://github.com/JakobRogstadius/MOSTACHI
[11] https://timesnordic.tokni.com/about

diversity and heterogeneity due to the need for aggregation of system variables (Köhler et al., 2018). This limits the ability to model fundamental aspects of freight transport, such as spatial characteristics and heterogeneity in freight flows, vehicle types, and transport chains. One option might be to combine system dynamics with a more detailed static model. In this setup, a high-level scenario model for dynamics related to technology uptake and freight demand could be modeled using system dynamics. This model could then be linked to a static, detailed freight model, such as Logmod-AE, which would be sampled at specific timesteps of the simulation with inputs from the dynamic model. Key aggregate results from the detailed model, such as modal split and transport costs, could then be fed back into the scenario model. This approach has been previously demonstrated for urban freight modeling (Thaller et al., 2016) and seems promising.

7.4.3 Case studies assessing driverless trucks from a logistics perspective

The research in this thesis is done at a relatively high abstraction level, with little detailed consideration of specific logistical contexts and constraints. In general, the literature would benefit from more detailed studies analyzing the prospects and potential impacts of driverless trucks to gain a nuanced understanding of real-world challenges and opportunities, which could also improve system-level modeling. Both qualitative and quantitative research would be of value. Several studies of this kind have been made for M-ETs (Bergqvist et al., 2023; Karlsson & Grauers, 2023a; Link & Plötz, 2022) but are largely lacking for driverless trucks except for a few initial, limited studies (Agrawal et al., 2023; Tjaden & Flämig, 2024). Potential issues to address include the management of non-driving driver-related tasks for various types of flows and logistical contexts. Other areas are analyses of truck fleet operations using realistic transport demand data to understand fleet-level truck utilization and remote operator utilization, as well as quantifying potential efficiency gains at the fleet level under various logistical constraints.

7.4.4 Relationships between electrification and automated driving and impacts on charging

Section 7.3.5 raised several open questions about the relationship between driverless trucks and truck electrification, particularly regarding charging planning and infrastructure. This is an important and largely unexplored area. In summary, research needs include the following questions:

1. How do various factors related to truck utilization, battery degradation, opportunity costs, grid impacts and charging infrastructure utilization interact and ultimately affect the cost efficiency of various types of charging strategies for driverless trucks?

2. How would various scenarios for charging supply (static vs dynamic, locations, available charging power) impact relative operational costs between manual and driverless trucks?
3. How would the spatiotemporal demand for different types of charging infrastructure be affected by an introduction of driverless trucks?

These areas may be relatively urgent to address, since future changes in charging demand could pose significant risks for short-term charging investment decisions due to the long economic planning horizons (over multiple decades), during which driverless trucks could start to impact charging demand.

7.4.5 Decarbonization impacts of driverless trucks

So far, the impacts of driverless trucks on the decarbonization of road freight has not been assessed from a life cycle and system-level perspective. Some effects and issues were raised in Section 7.2.2. Ideally, such studies should consider impacts on truck utilization, lifespan, fleet size and composition, energy consumption, and effects on transport demand and modal shifts. Given the many factors involved in such an analysis, one approach for obtaining policy-relevant insights could be to apply EMA and scenario discovery for identifying conditions under which driverless trucks have a positive net benefit (or the opposite) for freight decarbonization.

7.4.6 Exploring dynamic, adaptive policymaking

One perspective that is not addressed in this thesis is the issue of dynamic, adaptive policymaking (Walker, Haasnoot, et al., 2013), both for general transport climate policy and for transport planning in light of driverless trucks. Adaptive planning means that plans are designed to be adaptable over time in response to how the future unfolds, with the flexibility of the plan being a key means of achieving policy robustness over time (Maier et al., 2016).

Paper G takes a static perspective on transport climate policy analysis and focuses on static policy robustness. This approach is suitable for situations where investments or policies are costly or hard to change once implemented, such as infrastructure projects. However, several of the policy levers considered in Paper G, for instance, taxes or the share of mandated biofuel admixture, could, at least in principle, be varied dynamically and adjusted multiple times. A potential follow-up study for Paper G would be to extend the Scenario tool to multiple time steps and explore the use of dynamic adaptive policy pathways (Haasnoot et al., 2013).

Another research avenue is participatory studies with planners and policy makers to anticipate potential problems related to the deployment of driverless trucks at scale, and then devise hypothetical dynamic adaptive policies for managing them. Such studies have been used for several transport technology

related topics previously and may be valuable also for driverless trucks (Jittrapirom et al., 2018; Marchau et al., 2010; Walker & Marchau, 2017).

7.4.7 Understanding EMA and DMDU in the context of transport planning and policymaking practice

Potential challenges for implementing and adopting DMDU and EMA within current Swedish transport planning and policymaking are summarized in Section 6.4. These include the need to more fundamentally assess to what extent DMDU can, or cannot, be reconciled with the current practices which are based on a predictive approach and standardization in method, scenarios and valuation of outcomes. More work is required to answer whether DMDU can be integrated within transport planning practices to provide added value for decision-makers, and if so, for what types of analyses and how this can be achieved. Work also needs to be done to assess whether there are issues of perceived validity of DMDU methods and communication of DMDU results, and how these can be overcome.

One concrete first step for an introduction could be to pilot the addition of robustness and vulnerability analysis to the existing benefit-cost-analysis of infrastructure investment, potentially using a standardized set of parameter uncertainty ranges.

7.5 Concluding remarks

This thesis has explored potential long-term system-level impacts of driverless trucks on the Swedish freight transport system. The research contributes by developing future scenarios for the freight transport system, analyzing the Swedish innovation system for driverless trucks, and analyzing the system-level impacts of driverless trucks in a large number of scenarios using national-level freight modeling. Further, it demonstrates initial approaches for how driverless trucks can be modeled for cost performance analysis and system impact analysis on the national level, and how these modeling efforts can be enhanced through exploratory modeling and analysis and decision-making under deep uncertainty for improving model-based foresight and policy analysis.

The key findings indicate that driverless trucks could significantly reduce production costs for road freight transport, primarily through the elimination of in-vehicle driver costs. These cost reductions appear robust under various scenarios, even when accounting for increased acquisition costs and expenses related to remote operations. The change in cost structure could lead to increased road freight demand and modal shifts from rail and sea to road, which may have substantial implications for infrastructure planning and environmental sustainability. Moreover, the research shows that driverless trucks with limited operational domains could generate significant system-

level impacts, suggesting that full Level 5 automation is not required to realize substantial impacts although it would further increase the magnitude of effects.

The relationship between driverless trucks and electrification adds another layer of complexity. Findings suggest that driverless electric trucks may offer indirect benefits, such as reduced tare weight and energy consumption, potentially enhancing the attractiveness of electric trucks. In scenarios where electrification technologies are costly or less mature, driverless trucks could increase the electrification rate by compensating for higher costs through operational efficiencies. However, uncertainties remain regarding their overall impact on energy demand and decarbonization.

However, the modeling studies presented have several limitations. They do not account for feedback mechanisms between changes in transport supply and freight demand, lack integration with energy systems modeling, and do not consider fleet dynamics and potential technological development in other transport modes. Additionally, the analyses are specific to Sweden, and the findings may not be directly transferable to regions with different transport systems or regulatory environments.

While the literature on the impacts of driverless trucks on the transport system and society has developed in parallel with this thesis's development, it is still at an early stage, with this thesis being one of few studies having systematically assessed these questions.

To conclude this thesis a set of summarized takeaways for key stakeholders of this thesis is provided.

For the *freight modeling research community*:

- The future of road freight might be more diverse in that multiple types of trucks, differing in propulsion technology and automation level, may be operating in parallel. There is thus a need to reconsider how truck transport is modeled in large-scale freight models in terms of the level of detail of truck operations, cost functions, and the number of different truck types. The rapid technological development and uncertainty make it hard to predict future model needs. Thus, flexible models that can easily be adapted, modified and extended are preferable.
- The need for integration of multiple system levels. The impacts of driverless trucks occur on several system levels. On the truck level, there are impacts on operating costs and operational constraints. On the fleet and transport service level, there are impacts on utilization, planning and organization of freight services. On the freight system level, there are impacts on transport chain choices and other logistics decisions. There is a need for appropriate models of the impacts of driverless trucks on all these levels. However, equally important is to integrate findings from other levels, either through model integration

or through careful consideration when setting model inputs and designing scenarios.

- Support exploratory modeling. Uncertainty related to emerging technologies, such as driverless trucks, generates a need to model more scenarios to support exploratory modeling. Hence there is a need for fast, flexible models that are easy to interact with from external software to automate scenario generation and model execution. This is orthogonal to the two previous points: there is a need to work both on improvements of underlying freight transport models and to on tools supporting exploratory model use.

For the *freight transport industry*:

- Driverless trucks will likely have robustly lower total cost of ownership compared to manually driven trucks if the technology becomes sufficiently mature. The removal of an in-vehicle driver and, potentially, the driver's cabin may enable substantial labor reductions and possibly also manufacturing cost and weight benefits.
- The TCO approach is useful, but not sufficient, for understanding the economics and cost performance of driverless trucks. Driverless truck operations might incur costs in addition to vehicle ownership and operation costs such as additional costs for automated cargo handling, labor costs for operational support and, for some operating models, additional infrastructure such as "launch hubs" for the transfer hub model.
- Level 4 automation may be highly attractive if it enables driverless operations on key routes and infrastructure segments. While operating direct routes between production and consumption sites are preferable, operating models between terminals or transfer hubs can also be attractive options if sufficient operational efficiency is achieved. Having a sophisticated understanding of logistics and the freight market will be important for prioritizing ADS development to enable ODDs serving relevant freight flows.
- Fleet planning and efficient operations will become increasingly important and complex. With a higher share of fixed costs and lower marginal operating costs, efficient operations and high asset utilization of driverless trucks will be important to reduce total costs per unit of transport work. Achieving this will require sophisticated fleet planning. The removal of driver-related planning constraints increases the planning flexibility and allows, in theory, for a significant increase in truck utilization. However, more degrees of freedom increase the solution space and the need to also achieve a high utilization of associated resources such as remote operators and charging infrastructure further increases complexity. Additionally, with transport fleets using a mix of manually driven trucks and SAE level 4 driverless trucks, possibly with different ODDs and operational constraints, additional coordination and heterogeneity-related challenges may emerge.

For the *policy making and planning community*

- Driverless trucks at SAE level 4 might be deployed in a time-horizon that is relatively short compared to the timescales relevant for infrastructure planning. Such trucks could result in specific spatial characteristics of road freight by concentrating freight flows to certain parts of the infrastructure. It is hard to know whether such operating models would be a transient phenomenon or become more permanent.
- A multi-modal perspective is important for assessing system-level impacts of driverless trucks. This thesis suggests that driverless trucks may result in substantial changes not only to road freight, but also to the overall freight transport system. Driverless trucks may increase the relative cost performance of road freight, potentially rendering policies for achieving modal shift from road to rail and sea inefficient. However, these are early results that should be interpreted with caution. There is a need for more knowledge on the relationship between transport modes in a future with driverless trucks.
- Driverless trucks make decarbonization of road freight even more pressing. Their potential to increase road freight demand could increase emissions and an important area going forward is considering how driverless trucks can support an accelerated transition to ZETs.

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